Class Field Theory

— *EYNTKA* Series —

Nathanael Chwojko-Srawley

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Introduction

Class field theory is the classification of the abelian extensions of a local or global field K in terms of the arithmetic of K itself. It is the arithmetic capstone of the number-theory tower and the GL_1 case of the Langlands program: the place where the symmetries of polynomials (Galois theory) and the multiplicative arithmetic of a field (its ideal and idèle class groups) are revealed as two faces of one object. By the end of the book the abelian extensions of any global or local field will be classified, their primes matched to Galois elements through the *Artin reciprocity law*.

0.0.1 Prerequisites and Place in the Series This volume builds on *Algebraic Number Theory* [1], *Local Field Theory* [5], and *Group and Galois Cohomology* [4]. In turn, it is a prerequisite for *Shimura Varieties* [7] and *Langlands for* GL_2 [6]. The reader has access to every completed volume of the series but is not assumed to have read all of them; prerequisites are cited where they are invoked.

The book is organized in three chapters. Chapter 1 is a reminder of the group- and Galois-cohomology inputs the theory rests on (Tate cohomology and the Herbrand quotient, Hilbert 90, the Brauer group and its invariant map, and the fundamental class); these are stated as black boxes, proved in full in *Group and Galois Cohomology* [4], so that a reader willing to take them on faith can read the rest straight through. Chapter 2 develops the theory in the classical ideal-theoretic language of moduli, ray class groups, and conductors, culminating in Artin reciprocity. Chapter 3 recasts everything adelically, reaching the main theorem of global class field theory and its explicit payoffs: the Kronecker–Weber theorem, the Hilbert symbol and the reciprocity laws, the Hasse norm theorem, and the Chebotarev density theorem.

Consider a field extension L/K. There are many ways in which we may want to understand this extension. We may look at all the automorphisms $\mathrm{Aut}_K(L)$, which by galois theory shows that if L/K is finite this corresponds to permutations of roots of polynomials and hence it makes sense to consider the normal closure and separable polynomials, leading us to focus on Galois extensions. The galois group characterizes a lot of information about field extensions (and consequently, for polynomials and information about their solutions). There are still some open questions that an early exploration of Galois theory hasn't answered, for example:

1. What are the possible galois groups given a base field K? How ought they be classified?
2. Can we use solution mod primes to piece together more information about polynomials? How does this relate to the information about primes given by the galois group?
3. Can we also use this exploration to reveal more information about primes and how they relate (ex. are the more laws like quadratic reciprocity?)

Class Field Theory (CFT) provides the groundwork for answering exactly these questions in the case where L/K is *abelian*. The exploration of these results hinges on the fact that we may relate the primes of a Dedekind domain to galois groups via the Frobenius element in the case where L/K is an abelian extension. The nonabelian case is the generalization of these results and would be considered the beginning of Representation theory of galois groups and the Langland's program. By the end of this book, we shall fully classify all abelian extensions given a (global or local) base field K. We shall see that there is a bijective correspondence to (rational) primes[1] and the elements of the galois group via the *Artin Reciprocity Law*, which shall be the ultimate result of this book.

In the process, we shall fully classify all absolute abelian galois group $\mathrm{Gal}(K^{\mathrm{ab}}/K)$. As any abelian extension L/K has a galois group that is a quotient of the absolute galois group of K^{ab}/K, this group represents all (abelian) arithmetic information given base field K. The fact this corresponds to the (rational) prime and the units of K is an incredible feat. Understanding $\mathrm{Gal}(\overline{K}/K)$ is incredibly more difficult, and it is an active area of research. Due to the great increase in complexity, the abelian case can be thought of as a good first-case to study, especially that the non-abelian case is currently being approached through the tools of representation theory meaning that the abelian case can be thought of as the "1-dimensional representations".

[1] In particular, primes modulo principal primes

0.1 From Quadratic Reciprocity to Artin Reciprocity

When looking for solutions or irreducibility of integer polynomials, it is natural to want to find as much information as possible about the polynomial. One strategy is to find the solution's of a polynomials mod primes, and trying to piece together this information. As Galois groups holds information about the relation between the roots of polynomials, it would be nice if the "local" information provided by solving after modding can be linked to the "global" information provided by the Galois group. The Artin reciprocity can be interpreted as one part in achieving this result.

To demonstrate this, recall that p is a quadratic residue mod q ($n^2 \equiv p \bmod q$) if and only if $\left(\frac{p}{q}\right) = 1$ where $\left(\frac{p}{q}\right)$ is the Legendre symbol. This symbol satisfies the following equation:

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{q-1}{2}\frac{p-1}{2}}$$

Notice that we can consider the image of the Legendre symbol to be isomorphic to $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q}))$ (they are both $\cong \mathbb{Z}/2\mathbb{Z}$).It would more be interesting if there is a function from the primes of $\mathbb{Q}(\sqrt{q})/\mathbb{Q}$ to the galois group that commutes with the Legendre symbol:

$$\begin{tikzcd}
\{\text{primes of } \mathbb{Q}(\sqrt{q})/\mathbb{Q}\} \arrow[r, "\left(\frac{p}{q}\right)"] \arrow[d, "\mathrm{id}"] & \{-1,1\} \arrow[d, "\cong"] \\
\{\text{primes of } \mathbb{Q}(\sqrt{q})/\mathbb{Q}\} \arrow[r, "?"] & \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q}))
\end{tikzcd}$$

It would then be not too much a stretch to ask if we can replace the domain of these functions with the collection of all ideals of $\mathbb{Q}(\sqrt{q})/\mathbb{Q}$:

$$\begin{tikzcd}
\mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} \arrow[r, "\prod_p \left(\frac{p}{q}\right)"] \arrow[d, "\mathrm{id}"] & \{-1,1\} \arrow[d, "\cong"] \\
\mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} \arrow[r, "?"] & \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q}))
\end{tikzcd}$$

In fact, there is a connection: as $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q}))$ is abelian, then the Frobenius automorphism defined in [8, chapter 22.4][2] is exactly the map we can place at position "?", where the original condition of "$\left(\frac{p}{q}\right) = 1$ if and only if there is a quadratic residue" becomes "if and only if p splits completely":

$$\begin{tikzcd}
\mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} \arrow[r, "\prod_p \left(\frac{p}{q}\right)"] \arrow[d, "="] & \{-1,1\} \arrow[d, "\cong"] \\
\mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} \arrow[r, "\mathrm{Frob}_{\mathfrak{p}}"] & \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q}))
\end{tikzcd}$$

As nice as an observation this is, the question becomes whether this is a coincidence or a pattern; can $\mathrm{Gal}_{\mathbb{Q}}(K)$ (or even $\mathrm{Gal}_K(L)$) be replaced with another abelian galois group, where the abelian condition

[2]It shall be recalled in section 2.1

is necessary for the Frobenius automorphism to be well-defined, and we can find a corresponding "Legendre symbol" and a group in the codomain that would make the above diagram commute? Does this generalize when looking at cubic reciprocity, biquadratic reciprocity, and so forth (with the appropriate replacement of the Legendre symbol and its image)? This was a major endeavour in the 20th century: Hilbert's 23rd problem was

Find a proof of the most general reciprocity law for an arbitrary number field

As it turns out, such a law exists! The group that replaces $\{-1, 1\}$ will be the *ideal class group* Cl_K, and the Legendre symbol shall be generalized to the Artin symbol! The result is called *Artin Reciprocity*.

However, it turns out that most groups will not be isomorphic to the class group, but the *quotient* of the class group. The reader may rightfully wonder if there then possibly a maximal extension L/K where $\operatorname{Gal}_K(L)$ is isomorphic to the class group. As it will again turn out, the answer is yes and results from a generalization of Kronecker Weber's theorem! Thus, we shall spent a substantial amount of time showing there are maps that make this diagram commute:

$$\begin{tikzcd}
\{\text{primes of } K\} \arrow[r] \arrow[d, "="] & \operatorname{Cl}(K)/T_K \arrow[d, "\cong"] \\
\{\text{primes of } K\} \arrow[r] & \operatorname{Gal}_K(L)
\end{tikzcd}$$

and if $L = K_H$ is an appropriate "maximal" field, then:

$$\operatorname{Cl}(K) \cong \operatorname{Gal}_K(K_H)$$

This incredible unification of the properties of primes and the polynomial relationships that can be put to associate them can be thought of as a crowning achievement of number theory in the 20th century (where the polynomials relations are represented by the galois group, and the degree in which a prime ideal is away from being a prime element is represented in the class group). In the process of developing this theory, it will come out that there are two approaches to the problem: one done by focusing on primes and their generalizations (i.e. places), and the other focusing on *local fields* and putting together the information of local fields to find global results.

0.2 A Glimpse of the Nonabelian Horizon

The latter approach in the above will have the advantage of not needing to worry about ramifying primes and is be the beginnings of much research into finding a nonabelian version of *Artin Reciprocity* and is a very active area of research as of writing this book[3]. The non-abelian case requires new objects besides the primes of number fields as ideal class groups are abelian. Instead, an incredible observations in the 20th century was that we can capture the information given by primes by a generating function, and these shall be the appropriate replacement.

Though this is not part of the material of the book, it is nevertheless interesting to see how the content studied here will lead to modern number theory research, and so to illuminate a bit with a special case, take subset of polynomials $p(x, y)$ that are elliptic curves (polynomials of the form $y^2 = x^3 + ax + b$[4]). We can take the solution of this polynomial mod primes, and count the number of

[3] See *Langland's Program* for more details
[4] We are assuming we are over a field k where $\operatorname{char}(k) \neq 2, 3$

solutions, normalize them[5], and make them the coefficients of a generating function. This generating function will exhibit incredible symmetries and is hence given a name: a *modular form*. We may then associate each of these two *representation* of the galois group. This association has been done for elliptic curves, and it was enough to prove the infamous *Fermat's Last Theorem*, but the consensus is still out for the general Langland's correspondence!

[5]In particular take $p - \#S$ to normalize solution set to the as it is partly proportional to the size of the polynomial

0.3 The Two Languages

This is a graduate textbook in algebraic number theory; its prerequisites are the volumes listed above. Two consequences are worth previewing:

1. After proving the Chebotarev Density Theorem, we shall see that a galois extensions of number fields is uniquely determined by the primes of K that split completely in L. A fascinating consequence is that if a.e. primes of K split completely in L, then $L = K$.
2. We shall see something called the *Artin Reciprocity*. This is the step-1 of Langland's program.

Class field theory is about the study of finite galois extensions C_K of a field K with certain properties on their galois group:

1. It must be abelian
2. the primes must be unramified
3. all principal ideals split completely

The reader may be taking away from this that all the objects of interest will lay in the maximal abelian extensions of a field K, and this is correct. What will be the key object of study in class field theory will be the relations between these groups and the yet-to-be-defined generalized class group.

It will eventually be said that a *class field* over K is a field whose galois group is the quotient of the [generalized] class group of K, which will be an abelian group and whose quotient will always be finite. There will be multiple ways of describing the class group. From [1] it was defined for Dedekind domain to find how far away a Dedekind domain is from being principal. We shall see a way to describe them way *rays* and with *ideles*. The rays class groups are more algebraic in their definition, while the idele class is more topological in their definition. The ray class groups will be associated to fields which can be thought of as the generalization of the Kronecker-Weber Theorem.

The idelic language is useful, as it shall be shown in theorem 3.3.5 that *any* finite abelian extension L of K will have galois group isomorphic to a quotient of the idele class group of K, namely if I_K is the idele class group, then:

$$\mathrm{Gal}(L/K) \cong \frac{\mathbb{I}_K}{N}$$

If furthermore N has the property of being an open subgroup of finite index, then L will be a *class field.* In the case of number fields, for any finite field $K/\mathbb{Q}$, there will exist an extension of K known as the *Hilbert class field* of K where

$$\mathrm{Gal}_K(K_H) \cong \mathrm{Cl}\,(K)$$

where $\mathrm{Cl}\,(K)$ is the class group of K, namely the group representing how far away the ring $\overline{\mathbb{Z}}^K = \mathcal{O}_K$ is from being principal. This shows that there is a deep connection between the primes of the ring of integers and the symmetries of polynomials over fields. One of the main results that will be shown is that:

$$\begin{tikzcd}
\{\text{primes of } K\} \arrow[r] \arrow[d, dash, "="] & \mathrm{Gal}_K(K_H) \arrow[d, dash, "\cong"] \\
\{\text{primes of } K\} \arrow[r] & \mathrm{Cl}\,(K)
\end{tikzcd}$$

Note that the horizontal maps need not be isomorphism at all, however the right-vertical map will be an isomorphism!

A glimpse into motivation for why such maps can be interesting is by looking into the simpler case of $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ (where $\overline{\mathbb{Z}}^{\mathbb{Q}} = \mathbb{Z}$). In this case, every unramified prime $p \subseteq \mathbb{Z}$ corresponds to the Frobenius automorphism in $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$. The need for the prime to be unramified comes from the need for the inertia group to be trivial for the prime p to be uniquely associated to the Frobenius automorphism. Then recall that p is unramified in $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ if and only if $p \nmid n$. Thus, there is a natural map:

$$\{p \nmid n\} \to \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \qquad p \mapsto \mathrm{Frob}_p(\mathbb{Q}(\zeta_n)/\mathbb{Q}) = (\zeta_n \mapsto \zeta_n^p)$$

Or equivalently as the galois group is isomorphic to the $(\mathbb{Z}/n\mathbb{Z})^*$, it can be thought of the map sending $p \mapsto p \bmod n$.

As it shall turn out, an abelian extension is fully characterized by its splitting primes.

1

A Reminder of Galois Cohomology

Class field theory, in the cohomological development this book follows, rests on a short list of results from the cohomology of Galois groups. Each is constructed and proved in full in *Group and Galois Cohomology* [4]; this chapter collects the statements that the reciprocity theorems consume, so that the arithmetic can proceed without the cohomological machinery interrupting it. Nothing here is reproved: every boxed result carries a pointer to its proof upstream, and the body of the book treats these results as settled.

The chapter is meant to be read in either of two ways. A reader who has met this material can skim it as a notation reference and move on. A reader who has not can take the boxed statements as black boxes: each is stated precisely enough to be applied, and no later argument reopens the box. Taking them on faith costs only the proof, not the meaning, which the interpretive prose and the worked examples below supply. The one result to carry forward above all others is the reciprocity isomorphism of section 1.4: the entire book is the unfolding of its arithmetic content.

1.1 Tate Cohomology and the Herbrand Quotient

For a *finite* group G, ordinary cohomology and homology can be spliced into a single theory indexed by all of $\mathbb{Z}$. The device is the norm element $N_G = \sum_{\sigma \in G} \sigma \in \mathbb{Z}[G]$, which sends the coinvariants A_G into the invariants A^G; measuring its failure to be an isomorphism in degree zero glues the two half-lines together.

Definition 1.1.1: Tate Cohomology

Let G be a finite group and A a G-module. Write $N_G\colon A^G \to A^G$ for the map induced by the norm element and $I_G A$ for the submodule generated by $\{\sigma a - a\}$. The *Tate cohomology groups* are

$$\hat{H}^n(G,A) = \begin{cases} H^n(G,A) & n \geq 1 \\ A^G/N_G A & n = 0 \\ \ker N_G / I_G A & n = -1 \\ H_{-n-1}(G,A) & n \leq -2 \end{cases}$$

where $H_\bullet$ denotes group homology.

The positive degrees recover the cohomology already in hand and the negative degrees are the homology, with the norm map stitching them at $n = 0, -1$. The construction is exact in the strong sense that it converts short exact sequences into infinite long exact sequences with no endpoints, which is what makes it a computational tool rather than a bookkeeping convenience.

Theorem 1.1.2: Tate Cohomology Long Exact Sequence

Every short exact sequence $0 \to A \to B \to C \to 0$ of G-modules induces a long exact sequence

$$\cdots \to \hat{H}^n(G,A) \to \hat{H}^n(G,B) \to \hat{H}^n(G,C) \to \hat{H}^{n+1}(G,A) \to \cdots$$

infinite in both directions, and natural in the sequence.

The construction of Tate cohomology and the proof of the long exact sequence are given in the chapter on Tate cohomology of [4]. The following computation is the one used repeatedly in the reciprocity argument, and it already exhibits the two-sided periodicity that the next result explains.

Example 1.1.3: Tate Cohomology of $\mathbb{Z}$ with Trivial Action

Let G be cyclic of order n, acting trivially on $\mathbb{Z}$. Then $A^G = \mathbb{Z}$, the norm map is multiplication by n, and $I_G\mathbb{Z} = 0$, so

$$\hat{H}^0(G,\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}, \qquad \hat{H}^{-1}(G,\mathbb{Z}) = 0$$

In degree one $\hat{H}^1(G,\mathbb{Z}) = \operatorname{Hom}(G,\mathbb{Z}) = 0$, while in degree two $\hat{H}^2(G,\mathbb{Z}) \cong \operatorname{Hom}(G,\mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$. The pattern $\mathbb{Z}/n\mathbb{Z}, 0, \mathbb{Z}/n\mathbb{Z}, 0, \ldots$ repeats with period two in both directions.

The periodicity seen in the example is not special to $\mathbb{Z}$: for a cyclic group it holds for every module.

Theorem 1.1.4: Periodicity for Cyclic Groups

Let G be a finite cyclic group. For every G-module A and every $n \in \mathbb{Z}$ there is an isomorphism $\hat{H}^n(G,A) \cong \hat{H}^{n+2}(G,A)$, natural in A. In particular $\hat{H}^{-1}(G,A) \cong \hat{H}^1(G,A)$.

Periodicity collapses the whole $\mathbb{Z}$-graded theory of a cyclic group to two groups, $\hat{H}^0$ and $\hat{H}^1$. Their orders, when finite, combine into a single invariant that behaves multiplicatively and is therefore computable by reduction along exact sequences.

Definition 1.1.5: Herbrand Quotient

Let G be finite cyclic and A a G-module for which $\hat{H}^0(G,A)$ and $\hat{H}^1(G,A)$ are both finite. The *Herbrand quotient* is

$$h(A) = \frac{|\hat{H}^0(G,A)|}{|\hat{H}^1(G,A)|}$$

Proposition 1.1.6: Multiplicativity of the Herbrand Quotient

Let $0 \to A \to B \to C \to 0$ be a short exact sequence of G-modules. If any two of $h(A), h(B), h(C)$ are defined, so is the third, and $h(B) = h(A)\,h(C)$. Moreover $h(A) = 1$ whenever A is finite.

Multiplicativity plus vanishing on finite modules is exactly what lets one compute a Herbrand quotient by replacing a module with a commensurable one whose quotient is easier: the quotient is insensitive to finite perturbation. In the reciprocity proof this is applied to the unit groups and idèle class groups, where it produces the numerical index $[L : K]$ that the reciprocity map matches to a Galois group. The multiplicativity theorem and the finite-module vanishing are proved in [4].

Exercise 1.1.1

1. For G cyclic of order n acting trivially on $\mathbb{Z}$, confirm from definition 1.1.5 that $h(\mathbb{Z}) = n$, and explain why this is consistent with $h(A) = 1$ for finite A together with the sequence $0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$.
2. Using multiplicativity, compute $h(\mathbb{Q})$ for the trivial action of a cyclic group G of order n on $\mathbb{Q}$, and deduce $h(\mathbb{Q}/\mathbb{Z})$.

1.2 Hilbert's Theorem 90

The first structural input is the vanishing of a specific cohomology group of the multiplicative group of a Galois extension. It is the modern form of a theorem of Hilbert, itself a generalization of a result of Kummer, and it is what makes the Brauer group of the next section computable.

Theorem 1.2.1: Hilbert's Theorem 90

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$. Then

$$H^1(G, L^\times) = 0$$

In the additive form, $H^n(G, L^+) = 0$ for all $n \geq 1$, where L^+ is the additive group of L.

The multiplicative statement is the one used below: it is the vanishing that lets $H^2(G, L^\times)$ be identified with the relative Brauer group and, through the class formation axioms, underlies reciprocity. The additive statement is stronger in appearance but softer in content: by the normal basis theorem L is a free $K[G]$-module, hence cohomologically trivial, so all its higher additive cohomology vanishes. Both are proved in [4]. The classical face of the multiplicative statement is the description of norm-one elements.

Example 1.2.2: The Cyclic Norm Form

Suppose L/K is cyclic with $G = \langle\sigma\rangle$. For a cyclic group $\hat{H}^{-1}(G, L^\times) \cong H^1(G, L^\times) = 0$ by periodicity (theorem 1.1.4) and theorem 1.2.1. Unwinding $\hat{H}^{-1} = \ker \mathrm{Nm}_{L/K}/I_G L^\times$, this says an element $\alpha \in L^\times$ has $\mathrm{Nm}_{L/K}(\alpha) = 1$ if and only if $\alpha = \beta/\sigma(\beta)$ for some $\beta \in L^\times$. This is the original "Theorem 90": the norm-one elements are exactly the multiplicative σ-boundaries.

Exercise 1.2.1

1. Deduce from theorem 1.2.1 that for a cyclic extension L/K of degree n, the group $\mathrm{Br}(L/K) = H^2(\mathrm{Gal}(L/K), L^\times)$ is a quotient of $K^\times/\mathrm{Nm}_{L/K}(L^\times)$. (Use periodicity to relate $\hat{H}^2$ to $\hat{H}^0$.)

1.3 The Brauer Group and the Invariant Map

Hilbert 90 identifies the degree-two cohomology of $L^\times$ with an object of independent arithmetic meaning: the Brauer group, which classifies central simple algebras and, over a local field, is as simple as a cohomology group can be.

Definition 1.3.1: Brauer Group

The *Brauer group* of a field K is $\mathrm{Br}(K) = H^2(G_K, (K^{\mathrm{sep}})^\times)$, where $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$. For a finite Galois extension L/K the *relative Brauer group* is $\mathrm{Br}(L/K) = H^2(\mathrm{Gal}(L/K), L^\times)$, which is the kernel of the restriction $\mathrm{Br}(K) \to \mathrm{Br}(L)$; equivalently it classifies the central simple K-algebras split by L.

The two descriptions, cohomological and algebra-theoretic, agree because a central simple algebra split by L is a crossed product, encoded by a 2-cocycle. This book needs only the cohomological description, and only one computation of it: over a local field the Brauer group is a copy of $\mathbb{Q}/\mathbb{Z}$, recorded by a single rational number.

Theorem 1.3.2: The Invariant Map

Let K be a nonarchimedean local field. There is a canonical isomorphism

$$\mathrm{inv}_K : \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$$

Under it the relative Brauer group of the unramified extension of degree m is the subgroup $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$. For the archimedean local fields, $\mathrm{inv}_{\mathbb{R}} : \mathrm{Br}(\mathbb{R}) \xrightarrow{\sim} \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ and $\mathrm{Br}(\mathbb{C}) = 0$.

The invariant map is the single most substantive input from the cohomology side: it says that over a local field a central simple algebra is classified by one rational number modulo the integers, its *Hasse invariant*, and that unramified algebras occupy the predicted cyclic subgroup. The construction, via the cohomology of the maximal unramified extension, is in [4]. The generating example over $\mathbb{R}$ is the one every reader already knows.

Example 1.3.3: The Hamilton Quaternions
The Hamilton quaternions $\mathbb{H}$ form the unique nonsplit central simple algebra over $\mathbb{R}$, so they represent the nonzero class of $\mathrm{Br}(\mathbb{R}) = \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ and $\mathrm{inv}_{\mathbb{R}}(\mathbb{H}) = 1/2$. Every other central simple $\mathbb{R}$-algebra is Morita equivalent to either $\mathbb{R}$ (invariant 0) or $\mathbb{H}$ (invariant 1/2).

Exercise 1.3.1

1. Let K be a nonarchimedean local field and L/K the unramified extension of degree m. Using theorem 1.3.2, determine $\mathrm{Br}(L/K)$ as a subgroup of $\mathbb{Q}/\mathbb{Z}$ and its order, and reconcile the order with exercise 1.2.1 (1).

1.4 The Fundamental Class and the Cohomological Reciprocity Isomorphism

The previous three sections combine into the one result the whole book develops. The invariant map singles out a canonical degree-two class; cup product with it shifts Tate cohomology by two degrees; and in the degree where the shift lands on the abelianized Galois group, the result is an isomorphism between an arithmetic quotient and a Galois group. This is reciprocity, in its cohomological form.

Theorem 1.4.1: Local Fundamental Class and Tate–Nakayama

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. The *fundamental class* $u_{L/K} \in H^2(G, L^\times)$ is the unique class with $\mathrm{inv}_K(u_{L/K}) = 1/[L:K]$. Cup product with it is an isomorphism

$$\hat{H}^r(G,\mathbb{Z}) \xrightarrow{-\smile u_{L/K}} \hat{H}^{r+2}(G, L^\times)$$

for every $r \in \mathbb{Z}$.

Tate–Nakayama is the engine: it converts the cohomology of the trivial module $\mathbb{Z}$, which sees only the group G, into the cohomology of $L^\times$, which sees the arithmetic. Setting $r = -2$ lands on the abelianization, because $\hat{H}^{-2}(G,\mathbb{Z}) \cong G^{\mathrm{ab}}$, and produces the reciprocity isomorphism.

Corollary 1.4.2: The Local Reciprocity Isomorphism

With L/K and G as above, there is a canonical isomorphism

$$\mathrm{Gal}(L/K)^{\mathrm{ab}} \xrightarrow{\sim} K^\times/\mathrm{Nm}_{L/K}(L^\times)$$

the local Artin map, obtained from theorem 1.4.1 at $r = -2$ using $\hat{H}^{-2}(G,\mathbb{Z}) \cong G^{\mathrm{ab}}$ and $\hat{H}^0(G, L^\times) = K^\times/\mathrm{Nm}_{L/K}(L^\times)$.

The same mechanism runs globally, with the idèle class group in place of the multiplicative group. Here the role of the invariant map is played by the sum of local invariants, whose exactness is the theorem of Albert, Brauer, Hasse, and Noether.

Theorem 1.4.3: Albert–Brauer–Hasse–Noether

For a global field K the sequence

$$0 \to \mathrm{Br}(K) \to \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$$

is exact, the sum running over all places v of K. Consequently the pair $(G_K, C_{K^{\mathrm{sep}}})$ is a class formation, where $C_K = \mathbb{I}_K/K^\times$ is the idèle class group, and for every finite Galois L/K cup product with the global fundamental class gives the *global reciprocity isomorphism*

$$\mathrm{Gal}(L/K)^{\mathrm{ab}} \xrightarrow{\sim} C_K/\mathrm{Nm}_{L/K}(C_L)$$

The exact sequence says two things at once: a central simple algebra over a global field is determined by its local invariants (injectivity), and those invariants can be prescribed arbitrarily subject to the single constraint that they sum to zero (exactness in the middle and surjectivity). The class formation structure it produces is the exact global analogue of the local invariant map, and the reciprocity isomorphism it yields is the theorem this book exists to develop. Its proof, along with the global fundamental class, is in [4], with the one analytic input it requires supplied in section 2.6.

Example 1.4.4: Invariants Sum to Zero

Over $K = \mathbb{Q}$, the quaternion algebra $\left(\frac{-1,-1}{\mathbb{Q}}\right)$ ramifies exactly at the places 2 and ∞, with $\mathrm{inv}_2 = \mathrm{inv}_\infty = 1/2$ and $\mathrm{inv}_p = 0$ for all other p. The invariants sum to $1/2 + 1/2 = 1 = 0$ in $\mathbb{Q}/\mathbb{Z}$, as theorem 1.4.3 requires. No central simple $\mathbb{Q}$-algebra ramifies at a single place, because a single $1/2$ cannot sum to zero.

Exercise 1.4.1

1. Explain why the reciprocity isomorphism of corollary 1.4.2 is usually written as a map $K^\times \to \mathrm{Gal}(L/K)^{\mathrm{ab}}$ (rather than its inverse), and what the norm subgroup $\mathrm{Nm}_{L/K}(L^\times)$ corresponds to under it.

2. Determine which finite sets of places of $\mathbb{Q}$ can occur as the exact ramification set of a quaternion algebra over $\mathbb{Q}$, and justify your answer using theorem 1.4.3.

2

Class Field Theory: Places Approach

In the 19th century, Gauss discover that any square-root $\sqrt{p}$ can be written as a sum of roots, where the coefficients is the Legendre symbol:

$$\sqrt{p} = \sum_{n=0}^{p-1} \left(\frac{n}{p}\right) \zeta_p^n$$

He thus found another proof of quadratic reciprocity by seeing which $\sigma \in \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{p})$ split or are invariant[1]. This turns out to generalize reciprocity laws for higher powers (ex 3rd or 4th, different equations). However, the base field for these laws wasn't always $\mathbb{Q}$, for example for the biquadratic reciprocity law the base field was $\mathbb{Q}(i)$. Nevertheless, It was clear that in some way it was key to have these reciprocity laws. What was realized in the 20th century was that the extensions have to be *abelian* and *unramified*. As $\mathbb{Q}$ cannot have unramified extensions, and the result we need generalize the usual reciprocity cases, we have the *conductor* (which shall be denoted $\mathfrak{m}$) to represent the primes that ramify. This realization will tie together the class group and the galois group in what is one of the most important results of the 20th century.

2.1 Goal: Artin Map

Recall that if L/K is a finite abelian galois extension of global fields, and $\mathfrak{P}/\mathfrak{p}$, then there exists an element $\sigma \in \mathrm{Gal}_K(L)$ called he Frobenius element, which is the pre-image given by the map $D_{\mathfrak{p}}(\mathfrak{P}) \to \mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_q)$, or (as was proven to be equivalent) the element σ_p such that $\sigma_p(x) = x^{\mathfrak{N}(\mathfrak{p})} \mod \mathfrak{p}$. This

[1]Though he did not phrase it in this way, as Galois theory was not yet so well established

element is often denoted Frob_p, and if L/K where not abelian this would represent the conjugacy classes of σ_p. This can be represented another way.

Definition 2.1.1: Artin Symbol

Assume $AKLBG$ with finite residue fields. Then for each prime $\mathfrak{P}$ over $\mathfrak{p}$, define the *Artin symbol* to be:

$$\left(\frac{L/K}{\mathfrak{P}}\right) := \sigma_{\mathfrak{P}}$$

If L/K is abelian, then it is unambiguous to write:

$$\left(\frac{L/K}{\mathfrak{p}}\right) := \sigma_{\mathfrak{p}}$$

The simplest case for the Artin symbol if when $\mathfrak{p}$ splits completely, since then $D_{\mathfrak{p}}(\mathfrak{P}) = \{e\}$ and so

$$\left(\frac{L/K}{\mathfrak{P}}\right) = 1$$

Though this definition may seem a little strange with the notation, it shall turn out that an extension L/K of global fields can be completely determined by the set of primes $\mathfrak{p}$ that split completely in L! In terms of equations, if $f(x) \equiv 0 \bmod \mathfrak{p}$, then $\sigma = \left(\frac{\mathrm{Gal}_K(L)/K}{\mathfrak{p}}\right)$ must have at least one root, and if $f(x) \bmod \mathfrak{p}$ splits completely into linear terms, $\sigma = \mathrm{id}$.

A simple immediate thing we shall prove is:

Proposition 2.1.2: Artin Symbol Under Intermediate Fields

Le $AKLBG$ with finite residue field and $\mathfrak{P}/\mathfrak{p}$ unramified. Let E be an intermediate field extension of L and L, and let $\mathfrak{P}_E = \mathfrak{P} \cap E$. Then:

$$\left(\frac{L/E}{\mathfrak{P}}\right) = \left(\frac{L/K}{\mathfrak{P}}\right)^{[\mathbb{F}_{\mathfrak{P}_E}:\mathbb{F}_{\mathfrak{p}}]}$$

Furthermore, if E/K is galois, then $\left(\frac{E/K}{\mathfrak{P}_E}\right)$ is the restriction of $\left(\frac{E/K}{\mathfrak{P}_E}\right)$ to E.

***Proof*:**
The first comes from field theory, namely use $\#\mathbb{F}_{\mathfrak{P}_E} = (\#\mathbb{F}_{\mathfrak{p}})^{[\mathbb{F}_{\mathfrak{P}_E}:\mathbb{F}_{\mathfrak{p}}]}$. The second result come from the factorization in intermediary fields of of primes in galois extensions, see [1].

With this setup, we may define the following map. Let S be the set of ramified primes in L/K for global fields L and K. We then know that S is finite. If $\mathcal{I}_A$ is the free abelian group generated by the primes of a Dedekind domain A, let $\mathcal{I}_A^S$ be the free abelian group generated by all primes not in S. Then:

Definition 2.1.3: Artin Map

Let A be a Dedekind domain with finite residue fields. Let L/K be a finite abelian extension where $K = \mathrm{Frac}(A)$, and let S be the set of primes of A that ramify in L. Then there is the homomorphism:

$$\left(\frac{L/K}{\cdot}\right) : \mathcal{I}_A^S \to \mathrm{Gal}_K(L) \qquad \prod_i^m \mathfrak{p}_i^{e_i} \mapsto \prod_i^m \left(\frac{L/K}{\mathfrak{p}_i}\right)^{e_i}$$

called the *Artin map*.

A fascinating result we shall prove later is that this map is *surjective*, meaning all the elements of the galois group have a fiber of Frobenius elements (further motivating a generalization of K.W.). By the earlier discussion we have that

$$\ker\left(\frac{L/K}{\cdot}\right) = \text{all primes that split completely}$$

Example 2.1.4: Artin Map Practice

Consider $K = \mathbb{Q}$ and $L_q = \mathbb{Q}(\sqrt{q})$ where q is an odd prime. So that $\mathrm{Gal}_K(L_q) \cong \mathbb{Z}/2\mathbb{Z}$. The only prime that ramifies is q, hence $S = \{q\}$. Letting $A = \overline{\mathbb{Z}}_q^L$. Then for any other prime we have:

$$\left(\frac{L_q/K}{p}\right) = \pm 1$$

where $\left(\frac{L_q/K}{p}\right) = 1$ if p splits in L_q (and hence splits completely as the extension is only of degree 2). By [1], any prime $p\mathcal{O}_{L_q}$ factors if $q\mathcal{O}_{L_p}$ factors and p or q is equivalent to 1 mod 4.

For notational convenience, let us change about the notation. Let $\mathfrak{m} \subseteq \mathcal{O}_K$ be the minimal ideal divisible by every ramified prime of K, and let $\mathcal{I}_K^{\mathfrak{m}} \subseteq \mathcal{I}_K$ be the sub of the ideal class group where $\nu_{\mathfrak{p}}(I) = 0$ for all $\mathfrak{p}\big|\mathfrak{m}$. Then we can re-write the morphism as :

$$\psi_{L/K}^{\mathfrak{m}} : \mathcal{I}_K^{\mathfrak{m}} \to \mathrm{Gal}_K(L)$$
$$\prod_{\mathfrak{p}\nmid\mathfrak{m}} \mathfrak{p}^{n_{\mathfrak{p}}} \mapsto \prod_{\mathfrak{p}\nmid\mathfrak{m}} \left(\frac{L/K}{\mathfrak{p}}\right)^{n_{\mathfrak{p}}}$$

For a moment, let's say that $\psi_{L/K}^{\mathfrak{m}}$ is surjective. Then the kernel would be all unramified elements for which the Frobenius element is trivial, namely it will contain all primes that *split completely*. An important result that we shall need later is the following:

Proposition 2.1.5: Tower For Artin Map

Let $K \subseteq L \subseteq M$ be a tower of finite abelian extensions of global field and let $\mathfrak{m} \subseteq \mathcal{O}_K$ be an ideal divisible by all primes $\mathfrak{p} \subseteq \mathcal{O}_K$ that ramify in M. Then the following diagram commutes:

$$\begin{tikzcd} \mathcal{I}_K^{\mathfrak{m}} \arrow[r, "\psi_{M/K}^{\mathfrak{m}}"] \arrow[dr, "\psi_{L/K}^{\mathfrak{m}}"'] & \mathrm{Gal}_K(M) \arrow[d, "\sigma \mapsto \sigma|_L"] \\ & \mathrm{Gal}_K(L) \end{tikzcd}$$

Proof:
It follows from proposition 2.1.2.

Special case: $\mathbb{Q}$

Let us focus on the case where $K = \mathbb{Q}$ and a finite abelian extension $L/\mathbb{Q}$. In this case, the Kronecker-Weber Theorem shows that every abelian extension $L/\mathbb{Q}$ lies in a Cyclotomic extension $\mathbb{Q}(\zeta_m)$. By field theory we get a canonical map:

$$\omega : \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_m)) \xrightarrow{\cong} (\mathbb{Z}/m\mathbb{Z})^*$$

which can be characterized by $\sigma(\zeta_m) = \zeta_m^{\omega(\sigma)}$. In cyclic extensions, the primes that ramify are those where $p \big| m$. By definition, $(\mathbb{Z}/m\mathbb{Z})^*$ contains all elements that are coprime to m. Furthermore, for each prime $p \nmid m$, σ_p is the unique automorphism where $\sigma(x) \equiv x^p \mod \mathfrak{p}$ for all $\mathfrak{p}/p$ (well-defined as the galois group is abelian). In particular, for each prime p, for all $\mathfrak{p}/p$, the Decomposition group surjects onto the galois group of finite extensions, giving us a Frobenius element, and as we are abelian it in fact gives *the* Frobenius element for p. As we have characterized the galois group automorphisms who's power is coprime to m, we can explicitly write out ω as a map from Frobenius elements:

$$\omega(\sigma_p) = p \mod m$$

As a consequence, we can define an inverse map sending every integer a coprime to m to:

$$\begin{aligned} \omega^{-1} &: (\mathbb{Z}/m\mathbb{Z})^* \to \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_m)) \\ \omega^{-1}(\overline{a}) &= \left(\frac{\mathbb{Q}(\zeta_m)/\mathbb{Q}}{(a)} \right) \qquad (a) \in \mathcal{I}_{\mathbb{Q}}^m \end{aligned}$$

Next notice that for each $(a) \in \mathcal{I}_{\mathbb{Q}}^m$ a must be coprime to m, so that it is not too much work to show that the map extends to a surjective map:

$$\psi_{\mathbb{Q}(\zeta_m)/\mathbb{Q}}^m : \mathcal{I}_{\mathbb{Q}}^m \twoheadrightarrow \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_m))$$

Finally, by Kronecker-Weber we have that $L \subseteq \mathbb{Q}(\zeta_m)$. The galois group of L is a quotient of $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_m))$, and for any prime that ramifies in L certainly ramifies in $\mathbb{Q}(\zeta_m)$ and so we can keep the same m for the group $\mathcal{I}_{\mathbb{Q}}^L$. By proposition 2.1.5 we get the following surjective map:

$$\psi_{L/\mathbb{Q}}^m : \mathcal{I}_{\mathbb{Q}}^m \twoheadrightarrow \mathrm{Gal}_{\mathbb{Q}}(L)$$

completing the proof in the simplest case. To keep track of everything, here is a diagram:

$$\begin{tikzcd}
\mathcal{I}_{\mathbb{Q}}^m \arrow[r, shift left, "\psi:(a)\mapsto\left(\frac{\mathbb{Q}(\zeta_m)/\mathbb{Q}}{a}\right)"] \arrow[dr, "\operatorname{res}\circ\psi"'] & \operatorname{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_m)) \arrow[l, shift left, "\omega"] \arrow[d, "\sigma\mapsto\sigma_L"] \\
& \operatorname{Gal}_{\mathbb{Q}}(L)
\end{tikzcd}$$

Let us now introduce the key ideas and vocabulary that will be necessary to generalize this result to replace $\mathbb{Q}$ with an arbitrary global number field K. These terms shall be made more precise after we the need arises:

- **Existence**: For each integer m, we have a *ray class field* $\mathbb{Q}(\zeta_m)$ which is an abelian extension that unramified only at $p\big|m$ (or equivalently a maximally unramified extension up to the primes $p\big|m$) with galois group isomorphic to the *ray class group* $(\mathbb{Z}/m\mathbb{Z})^*$
- **Completeness**: Every abelian extension of $\mathbb{Q}$ lies in a ray class field
- **Reciprocity**: If $L/\mathbb{Q}$ is abelian and contains in a ray class field $\mathbb{Q}(\zeta_m)$, the Artin map $\mathcal{I}_{\mathbb{Q}}^m \to \operatorname{Gal}_{\mathbb{Q}}(L)$ induces a surjective homomorphism from the ray class group to the galois group $\operatorname{Gal}_{\mathbb{Q}}(L)$, letting us view $\operatorname{Gal}_{\mathbb{Q}}(L)$ as a quotient of the ray class group, in this case $(\mathbb{Z}/m\mathbb{Z})^*$.

2.1.1 Moduli and Ray class Groups

Let us first generalize the role of m, which kept track of the primes that ramify. These primes are all associated to a valuation, which themselves are associated to absolute values. Recall that there are two absolute value not associated to any prime: the trivial and the Archimedean. We ignore the trivial and consider the equivalence classes of absolute values, in other words we are going to be considering *places*.

Definition 2.1.6: Modulus

Let K be a number field. Then a *modulus* or *cycle* $\mathfrak{m}$ for K is a function $M_K \to \mathbb{N}$ with finite support such that if $\nu|\infty$ we have $\mathfrak{m}(\nu) \leq 1$ with $\mathfrak{m}(\nu) = 0$ unless ν is a real place. The modulus is often seen as a formal product $\prod \nu^{\mathfrak{m}(\nu)}$ over M_k which can factor as:

$$\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty \qquad \mathfrak{m}_0 = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{\mathfrak{m}(\mathfrak{p})} \qquad \mathfrak{m}_\infty = \prod_{\nu|\infty} \nu^{\mathfrak{m}(\nu)}$$

The modulus $\mathfrak{m}_0$ can be interpreted as an $\mathcal{O}_K$ ideal, and $\mathfrak{m}_\infty$ represents the real places of K. We often write $\#\mathfrak{m}_\infty$ to denote the number of real places in the support of $\mathfrak{m}$. If $\mathfrak{m}, \mathfrak{n}$ are moduli for K, we say that $\mathfrak{m}$ *divides* $\mathfrak{n}$ if $\mathfrak{m}(\nu) \leq \mathfrak{n}(\nu)$ for all $\nu \in M_K$. Similarly, the product is:

$$\mathfrak{mn}(\nu) = \mathfrak{m}(\nu) + \mathfrak{n}(\nu) \qquad \mathfrak{mn}(\nu) = \max(\mathfrak{m}(\nu) + \mathfrak{n}(\nu), 1)$$

where the first is for $\nu \nmid \infty$ and the second for $\nu \big| \infty$. Define also:

$$\gcd(\mathfrak{m},\mathfrak{n})(\nu) = \min(\mathfrak{m}(\nu),\mathfrak{n}(\nu)) \qquad \gcd(\mathfrak{m},\mathfrak{n})(\nu) = \max(\mathfrak{m}(\nu),\mathfrak{n}(\nu))$$

The trivial modulus is the modulus whose $\mathcal{O}_K$ ideal is (1), i.e. $\mathfrak{m}_0 = (1)$,and $\#\mathfrak{m}_\infty = 0$. The modulus shall be similar to the discriminant in it shall determine ramification, however unlike the discriminant of a field extension $K/\mathbb{Q}$ which determine which primes $(p) \subseteq \mathbb{Q}$ ramify in K, the modulus of K will determine which primes in K ramify in the Ray class field (which will be denoted $K(\mathfrak{m})$). The modulus will insure that these primes are ignored. We shall see that there are multiple choices of modulus, and that there is a natural minimal choice; this modulus shall be called the *conductor* and is related to the conductor as presented in [1].

Given this requirement, we have the following series of definitions:

Definition 2.1.7: Coprime Fractional Ideal and modulus

Let $I \in \mathcal{I}_K$ be a fractional ideal. Then $\mathcal{I}$ is said to be *coprime* if

$$\nu_\mathfrak{p}(I) = 0 \qquad \forall \mathfrak{p}\big|\mathfrak{m}_0$$

Definition 2.1.8: Ray Group and Ray Class Group

Let $K/\mathbb{Q}$ be a number field. Then:

1. $\mathcal{I}_K^\mathfrak{m} \subseteq \mathcal{I}_K$ is the subgroup of fractional ideals coprime to $\mathfrak{m}$
2. $K^\mathfrak{m} \subseteq K^\times$ is the subgroup of elements $\alpha \in K^\times$ where $(a) \in \mathcal{I}_K^\mathfrak{m}$
3. $K^{\mathfrak{m},1} \subseteq K^\mathfrak{m}$ is the subgroup of elements $\alpha \in K^\mathfrak{m}$ where

$$\nu_\mathfrak{p}(\alpha - 1) \geq \nu_\mathfrak{p}(\mathfrak{m}_0)$$

   for all $\mathfrak{p}\big|\mathfrak{m}_0$ and $\alpha_\nu > 0$ for $\nu\big|\mathfrak{m}_\infty$ (where α_ν is the image of $K \hookrightarrow K_\nu \cong \mathbb{R}$)
4. $\mathcal{R}_K^\mathfrak{m} \subseteq \mathcal{I}_K^\mathfrak{m}$ is the subgroup of principal fractional ideals $(\alpha) \in \mathcal{I}_K^\mathfrak{m}$ with $\alpha \in K^{\mathfrak{m},1}$

The group $\mathcal{R}_K^\mathfrak{m}$ is called the *ray group* with modulus $\mathfrak{m}$. The *ray class group* of modulus $\mathfrak{m}$ is the quotient:

$$\mathrm{Cl}_K^\mathfrak{m} = \frac{\mathcal{I}_K^\mathfrak{m}}{\mathcal{R}_K^\mathfrak{m}}$$

Definition 2.1.9: Wide and Narrow Ray Class Group

Let $\mathrm{Cl}_K^\mathfrak{m}$ be a ray class group. Then if $\mathfrak{m}(\nu) = 1$ for every real place, then $\mathrm{Cl}_K^\mathfrak{m}$ is called a *narrow ray class group*. A narrow ray class group where $\mathfrak{m}_0 = (1)$ is calle a *narrow class group*.

In this context, the usual class group is sometimes called the *wide class group*.

If $\mathfrak{m}$ is trivial, namely $\mathfrak{m}_0 = (1)$ and $\mathfrak{m}_\infty = 0$ the ray class group is the class group. The $K^{\mathfrak{m},1}$ group given by $\nu_\mathfrak{p}(\alpha - 1) \geq \nu_\mathfrak{p}(\mathfrak{m}_0)$ (for each prime) translates to in the case where we are only working with finite primes with the condition on representative:

$$(\alpha - 1) = \mathfrak{p}^{\nu_{\mathfrak{p}}(\alpha-1)}\mathfrak{q} \qquad \Rightarrow \qquad \alpha \equiv 1 \bmod \mathfrak{p}^{\nu_{\mathfrak{p}}(\mathfrak{m}_0)}$$

In the case with fractional ideals, we require the generalization. We also require that if there is any real places that α be positive with respect to to the embedding. In the concrete case where $K = \mathbb{Q}$ and $\mathfrak{m}_0$ contains of all ramified primes, we are asking that for each $\mathfrak{p} \in S$, we take the $\alpha \in K^{\mathfrak{m}}$ such that

$$\alpha \equiv 1 \bmod \mathfrak{p} \qquad \forall \mathfrak{p} \in S$$

that is we are taking all the elements that map to 1 mod $\mathfrak{p}$. This can be thought of as taking the "units" of the ring of integers and adding the infinite place is because p mod m is not the same as $-p$ mod m and so we would only want to sick to p mod m (adding $-p$ mod m will usually make us only get the real field associated to the modulus). The fact $\mathcal{O}_K^\times \cap K^{\mathfrak{m},1}$ injects into $\mathcal{O}_K^{\times}$[2]. This group shall end up being the kernel of the Artin map.

Definition 2.1.10: Ray Class Field

Let $\mathfrak{m}$ be a modulus. Then a finite abelian extension L/K that is unramified at all places not in $\mathfrak{m}$ for which the kernel of the Artin map:

$$\psi_{L/K}^{\mathfrak{m}} : \mathcal{I}_K^{\mathfrak{m}} \to \mathrm{Gal}_K(L)$$

is equal to the ray group $\mathcal{R}_K^{\mathfrak{m}}$ ($\ker \psi_{L/K}^{m} = \mathcal{R}_K^{\mathfrak{m}}$) is called the *ray class field* for modulus $\mathfrak{m}$.

In the definition, we say "the" ray class field, though we have not yet proved there is a unique modulus $\mathfrak{m}$ associated to each such field. This will be given soon.

Example 2.1.11: Ray Class Groups

Let $K = \mathbb{Q}$ and $\mathfrak{m} = (5)$. Then $K^{\mathfrak{m}}$ contains all elements $a/b \in \mathbb{Q}$ (with representative where $\gcd(a, b) = 1$) such that there is no factor of 5. That gives

$$\mathbb{Q}^{\mathfrak{m}} = \left\{ \frac{a}{b} \;\middle|\; a, b \not\equiv 0 \bmod 5 \right\}$$

For $\mathbb{Q}^{\mathfrak{m},1}$, we need need to include all elements such that $a/b \equiv 1 \bmod 5$, or $a \equiv b \bmod 5$. Coupled with the above condition, that gives:

$$\mathbb{Q}^{\mathfrak{m},1} = \left\{ \frac{a}{b} \;\middle|\; a \equiv b \not\equiv 0 \bmod 5 \right\}$$

Then:

$$\begin{aligned} \mathcal{I}_K^{\mathfrak{m}} &= \{(1), (2), (3), (4), (1/2), (1/3), (2/3), (3/2), (1/4), (3/4), (4/3), (1/6), (6), ...\} \\ \mathcal{R}_K^{\mathfrak{m}} &= \{(1), (4), (2/3), (3/2), (1/4), (6), (2/7), (7/2), ...\} \end{aligned}$$

Note that $(2/3) \in \mathcal{R}_K^{\mathfrak{m}}$ even though $2/3 \notin K^{\mathfrak{m},1}$ since $-2/3 \in K^{\mathfrak{m},1}$ and $(-2/3) = (2/3)$. Similarly, $4 \in \mathcal{R}_K^{\mathfrak{m}}$ as $-4 = -4/1$ satisfies the congruence. This shall add all elements that are "positive conjugate". Hence:

$$\mathrm{Cl}_K^{\mathfrak{m}} = \mathcal{I}_K^{\mathfrak{m}}/\mathcal{R}_K^{\mathfrak{m}} = \{[(1)], [(2)]\} \cong (\mathbb{Z}/5\mathbb{Z})^\times/\{\pm 1\}$$

[2] More on shall be made precise in theorem 2.1.13

This group is isomorphic to the galois group of the totally real subfield $\mathbb{Q}(\zeta_5)^+ \subseteq \mathbb{Q}(\zeta_5)$, which is the ray class for this modulus (as you can verify). If we let $\mathfrak{m} = (5)\infty$, then all the ideals whose congruence was achieved by going into the negatives disappear $(-2/3, -4/1, ...)$ and so we change $\mathcal{R}_K^{\mathfrak{m}}$ to:

$$\mathcal{R}_K^{\mathfrak{m}} = \{(1), (6), (1/6), (2/7), (7/2), ...\}$$

and $\mathrm{Cl}_K^{\mathfrak{m}} \cong (\mathbb{Z}/5\mathbb{Z})^\times$ and the ray class field becomes $\mathbb{Q}(\zeta_5)$. This motivates keeping the real places, as without them we shall get many instances where we have real subfields. Furthermore, notice the resemblance of the ray class group and the modular condition on $K^{\mathfrak{m}}$, namely they are strongly related.

As we see in the example, there is often a difference between $K^{\mathfrak{m},1}$ and $K^{\mathfrak{m}}$. We can quantify exactly how they are different through an appropriate exact sequence. To define it, we first require the following lemma

Lemma 2.1.12: Coprime Ideals in Class Group

Let A be a Dedekind domain and let $\mathfrak{a} \subseteq A$ be an ideal. Then every ideal class in $\mathrm{Cl}(A)$ contains an A-ideal coprime to $\mathfrak{a}$.

***Proof*:**

Let I be a nonzero fractional ideal of A. Then for each $\mathfrak{p}|\mathfrak{a}$, pick $\pi_\mathfrak{p} \in \mathfrak{p}$ such that

$$\nu_\mathfrak{q}(\pi_\mathfrak{p}) = \nu_\mathfrak{q}(\mathfrak{p}) \qquad \forall\ \mathfrak{q}|\mathfrak{a}$$

Then by some number theory we can put $\alpha = \prod_{\mathfrak{p}|\mathfrak{a}} \pi_\mathfrak{p}^{-\nu_\mathfrak{p}(I)}$, so that

$$\nu_\mathfrak{p}(\alpha I) = 0 \qquad \forall\ \mathfrak{p}|\mathfrak{a}$$

thus αI is coprime to $\mathfrak{a}$ and $[\alpha I] = [I]$. Now ,let S be the (finite) set of primes $\mathfrak{p}$ where $\nu_\mathfrak{p}(\alpha I) < 0$. Pick $\pi_\mathfrak{p} \in \mathfrak{p}$ such that $\nu_\mathfrak{q}(\pi_p) = \nu_\mathfrak{q}(\mathfrak{p})$ for all $fq \in S$ and $\mathfrak{q}|\mathfrak{a}$ (which can be done again by number theory, see [8, chapter 22]). Now, set

$$a = \prod_{\mathfrak{p}\in S} \pi_\mathfrak{p}^{-\nu_\mathfrak{p}(\alpha I)} \in A$$

Then $\nu_\mathfrak{p}(a\alpha I) \geq 0$ for all $\mathfrak{p}$, and $\nu_\mathfrak{p}(a\alpha I) = 0$ for all $\mathfrak{p}|\mathfrak{a}$. Then $a\alpha I$ is an A-ideal coprime to $\mathfrak{a}$, and $[a\alpha I] = [I]$, showing they are in the same equivalence class in the class group, as we sought to show.

Theorem 2.1.13: Canonical Exact Sequence for Modulus

Let $\mathfrak{m}$ be a modulus for a number field K. Then the following is an exact sequence:

$$1 \to \mathcal{O}_K^\times \cap K^{\mathfrak{m},1} \to \mathcal{O}_K^\times \to K^{\mathfrak{m}}/K^{\mathfrak{m},1} \to \mathrm{Cl}_K^{\mathfrak{m}} \to \mathrm{Cl}_K \to 1$$

where $K^{\mathfrak{m}}/K^{\mathfrak{m},1}$ is canonically isomorphic to:

$$K^{\mathfrak{m}}/K^{\mathfrak{m},1} \cong \{\pm 1\}^{\#\mathfrak{m}_\infty} \times (\mathcal{O}_K/\mathfrak{m}_0)^\times$$

***Proof*:**
Consider

$$K^{\mathfrak{m},1} \overset{f}{\hookrightarrow} K^{\mathfrak{m}} \xrightarrow{g:\alpha\mapsto(\alpha)} \mathcal{I}_K^{\mathfrak{m}}$$

This gives us some immediate results:

1. $\ker(f) = 0$ and $\ker g \circ f = \mathcal{O}_K^\times \cap K^{\mathfrak{m},1}$ (as $(\alpha) = (1)$ if and only if $\alpha \in \mathcal{O}_K^\times$),
2. $\ker g = \mathcal{O}_K^\times$, $\mathrm{coker} f = K^{\mathfrak{m}}/K^{\mathfrak{m},1}$,
3. $\mathrm{coker} g \circ f = \mathrm{Cl}_K^{\mathfrak{m}} = \mathcal{I}_K^{\mathfrak{m}}/\mathcal{R}_K^{\mathfrak{m}}$ and $\mathrm{coker} g = \mathrm{Cl}_K$ (by lemma 2.1.12).

Thus, by the snake lemma, the following commutative diagram with exact arrows:

$$\begin{tikzcd}
1 \arrow[r] & K^{\mathfrak{m},1} \arrow[r, hook, "f"] \arrow[d, "g\circ f"] & K^{\mathfrak{m}} \arrow[r] \arrow[d, "g"] & K^{\mathfrak{m}}/K^{\mathfrak{m},1} \arrow[r] \arrow[d, "\pi"] & 1 \\
1 \arrow[r] & \mathcal{I}_K^{\mathfrak{m}} \arrow[r, "\cong"] & CI_K^{\mathfrak{m}} \arrow[r] & 1 \arrow[r] & 1
\end{tikzcd}$$

gives a long exact sequence:

$$1 \to \mathcal{O}_K^\times \cap K^{\mathfrak{m},1} \to \mathcal{O}_K^\times \to K^{\mathfrak{m}}/K^{\mathfrak{m},1} \to \mathrm{Cl}_K^{\mathfrak{m}} \to \mathrm{Cl}_K \to 1$$

For the isomorphism, each $\alpha \in K^{\mathfrak{m}}$ can be written as a/b, $a, b \in \mathcal{O}_K$. Hence, by definition of $K^{\mathfrak{m}}$, choose $(a), (b)$ coprime to $\mathfrak{m}_0$. Define the homomorphism:

$$\varphi : K^{\mathfrak{m}} \to \left(\prod_{\nu|\mathfrak{m}_\infty} \{\pm 1\}\right) \times (\mathcal{O}_K/\mathfrak{m}_0)^\times$$

$$\alpha \mapsto \left(\prod_{\nu|\mathfrak{m}_\infty} \mathrm{sgn}(\alpha_\nu)\right) \times (\overline{\alpha})$$

where the map is well-defined since $\overline{\alpha} = \overline{a}\overline{b}^{-1} \in (\mathcal{O}_K/\mathfrak{m}_0)^\times$ is well-defined. Now, by the Chinese Remainder Theorem:

$$(\mathcal{O}_K/\mathfrak{m}_0)^\times \cong \prod_{\mathfrak{p}|\mathfrak{m}_0} (\mathcal{O}_K/\mathfrak{p}^{\mathfrak{m}(\mathfrak{p})})^\times$$

and by the weak approximation theorem φ is surjective. Then the kernel is $K^{\mathfrak{m},1}$, thus φ induces the isomorphism:

$$K^{\mathfrak{m}}/K^{\mathfrak{m},1} \cong \{\pm 1\}^{\#\mathfrak{m}_\infty} \times (\mathcal{O}_K/\mathfrak{m}_0)^\times$$

as we sought to show.

Corollary 2.1.14: Size of Ray Class Group

Let K be a number field and let $\mathfrak{m}$ be a modulus for K. The ray class group $\mathrm{Cl}_K^{\mathfrak{m}}$ is a finite abelian group whose cardinality $h_K^{\mathfrak{m}} := \#\mathrm{Cl}_K^{\mathfrak{m}}$ is given by:

$$h_k^{\mathfrak{m}} = \frac{\varphi(\mathfrak{m}) h_K}{[\mathcal{O}_K^\times : \mathcal{O}_K^\times \cap K^{\mathfrak{m},1}]}$$

where $h_K = \#\mathrm{Cl}_K$, $\varphi(\mathfrak{m}) = \#(K^{\mathfrak{m}}/K^{\mathfrak{m},1}) = \varphi(\mathfrak{m}_\infty)\varphi(\mathfrak{m}_0)$ with:

$$\varphi(\mathfrak{m}_\infty) = 2^{\#\mathfrak{m}_\infty} \qquad \varphi(\mathfrak{m}_n) = \#(\mathcal{O}_K/\mathfrak{m}_0)^\times = N(\mathfrak{m}_0)\prod_{\mathfrak{p}|\mathfrak{m}_0}(1 - N(\mathfrak{p})^{-1})$$

In particular, h_K divides $h_K^{\mathfrak{m}}$ and $h_K^{\mathfrak{m}}$ divides $h_K\varphi(\mathfrak{m})$. The number h_K is called the *ray class number*.

***Proof*:**
The exact sequence in theorem 2.1.13 implies:

$$\frac{\varphi(\mathfrak{m})}{[\mathcal{O}_K^\times : \mathcal{O}_K^\times \cap K^{\mathfrak{m},1}]} = \frac{h_K^{\mathfrak{m}}}{h_K}$$

Computing the ray class number is not easy in general, however there are some algorithms for doing so (MIT notes quote Henri Cohen, Advanced topics in computational number theory)

2.1.2 Polar Density

We shall now build up to proving surjectivity of the Artin map for finite abelian extensions L/K of number fields (namely for any number field K, not just $K = \mathbb{Q}$). The key result needed will be a density result about primes. In this section, all numbers fields shall lie in some fixed algebraic clsorue of $\mathbb{Q}$

Definition 2.1.15: Partial Dedekind Zeta Function

Let K be a number field and let S be a set of primes of K. Then the *partial Dedekind zeta function* associated to S is the holomorphic function on $\mathrm{Re}(s) > 1$[a].

$$\zeta_{K,S}(s) = \prod_{\mathfrak{p}\in S}\frac{1}{(1 - \mathfrak{N}(\mathfrak{p})^{-s})}$$

[a]For the same reason as the Dedekind-zeta function

If S is finite, Then $\zeta_{K,S}(s)$ is certainly holomorphic and nonzero on a neighborhood of 1. If S is cofinite, then it will differ from $\zeta_K(s)$ by a holomorphic factor, and hence extends to a meromorphic function with a simple pole at $s = 1$. If S an infinite collection that is not co-finite, then the situation is in general more complicated; $\zeta_{K,S}(s)$ may or may not extend to a function that is meromorphic on a neighborhood of 1. If it does (or if some power of it does), then we can use the order of the pole at 1 to measure the density of S:

Definition 2.1.16: Polar Density

Let $\zeta_{K,S}$ be the partial Dedekind-zeta function. Then if for some integer $n \geq 1$ the function $\zeta_{K,S}^n$ extends to a meromorphic function on a neighborhood of 1, then the *polar density* of S is defined to be:

$$\rho(S) = \frac{m}{n} \qquad m = -\text{ord}_{s=1}\zeta_{K,S}^n(s)$$

Note that m is the order of the pole if there is a pole at 1. If $\zeta_{K,S}^{n_1}$ and $\zeta_{K,S}^{n_2}$ both extend to a meromorphic function on a neighborhood of 1, then necessarily:

$$n_2\text{ord}_{s=1}\zeta_{K,S}^{n_1}(s) = \text{ord}_{s=1}\zeta_{K,S}^{n_1 n_2} = n_1\text{ord}_{s=1}\zeta_{K,S}^{n_2}(s)$$

hence, $\rho(S)$ does not depend on the choice of n. We shall soon show that $\rho(S) \in [0,1]$, motivating the name *density*. Recall that we have seen something very similar before when we talked about natural density and Dirichlet density. These two are in fact related:

Proposition 2.1.17: Polar and Dirichlet Density

Let S be a set of primes of a number field K. Then if S has a polar density, it has a Dirichlet density, and the two are equal. As a consequence, $\rho(S) \in [0,1]$ whenever it is defined

Something is off here. In the original notes, it references theorem 19.12, and when I wrote this I did not write theorem 19.12; so something is amiss here

***Proof*:**
Suppose S has a polar density $\rho(S) = \frac{m}{n}$. Taking the Laurent series expansion of $\zeta_{K,S}^n(s)$ at $s = 1$, and factoring out the leading nonzero term we get:

$$\zeta_{K,S}(s)^n = \frac{a}{(s-1)^m}\left(1 + \sum_{r\geq 1} a_r(s-1)^r\right)$$

for some constant $a \in \mathbb{C}^\times$. In fact, as $\zeta_{K,S}(s) \in \mathbb{R}_{>0}$ for $s \in \mathbb{R}_{>1}$, we must have $a \in \mathbb{R}_{>0}$, and hence $\lim_{s\to 1^+}(s-1)^m\zeta_{K,S}(s)^n \in \mathbb{R}_{>0}$. Taking the log of both sides, we get:

$$n\sum_{\mathfrak{p}\in S}\mathfrak{N}(\mathfrak{p})^{-s} \sim m\log\frac{1}{s-1} \qquad (\text{as } s \to 1^+)$$

Then as $\log(a) = O(1)$, it has no effect since $-m\log(s_1) \to \infty$ as $s \to 1^+$. It then follow that S has Dirichlet density

$$d(S) = \frac{m}{n}$$

Corollary 2.1.18: Polar Density and Natural Density

Let S be a set of primes of a number field K. Then if S has both has polar and natural density, the two coincide

***Proof*:**
As the polar and Dirichlet density coincide, and if the Dirichlet density exists it must coincide with the natural density, all these densities are equal

Note that not all sets of primes with a natural density have a polar density (the latter is always a rational number, the former need not be).

Polar density is pretty well-behaved. For the following list of properties, we shall need the notion of a degree-1 prime, which is a prime in a number field K such that the degree of its residue field is 1 over $\mathbb{Q}$, i.e. $\mathfrak{N}(\mathfrak{p}) = [\mathcal{O}_K : \mathfrak{p}] = \#\mathbb{F}_{\mathfrak{p}}$.

Proposition 2.1.19: Properties of Polar Density

Let S, T dentoe sets of primes in a number field K, and let $\mathcal{P}$ denote the set of all primes of K, and let $\mathcal{P}_1$ be the set of all degree-1 primes of K. Then:

1. If S is finite, $\rho(S) = 0$. If $\mathcal{P} - S$ is finite, then $\rho(S) = 1$
2. If $S \subseteq T$, and both have polar densities, then $\rho(S) \leq \rho(T)$
3. If two set s S, T have finite intersection and if any two of $S, T, S \cup T$ have polar density, so does the third, and
$$\rho(S \cup T) = \rho(S) + \rho(T)$$
4. $\rho(\mathcal{P}_1) = 1$ and $\rho(S \cap \mathcal{P}_1) = \rho(S)$ whenever S has a polar density

***Proof*:**
First, if S is a finite set , $\zeta_{K,S}(s)$ is a finite product of nonvanishing entire functions, and hence is entire (including at $s = 1$!). If the symmetric different between S and T is finite,

$$\zeta_{K,S}(s) f(s) = \zeta_{K,T}(s) g(s)$$

where f, g are non vanishing entire functions. Hence, if S, T differ by a finite set

$$\rho(S) = \rho(T)$$

whenever either have polar density. Then:

1. If $\rho(\emptyset) = 0$, and $\rho(\mathcal{P}) = 1$ (This will follow from the analytic class number formula, which I have will certainly put down as soon as possible!
2. Follows from results of Dirichlet Density
3. If the sets have finite intersection, by the argument made at the beginning of the proof we may take them to be disjoint. Then
$$\zeta_{K,S \cup T}(s)^n = \zeta_{K,S}(s)^n \zeta_{K,T}(s)^n$$
for all $n \geq 1$, which gives the result
4. Take $\mathcal{P}_2 = \mathcal{P} \setminus \mathcal{P}_1$ so that $\mathcal{P} = \mathcal{P}_1 \sqcup \mathcal{P}_2$. The for each rational prime p, we have at most $[K; \mathbb{Q}] = n$ primes $\mathfrak{p}/p$ in $\mathcal{P}_2$ (in particular $n/2$), wheer each have absolute norm

$\mathrm{Nm}(\mathfrak{p}) \geq p^2$. Now, following by a comparison with $\zeta(2s)^n$, the product $\zeta_{K,\mathcal{P}_2}(s)$ converges absolutely to a holomorphic function on $\mathrm{Re}(s) > 1/2$, and hence is holomorphic and nonvanishing on a neighborhood of 1 (where it is nonvanishing by the Euler product). Then by definition,

$$\rho(S \cap \mathcal{P}_2) = 0$$

Which then by part (c), when $\rho(S)$ exists, $\rho(S) = \rho(S \cap \mathcal{P}_1)$, completing the proof.

Let us now take a particular S. We will require the following important subset of primes that twill come up again and again:

Definition 2.1.20: Splitting Primes Set

Let L/K be a galois extension, and let $\mathrm{Spl}_K(L)$ denote the set of all primes in K that split completely in L; if K is clear the set will be written $\mathrm{Spl}(L)$

The polar density of this set always exists, and is strongly related to the degree of the extension:

Theorem 2.1.21: Density of Completely Split Prime in Galois Extension

Let L/K be a galois extension of number fields of degree n. Then:

$$\rho(\mathrm{Spl}(L)) = \frac{1}{n}$$

***Proof*:**
By proposition 2.1.19, it suffices to define the set S to be set of degree-1 prime ideal of K that split completely in L and show that:

$$\rho(S) = 1/n$$

Let T be the set of all primes $\mathfrak{q} \subseteq L$ that lie above some $\mathfrak{p} \in S$. As $\mathfrak{p}$ splits completely if and only if $e_\mathfrak{p} = f_\mathfrak{p} = 1$. Then for each $\mathfrak{q} \in T$,

$$N_{L/K}(\mathfrak{q}) = \mathfrak{p}^{f_\mathfrak{p}} = \mathfrak{p} \qquad \Rightarrow \qquad \mathfrak{N}(\mathbb{N}_{L/K}(\mathfrak{q})) = \mathfrak{N}(\mathfrak{p})$$

and hence $\mathfrak{q}$ is a degree-1 polynomial. Furthermore, if $\mathfrak{q}$ is any unramified degree-1 prime of L and $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}_K$, then:

$$\mathfrak{N}(\mathfrak{q}) = \mathfrak{N}(N_{L/K}(\mathfrak{q})) = \mathfrak{N}(\mathfrak{p}^{f_\mathfrak{p}})$$

and hence $f_\mathfrak{p} = e_\mathfrak{p} = 1$, implying that $\mathfrak{p}$ would be a degree-1 prime that splits completely in L and hence is an element of S. As only finitely many primes ramify, all but finitely many of the degree-1 primes in L lie in T, and hence

$$\rho(T) = 1$$

Finally, using proposition 2.1.19, each $\mathfrak{p} \in S$ has exactly n primes $\mathfrak{q} \in T$ lying above it (as $\mathfrak{p}$ splits completely). Thus:

$$\zeta_{L,T}(s) = \prod_{\mathfrak{q} \in T} (1 - \mathfrak{N}(\mathfrak{q})^{-s})^{-1} = \prod_{\mathfrak{p} \in S} (1 - \mathfrak{N}(\mathfrak{p})^{-s})^{-n} = \zeta_{K,S}(s)^n$$

which wrapping up gives:

$$\rho(S) = \frac{1}{n}\rho(T) = \frac{1}{n}$$

as we sought to show.

Corollary 2.1.22: Density of Completely Split Prime in Finite Extension

Let L/K be a finite extension of number fields, M/K it's galois closure of degree n. Then:

$$\rho(\mathrm{Spl}(L)) = \rho(\mathrm{Spl}(M)) = \frac{1}{n}$$

***Proof*:**
A prime $\mathfrak{p} \subseteq K$ splits completely in L if and only if it splits completely in all conjugates of L in M. The galois closure of M is the compositum of all conjugates of L, so $\mathfrak{p}$ splits completely in L if and only if it splits completely in M, completing the proof.

Corollary 2.1.23: Polar Density in Sub-extensions

Let L/K be a finite galois extension of number fields with galois group $G = \mathrm{Gal}_K(L)$ and let $H \trianglelefteq G$. Then the set S of primes for which $\mathrm{Frob}_{\mathfrak{p}} \subseteq H$ has polar density

$$\rho(S) = \frac{\#H}{\#G}$$

***Proof*:**
Let $F = L^H$. Then F/K is galois (as H is normal) and $\mathrm{Gal}_K(F \cong G/H$ by galois theory. Then for each unramified prime $\mathfrak{p}$ of K, the Frobenius class $\mathrm{Frob}_{\mathfrak{p}}$ lies in H if and only if every $\sigma_{\mathfrak{P}} \in \mathrm{Frob}_{\mathfrak{p}}$ acts trivially on $L^H = F$. This occurs if and only if $\mathfrak{p}$ splits completely in F. Then by theorem 2.1.21, the density of this set of primes is:

$$\frac{1}{[F:K]} = \frac{\#H}{\#G}$$

as we sought to show.

We shall now show that given a fixed field K, we may characterize two number extensions L/K and M/K by only looking at the split primes! To that end, let S, T be sets of primes whose symmetric difference is finite[3]. Then either $\rho(S) = \rho(T)$ or neither sets have a polar density. This forms an equivalence class $S \sim T$. Partially orders the sets of primes via:

$$S \preceq T \qquad \text{if and only if} \qquad S \sim S \cap T$$

that is, $S - T$ is finite. If S, T have polar densities, then by proposition 2.1.19 $S \preceq T$ implies $\rho(S) \leq \rho(T)$. The reason for the direction of the $\preceq$ is clear in the following theorem:

[3] The symmetric distance is $(A \setminus B) \cup (B \setminus A)$

Theorem 2.1.24: Characterizing Galois Extensions Using Polar Density

Let L/K, M/K be two finite galois extensions of number fields. Then:

$$L \subseteq M \Leftrightarrow \mathrm{Spl}(M) \preceq \mathrm{Spl}(L) \Leftrightarrow \mathrm{Spl}(M) \subseteq \mathrm{Spl}(L)$$
$$L = M \Leftrightarrow \mathrm{Spl}(M) \sim \mathrm{Spl}(L) \Leftrightarrow \mathrm{Spl}(M) = \mathrm{Spl}(L)$$

and the map $L \mapsto \mathrm{Spl}(L)$ is an injection from the set of finite galois extension of K (in some fixed algebraic closure) to the sets of primes of K that have a positive polar density

Recall too that the kernel of the Artin map is all primes that split completely. This should give an idea behind why the Artin map is naturally connected to galois group.

***Proof*:**
Certainly:

$$L \subseteq M \Rightarrow \mathrm{Spl}(M) \subseteq \mathrm{Spl}(L) \Rightarrow \mathrm{Spl}(M) \preceq \mathrm{Spl}(L)$$

Thus, it suffices to show $\mathrm{Spl}(M) \preceq \mathrm{Spl}(L) \Rightarrow L \subseteq M$. Recall that $\mathfrak{p} \subseteq K$ splits completely in LM if and only if it splits in both L and M (this is some galois theory and algebraic number theory). Then:

$$\mathrm{Spl}(M) \preceq \mathrm{Spl}(L) \Rightarrow \mathrm{Spl}(LM) \sim \mathrm{Spl}(M)$$

Then as we have the following by theorem 2.1.21,

$$\rho(\mathrm{Spl}(M)) = \frac{1}{[M:K]} \qquad \rho(\mathrm{Spl}(LM)) = \frac{1}{[LM:K]}$$

we get that

$$[LM:K] = \rho(\mathrm{Spl}(LM))^{-1} = \rho(\mathrm{Spl}(M))^{-1} = [M:K]$$

which shows that $LM = M$, and so $L \subseteq M$, giving the desired result. This also implies the equality immediately.

Finally, the map is certainly now an injection if it is well defined, which since $\mathrm{Spl}(L)$ always has positive density by theorem 2.1.21, this completes the proof.

2.1.3 Surjectivity of Artin Map

Theorem 2.1.25: Artin Map is Surjective

Let L/K be an abelian extension of number fields and $\mathfrak{m}$ a modulus divisible by all ramified primes. Then the Artin map

$$\psi^{\mathfrak{m}}_{L/K} : \mathcal{I}^{\mathfrak{m}}_K \twoheadrightarrow \mathrm{Gal}_K(L)$$

is surjective

***Proof*:**
Take $\text{im}(\psi^{\mathfrak{m}}_{L/K}) = H \subseteq \text{Gal}_K(L)$. Let $F = L^H$. As L/K is abelian, F/K is a galois extension. For each prime $\mathfrak{p} \in \mathcal{I}^{\mathfrak{m}}_K$, the automorphism $\psi^{\mathfrak{m}}_{L/K}(\mathfrak{p}) \in H$ acts trivially on $F = L^H$, therefore $\mathfrak{p}$ splits completely in F. The group $\mathcal{I}^{\mathfrak{m}}_K$ contains all but finitely many primes $\mathfrak{p}$ of K, so the polar density of the primes in K that split completely is 1. But then $[F : K] = 1$, and $H = \text{Gal}_K(L)$, showing surjectivity

Theorem 2.1.26: Kernel of Artin Map

Let $\mathfrak{m}$ be a modulus for a number field K and let L, M be finite abelian extension of K which are unramified at all primes not in the support of $\mathfrak{m}$. If

$$\ker \psi^{\mathfrak{m}}_{L/K} = \ker \psi^{\mathfrak{m}}_{M/K}$$

Then

$$L = M$$

Hence, the ray class field is unique when it exists

***Proof*:**
Let S be the set of primes of K that do not divide $\mathfrak{m}$, The each prime $\mathfrak{p}$ in S is unramified in both L and M, and split completely in L (resp. M) if and only if it lies in the kernel of $\psi^{\mathfrak{m}}_{L/K}$ (look back at the definition of Artin map if this is unclear). If $\ker \psi^{\mathfrak{m}}_{L/K} = \ker \psi^{\mathfrak{m}}_{M/K}$, then:

$$\text{Spl}(L) \sim (S \cap \ker \psi^{\mathfrak{m}}_{L/K}) = (S \cap \ker^{\mathfrak{m}}_{M/K}) \sim \text{Spl}(M)$$

Hence, by theorem 2.1.24, $L = M$, as we sought to show.

Global Class Field Theory

So a lot of the work here is generalizing The stuff the analytic of Dedekind-Zeta and Dirichlet Theorem For primes in number fields. I really want to make it so that when you reach that point you go "ah yes, that *is* what I want to do. I don't have that feeling rn

Theorem 2.1.26 let us conclude that there is at most 1 ray class field. Let us now return to finding the Ray Class field (definition 2.1.10). Recall that the Ray class field is when the class group $\mathcal{R}^{\mathfrak{m}}_K$ equals the kernel of the Artin map. By theorem 2.1.25, we have an exact sequence:

$$1 \to \ker \psi^{\mathfrak{m}}_{L/K} \to \mathcal{I}^{\mathfrak{m}}_K \to \text{Gal}_K(L) \to 1$$

We may then ask if for a suitable choice of modulus $\mathfrak{m}$ we get the equality:

$$\mathcal{R}^{\mathfrak{m}}_K \cong \ker \psi^{\mathfrak{m}}_{K(\mathfrak{m})/K}$$

As any intermediate field should be a quotient of the galois group, the kernel of the Artin map would be a subgroup $\mathcal{C} \subseteq \mathcal{I}^{\mathfrak{m}}_K$ such that

$$\mathcal{R}^{\mathfrak{m}}_K \subseteq \mathcal{C} \subseteq \mathcal{I}^{\mathfrak{m}}_K$$

where

$$\mathcal{I}_K^{\mathfrak{m}}/\mathcal{C} \cong \mathrm{Cl}\,_K^{\mathfrak{m}}/\overline{C} \cong \mathrm{Gal}_K(L)$$

Note further that any finite abelian extension L/K is indeed an intermediate field of a ray class field for a modulus $\mathfrak{m}$ $K(\mathfrak{m})$ by showing that

$$\ker \psi_{L/K}^{\mathfrak{m}} \subseteq \mathcal{R}_K^{\mathfrak{m}}$$

For by theorem 2.1.24 this implies $\mathrm{Spl}(K(\mathfrak{m}) \preceq \mathrm{Spl}(L)$ and $L \subseteq K(\mathfrak{m})$. A natural question becomes to ask for a *minimal* such modulus. This shall be called the *conductor*. This shall be an interesting question even for the case of $K = \mathbb{Q}$ (for example, given $K/\mathbb{Q}$, what is the minimal m for which K embeds into $\mathbb{Q}(\zeta_m)$).

2.2 Congruence Subgroup and Conductor

Definition 2.2.1: Congruence Subgroup

Let K be a number field and let $\mathfrak{m}$ be a modulus for K. Then a *congruence subgroup* for the modulus $\mathfrak{m}$ is a subgroup $\mathcal{C} \leqslant \mathcal{I}_K^{\mathfrak{m}}$ such that $\mathcal{R}_K^{\mathfrak{m}} \leqslant \mathcal{C}$. The image of this group in $\mathcal{I}_K^{\mathfrak{m}}/\mathcal{R}_K^{\mathfrak{m}}$ will be denoted $\overline{\mathcal{C}}$.

We would like to say that the congruence subgroups will be the kernel of an Artin map $\psi_{L/K}^{\mathfrak{m}}$ given a choice of $\mathfrak{m}$. However, not every $\psi_{L/K}^{\mathfrak{m}}$ is is a congruence subgroup:

Example 2.2.2: Wrong Choice of Modulus

Take $L/\mathbb{Q}$ where $L = \mathbb{Q}[x]/(x^3 - 3x - 1)$. Then you can check that only 3 ramifies, hence the Artin map $\psi_{L/\mathbb{Q}}^{\mathfrak{m}}$ is well-defined for any $\mathfrak{m}$ divisible by 3. The simplest choice would be $\mathfrak{m} = (3)$. But computing the Ray class field, we get:

$$\mathbb{Q}(\zeta_3)^+ = \mathbb{Q}$$

and for $\mathfrak{m} = (3)\infty$ it is:

$$\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$$

Note that neither contain L (they are both degree 2 extensions!), and so $\ker \psi_{L/K}^{(3)}$ and $\ker \psi_{L/K}^{(3)\infty}$ does *not* contain $\mathcal{R}_K^{\mathfrak{m}}$. If we choose $\mathfrak{m} = (9)$, then the corresponding field is $\mathbb{Q}(\zeta_9)^+$, and in fact $L \cong \mathbb{Q}(\zeta_9)^+$, and so $\ker \psi_{L/K}^{(9)}$ is a congruence subgroup for the modulus $\mathfrak{m} = (9)$.

Thus, let us look at the relations between congruence subgroups and moduli. If $\ker \psi_{L/K}^{\mathfrak{m}}$ is a congruence subgroup for a modulus $\mathfrak{m}$, then for each $\mathfrak{m} \big| \mathfrak{n}$, $\ker \psi_{L/K}^{\mathfrak{n}}$ is a congruence subgroup. If $\mathfrak{m} \big| \mathfrak{n}$, $\mathcal{R}_K^{\mathfrak{n}} \subseteq \mathcal{R}_K^{\mathfrak{m}}$ and $\psi_{L/K}^{\mathfrak{n}}$ is the restriction of $\psi_{L/K}^{\mathfrak{m}}$ to $\mathcal{I}_K^{\mathfrak{n}}$ which contains $\mathcal{R}_K^{\mathfrak{n}}$, and so $\ker \psi_{L/K}^{\mathfrak{n}}$ is a congruence subgroup for the modulus $\mathfrak{n}$. If the support of $\mathfrak{m}$ and $\mathfrak{n}$ are the same, then

$$\mathcal{I}_K^{\mathfrak{m}} = \mathcal{I}_K^{\mathfrak{n}} \qquad \psi_{L/K}^{\mathfrak{m}} = \psi_{L/K}^{\mathfrak{n}}$$

however, the ray groups $\mathcal{R}_K^{\mathfrak{m}}$ and $\mathcal{R}_K^{\mathfrak{n}}$ may differ (as demonstrated in example 2.2.2). Thus, which modules on which congruence subgroup induces the "equivalent" structure must be tracked:

Definition 2.2.3: Equivalence Congruence Subgroups

Let K be a number field with moduli $\mathfrak{m}_1, \mathfrak{m}_2$. If $\mathcal{C}_1$ is a congruence subgroup for $\mathfrak{m}_1$ and $\mathcal{C}_2$ is a congruence subgroup for $\mathfrak{m}_2$, then we say that $(\mathcal{C}_1, \mathfrak{m}_1)$ and $(\mathcal{C}_2, \mathfrak{m}_2)$ are *equivalent* if

$$\mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_2 = \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}_1$$

If $\mathfrak{m}_1 = \mathfrak{m}_2$, then this becomes $\mathcal{C}_1 = \mathcal{C}_2$

Proposition 2.2.4: Equivalence relation of congruence subgroups

Let K be a number field. Then the relation in definition 2.2.3 is an equivalence relation. If $(\mathcal{C}_1, \mathfrak{m}_1) \sim (\mathcal{C}_2, \mathfrak{m}_2)$, then

$$\mathcal{I}_K^{\mathfrak{m}_1}/\mathcal{C}_1 \cong \mathcal{I}_K^{\mathfrak{m}_2}/\mathcal{C}_2$$

via canonical isomorphism that preserves cosets of fractional ideals prime to both $\mathfrak{m}_1$ and $\mathfrak{m}_2$

***Proof*:**
Let $\sim$ be the relationship defined above. Certainly $\sim$ is symmetric and reflexive. For transitivity, take $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ with $\mathfrak{m}_1, \mathfrak{m}_2, \mathfrak{m}_3$ and assume:

$$(\mathcal{C}_1, \mathfrak{m}_1) \sim (\mathcal{C}_2, \mathfrak{m}_2) \qquad (\mathcal{C}_2, \mathfrak{m}_2) \sim (\mathcal{C}_3, \mathfrak{m}_3)$$

Let $I \in \mathcal{I}_K^{\mathfrak{m}_3} \cap \mathcal{C}_1$. Then by lemma 2.1.12 and the weak approximation theorem, pick $\alpha \in K^{\mathfrak{m}_1\mathfrak{m}_3,1}$ such that $\alpha I \in \mathcal{I}_K^{\mathfrak{m}_1\mathfrak{m}_2\mathfrak{m}_3}$. Then $(\alpha) \in \mathcal{R}_K^{\mathfrak{m}_1\mathfrak{m}_3} \subseteq \mathcal{R}_K^{\mathfrak{m}_1} \subseteq \mathcal{C}_1$, and $I \subseteq \mathcal{C}_1$, hence $\alpha I \in \mathcal{C}_1$ and $\alpha I \in \mathcal{I}_K^{\mathfrak{m}_1\mathfrak{m}_2\mathfrak{m}_3} \subseteq \mathcal{I}_K^{\mathfrak{m}_2}$. Therefore, as $\mathcal{C}_1 \sim \mathcal{C}_2$:

$$\alpha I \in \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}_1 = \mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_2 \subseteq \mathcal{C}_2$$

Next, $\alpha I \in \mathcal{I}_K^{\mathfrak{m}_1\mathfrak{m}_2\mathfrak{m}_3} \subseteq \mathcal{I}_K^{\mathfrak{m}_2}$, and as $\mathcal{C}_2 \sim \mathcal{C}_3$:

$$\alpha I \in \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}_2 = \mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_3 \subseteq \mathcal{C}_3$$

Now, as $(\alpha) \in \mathcal{R}_K^{\mathfrak{m}_1\mathfrak{m}_3} \subseteq \mathcal{R}_K^{\mathfrak{m}_3}$, $(\alpha) \in \mathcal{C}_3$, and so $(\alpha)^{-1} \in \mathcal{C}_3$, and so $\alpha^{-1}\alpha I = I \in \mathcal{C}_3$. Then as we have $I \in \mathcal{C}_1 \subseteq \mathcal{I}_K^{\mathfrak{m}_1}$, $I \in \mathcal{I}_K^{\mathfrak{m}_1} ca[\mathcal{C}_3$. As $I \in \mathcal{I}_K^{\mathfrak{m}_3} \cap \mathcal{C}_1$ was arbitrary,we have:

$$I_K^{\mathfrak{m}_3} \cap \mathcal{C}_1 \subseteq \mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_3$$

Then by symmetry we get the reverse, and hence

$$(\mathcal{C}_1, \mathfrak{m}_1) \sim (\mathcal{C}_3, \mathfrak{m}_3)$$

For the isomorphism, let us define the homomorphism, and then show it is an isomorphism. Take $I \in \mathcal{I}_K^{\mathfrak{m}_1}$ and again pick $\alpha \in K^{\mathfrak{m}_1,1}$ so that $\alpha I \in \mathcal{I}_K^{\mathfrak{m}_2}$. Then the image of αI in $\mathcal{I}_K^{\mathfrak{m}_2}/\mathcal{C}_2$ is independent of choice of α, as for any other $\alpha' \in K^{\mathfrak{m}_1,1}$ where $\alpha' I \in \mathcal{I}_K^{\mathfrak{m}_2}$, $(\alpha I)/(\alpha' I) = (\alpha/\alpha') \in \mathcal{I}_K^{\mathfrak{m}_2}$ and $(\alpha/\alpha' \in \mathcal{R}_K^{\mathfrak{m}_1}$, and so $(\alpha/\alpha') \in \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{R}_K^{\mathfrak{m}_1} = \mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{R}_K^{\mathfrak{m}_2} \subseteq \mathcal{R}_K^{\mathfrak{m}_2}$. We thus have a group homomorphism $\varphi : \mathcal{I}_K^{\mathfrak{m}_1} \to \mathcal{I}_K^{z\mathfrak{m}_2}/\mathcal{C}_2$.
Now, for $I \in \mathcal{C}_1$, $\alpha I \in \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}_1 = \mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_2 \subseteq \mathcal{C}_2$. As $I \in \mathcal{I}_K^{\mathfrak{m}_1} \setminus \mathcal{C}_1$, $\alpha I \in \mathcal{I}_K^{\mathfrak{m}_2} \setminus \mathcal{C}_1$, and so $\alpha I \notin \mathcal{C}_2$, so

$$\ker \varphi = \mathcal{C}_1$$

and so φ is an injective homomorphism:

$$\mathcal{I}_K^{\mathfrak{m}_1}/\mathcal{C}_1 \to \mathcal{I}_K^{\mathfrak{m}_2}/\mathcal{C}_2$$

Then by symmetry we have an injective homomorphism going the other way as well and so

$$\mathcal{I}_K^{\mathfrak{m}_1}/\mathcal{C}_1 \cong \mathcal{I}_K^{\mathfrak{m}_2}/\mathcal{C}_2$$

Note that the isomorphism is independent of choice of α, making it canonical, and for any fractional ideal I coprime to $\mathfrak{m}_1$ and $\mathfrak{m}_2$, we may as well choose $\alpha = 1$, in which use the coset of I in $\mathcal{I}_K^{\mathfrak{m}_1}/\mathcal{C}_1$ will be identified with the coset of I in $\mathcal{I}_K^{\mathfrak{m}_2}/\mathcal{C}_2$, completing the proof.

Corollary 2.2.5: Congruence Subgroup For Different Moduli

Let $\mathcal{C}$ be a congruence subgroup for two moduli $\mathfrak{m}_1, \mathfrak{m}_2$. Then

$$(\mathcal{C}, \mathfrak{m}_1) \sim (\mathcal{C}, \mathfrak{m}_2)$$

***Proof*:**
By definition of a congruence subgroup for a modulus, $\mathcal{C} \subseteq \mathcal{I}_K^{\mathfrak{m}_1}$ and $\mathcal{C} \subseteq \mathcal{I}_K^{\mathfrak{m}_2}$. Hence $\mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C} = \mathcal{C} = \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}$, which is precisely the condition $(\mathcal{C}, \mathfrak{m}_1) \sim (\mathcal{C}, \mathfrak{m}_2)$ from definition 2.2.3, completing the proof.

As a consequence, notice that each subgroup $\mathcal{I}_K$ lies in at most one equivalence class of congruence subgroup, and so we can view the equivalence relation $(\mathcal{C}_1, \mathfrak{m}_1) \sim (\mathcal{C}_2, \mathfrak{m}_2)$ as an equivalence relation on the congruence subgroups of $\mathcal{I}_K$, and write $\mathcal{C}_1 \sim \mathcal{C}_2$ unambiguously. Then each equivalence class of congruence subgroups uniquely determines a finite abelian group that is the quotient of a ray class group.

Now, we have that there can be at most one congruence subgroup for each modulus (as $\mathcal{C}_1 \sim \mathcal{C}_2 \Leftrightarrow \mathcal{C}_1 = \mathcal{C}_2$ when $\mathcal{C}_1, \mathcal{C}_2$ are congruence subgroups for the same modulus). Let us now find when there exists a congruence subgroup of a given modulus within an equivalence class

Lemma 2.2.6: Divisibility of Congruence subgroup

Let $\mathcal{C}_1$ be a congruence subgroup of modulus $\mathfrak{m}_1$ for a number field $K>$ Then there exits a congruence subgroup $\mathcal{C}_2$ of modulus $\mathfrak{m}_2 \big| \mathfrak{m}_1$ equivalent to $\mathcal{C}_1$ if and only if

$$\mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{R}_K^{\mathfrak{m}_2} \subseteq \mathcal{C}_1$$

in which case:

$$\mathcal{C}_2 = \mathcal{C}_1 \mathcal{R}_K^{\mathfrak{m}_2}$$

***Proof*:**
As $\mathfrak{m}_2 \big| \mathfrak{m}_1$, $\mathcal{I}_K^{\mathfrak{m}_1} \subseteq \mathcal{I}_K^{\mathfrak{m}_2}$, so $\mathcal{C}_1 \subseteq \mathcal{I}_K^{\mathfrak{m}_1} \subseteq \mathcal{I}_K^{\mathfrak{m}_2}$. Now suppose $\mathcal{C}_1 \sim \mathcal{C}_1$ has modulus $\mathfrak{m}_2$. Then $\mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_2 = \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}_1 = \mathcal{C}_1$ and $\mathcal{R}_K^{\mathfrak{m}_2} \subseteq \mathcal{C}_2$, so

$$\mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{R}_K^{\mathfrak{m}_2} \subseteq \mathcal{C}_1$$

Giving the first claim. Now given the above subset, let $\mathcal{C}_2 := \mathcal{C}_1\mathcal{R}_K^{\mathfrak{m}_2}$. Then $\mathcal{C}_2$ is a congruence subgroup of modulus $\mathfrak{m}_2$. Then:

$$\mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}_1 = \mathcal{C}_1(\mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{R}_K^{\mathfrak{m}_2}) = \mathcal{I}_K^{\mathfrak{m}_1}\mathcal{C}_1\mathcal{R}_K^{\mathfrak{m}_2} = \mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_2$$

Hence $\mathcal{C}_1 \sim \mathcal{C}_2$. Then as the equivalence class of $\mathcal{C}_1$ contains *at most one* congruence subgroup of modulus $\mathfrak{m}_2$, if one exists it in fact *has* to be $\mathcal{C}_1 = \mathcal{C}_1\mathcal{R}^{\mathfrak{m}_2}$, as we sought to show.

Lemma 2.2.7: Congruence Subgroup and GCD

Let $\mathcal{C}_1 \sim \mathcal{C}_2$ be congruence subgroups of modulus $\mathfrak{m}_1, \mathfrak{m}_2$ respectively. Then there exists a congruence subgroup $\mathcal{C} \sim \mathcal{C}_1 \sim \mathcal{C}_2$ with $\mathfrak{n} = \gcd(\mathfrak{m}_1, \mathfrak{m}_2)$

***Proof*:**
Let $\mathfrak{m} = \operatorname{lcm}(\mathfrak{m}_1, \mathfrak{m}_2)$ and take $\mathcal{D} = \mathcal{I}_K^{\mathfrak{m}_2} \cap \mathcal{C}_1 = \mathcal{I}_K^{\mathfrak{m}_1} \cap \mathcal{C}_2$. Then:

$$\mathcal{R}_K^{\mathfrak{m}} = CR_K^{\mathfrak{m}_1} \cap \mathcal{R}_K^{\mathfrak{m}_2} \subseteq \mathcal{D} \subseteq \mathcal{I}_K^{\mathfrak{m}}$$

Hence $\mathcal{D}$ is a congruence subgroup of modulus $\mathfrak{m}$. As:

$$\mathcal{I}_K^{\mathfrak{m}} \cap \mathcal{R}_K^{\mathfrak{m}_1} \subseteq \mathcal{D} \qquad \mathcal{I}_K^{\mathfrak{m}} \cap \mathcal{R}_K^{\mathfrak{m}_2} \subseteq \mathcal{D}$$

by lemma 2.2.6 we have

$$\mathcal{D} \sim \mathcal{C}_1 \sim \mathcal{C}_2$$

Now, letting $\mathfrak{n} = \gcd(\mathfrak{m} - 1, \mathfrak{m}_2)$, if we show $\mathcal{I}_K^{\mathfrak{m}} \cap \mathcal{R}_K^{|} fn \subseteq \mathcal{D}$, by lemma 2.2.6 we are done.
Let $\mathfrak{a} = (\alpha) \in \mathcal{I}_K^{\mathfrak{m}} \cap \mathcal{R}_K^{\mathfrak{n}}$. Choose $\beta \in K^{\mathfrak{m}} \cap K^{\mathfrak{m}_2,1}$ such that $\alpha\beta \in K^{\mathfrak{m}_1,1}$. Then $(\beta) \in \mathcal{D}$ and $\beta\mathfrak{a} \in \mathcal{I}_K^{\mathfrak{m}} \cap \mathcal{R}_K^{\mathfrak{m}_1} \subseteq \mathcal{D}$, and so $\beta^{-1}\beta\mathfrak{a} = \mathfrak{a} \in \mathcal{D}$. Thus,

$$\mathcal{I}_K^{\mathfrak{m}} \cap \mathcal{R}_K^{\mathfrak{n}} \subseteq \mathcal{D}$$

Thus $\mathcal{C} = \mathcal{D}\mathcal{R}_K^{\mathfrak{n}}$ is a congruence subgroup with modulus $\mathfrak{n}$, as we sought to show.

Corollary 2.2.8: Minimal Congruence Subgroup: Conductor

Let $\mathcal{C}$ be a congruence subgroup of modulus $\mathfrak{m}$ for a number field K. Then here exists a unique congruence subgroup in the equivalence class of $\mathcal{C}$ whose modulus $\mathfrak{c}$ divides the modulus of every congruence subgroup equivalent to $\mathcal{C}$

***Proof*:**
Direct application of the above lemma

The $\mathfrak{c}$ is given a name:

Definition 2.2.9: Conductor of Congruence Subgroup

Let $\mathcal{C}$ be a congruence subgroup of a number field K. Then the unique $\mathfrak{c} = \mathfrak{c}(\mathcal{C})$ is called the *conductor* of $\mathcal{C}$. We say that $\mathcal{C}$ is *primitive* if $\mathcal{C} = \mathcal{C}\mathcal{R}_K^{\mathfrak{c}}$.

Proposition 2.2.10: Primitive Congruence Subgroup and Conductor

Let $\mathcal{C}$ be a primitive congruence subgroup for a modulus $\mathfrak{m}$ for a number field K. Then $\mathfrak{m}$ is the conductor for every congruence subgroup of $\mathfrak{m}$ contained in $\mathcal{C}$, in particular, $\mathfrak{m}$ is the conductor of $\mathcal{R}_K^{\mathfrak{m}}$

***Proof*:**
Let $\mathcal{C}_0 \subseteq \mathcal{C}$ be a congruence subgroup of modulus $\mathfrak{m}$ and let $\mathfrak{c}$ be its conductor. Then $\mathfrak{c}\big|\mathfrak{m}$ and

$$\mathcal{I}_K^{\mathfrak{m}} \cap \mathcal{R}_K^{\mathfrak{c}} \subseteq \mathcal{C}_0 \subseteq \mathcal{C}$$

Then by lemma 2.2.6 there is a congruence subgroup of modulus $\mathfrak{c}$ equivalent to $\mathcal{C}$, hence $\mathfrak{m}\big|\mathfrak{c}$, but then $\mathfrak{c} = \mathfrak{m}$, completing the proof.

Note that the modulus $\mathfrak{m}$ occurs as a conductor if and only if $\mathcal{R}_K^{\mathfrak{m}}$ is primitive. This need not hold in general

Example 2.2.11: Counter Example
Take $K = \mathbb{Q}$, $\mathfrak{m} = (2)$. Then the conductor of $\mathcal{R}_{\mathbb{Q}}^{(2)} = \mathcal{I}_{\mathbb{Q}}^{(2)}$ is (1) as

$$\mathcal{R}_{\mathbb{Q}}^{(2)} \cap \mathcal{I}_{\mathbb{Q}}^{(1)} = \mathcal{I}_{\mathbb{Q}}^{(1)} \cap \mathcal{I}_{\mathbb{Q}}^{(2)} \qquad \Rightarrow \qquad \mathcal{R}_{\mathbb{Q}}^{(2)} \sim \mathcal{I}_{\mathbb{Q}}^{(1)}$$

Thus, (2) is *not* the conductor of any congruence subgroup of $\mathbb{Q}$.

2.3 Ray Class Character

The next goal is to show that we can link a congruence subgroup to the kernel of an Artin map. As stated at the beginning of section 2.2, this can be deduced by looking at the primes that split. This can be done by properly generalizing the Dirichlet density for number fields Recall the Dirichlet density gave us a way of measuring the density of certain sets of primes. Given a congruence of subgroups $\mathcal{C}$ for a modulus $\mathfrak{m}$, when it exists, we will compute a new Dirichlet density $d(\mathcal{C})$ of the set of primes ideals in $\mathfrak{p} \in \mathcal{I}_K^{\mathfrak{m}}$ that lie in $\mathcal{C}$. To that end:

Definition 2.3.1: Ray Class Character

Let K be a number fields, $\chi : \mathcal{I}_K \to \mathbb{C}$ a totally multiplicative function with finite image (so $\chi((\mathcal{O}_K)) = 1$, $\chi(IJ) = \chi(I)\chi(J)$ and χ restricts to a homomorphism from a subgroup of $\mathcal{I}_K$ to a finite subgroup of $U(1)$). If $\mathfrak{m}$ is a modulus for K such that $\chi^{-1}(U(1)) = \mathcal{I}_K^{\mathfrak{m}}$ and $\mathcal{R}_K^{\mathfrak{m}} \subseteq \ker \chi$, we say that χ is a *ray class character* of modulus $\mathfrak{m}$, and its kernel is the congruence subgroup of modulus $\mathfrak{m}$. The extension by zero f a character is then the finite abelian group $\mathrm{Cl}_K^{\mathfrak{m}} = \mathcal{I}_K^{\mathfrak{m}}/\mathcal{R}_K^{\mathfrak{m}}$ defined by setting $\chi(I) = 0$ for $I \notin \mathcal{I}_K^{\mathfrak{m}}$.

Example 2.3.2: Ray Class Character For Rationals

Let $K = \mathbb{Q}$. Then there is a one-to-one correspondence between Dirichlet characters $\chi : \mathbb{Z} \to \mathbb{C}$ and ray class characters $\chi' : \mathcal{I}_{\mathbb{Q}} \to \mathbb{C}$ given by

$$\chi(a) = \chi'((a))$$

for al l$a \in \mathbb{Z}_{\geq 1}$. Then each Dirichlet character χ of modulus m correspond to a ray class character of module $\mathfrak{m} = (m)\infty$ whose conductor divides (m) if and only if χ is an *even* Dirichlet character, i.e.

$$\chi(-1) = 1$$

Definition 2.3.3: Induced Ray Class Character and Conductor

Let χ_1, χ_2 be two ray class characters of moduli $\mathfrak{m}_1, \mathfrak{m}_2$ of a number field K, where $\mathfrak{m}_1 \big| \mathfrak{m}_2$. If $\chi - 2(I) = \chi_1(I)$ for all $I \in \mathcal{I}_K^{\mathfrak{m}_2}$, then $\chi - 2$ is *induced* by $\chi - 1$. A ray class character is *primitive* if it is not induced by any ray class character other than itself.
The *conductor* of a ray class character χ is the conductor of its kernel as a congruence subgroup:

$$\mathfrak{c}(\chi) = \mathfrak{c}(\ker(\chi))$$

The following is a direct generalization of the characterization of Dirichlet characters [2]:

Theorem 2.3.4: Primitive Ray Class Character

A ray class character is primitive if and only if its kernel is primitive. Every ray class character χ is induced by a unique primitive ray class character $\widetilde{\chi}$

***Proof*:**
Let χ be a ray class character of modulus $\mathfrak{m}$, let $\kappa : \mathcal{I}_K^{\mathfrak{m}}/(\ker \chi) \to \mathrm{U}(1)$ be the group character induced by χ, and let $\mathcal{C}$ be the primitive congruence subgroup equivalent to $\ker \chi$ with modulus $\mathfrak{c} = \mathfrak{c}(\chi)$ dividing $\mathfrak{m}$ given by corollary 2.2.8. By proposition 2.2.4, let $\varphi : \mathcal{I}_K^{\mathfrak{c}}/\mathcal{C} \xrightarrow{\sim} \mathcal{I}_K^{\mathfrak{m}}/(\ker \chi)$ be the canonical isomorphism. Then we can use φ to define a ray class character $\widetilde{\chi}$ of modulus $\mathfrak{c}$ as the extension by zero of the character $\kappa \circ \varphi$ of $\mathcal{I}_K^{\mathfrak{c}}/\mathcal{C}$. The isomorphism φ preserves cosets of fractional ideals in $\mathcal{I}_K^{\mathfrak{m}} \subseteq \mathcal{I}_K^{\mathfrak{c}}$, so $\widetilde{\chi}(I) = \chi(I)$ for all $I \in \mathcal{I}_K^{\mathfrak{m}}$ and χ is induced by $\widetilde{\chi}$.

Now, if χ_2 is a ray class character of conductor $\mathfrak{m}_2$ induced by a ray class character χ_1 of conductor $\mathfrak{m}_1$, then

$$\ker \chi_1 \cap \mathcal{I}_K^{\mathfrak{m}_2} = \ker \chi_2 = \ker \chi_2 \cap \mathcal{I}_K^{\mathfrak{m}_1}$$

and $\ker\chi_1 \sim \ker\chi_2$. Note that if $\chi_1 \neq \chi_2$ then $\mathcal{I}_K^{\mathfrak{m}_1} \neq \mathcal{I}_K^{\mathfrak{m}_2}$ and $\mathfrak{m}_1 \neq \mathfrak{m}_2$. It follows that $\widetilde{\chi}$ is primitive, it is the unique primitive ray class character that induces χ, and so χ is primitive if and only if it is equal to $\widetilde{\chi}$, which holds if and only if $\ker\chi = \ker\widetilde{\chi}$ is primitive, as we sought to show.

Definition 2.3.5: Principal Ray Class Character

A ray class character χ is called *principal* if $\ker\chi = \chi^{-1}(U(1))$. Let $\mathbb{1}$ denote the unique primitive principal ray class character.

Note that for Dirichlet characters, $\mathbb{1}$ is the unique character of conductor 1, but for Ray class characters this is only the case when Cl_K is trivial (ex. when $K = \mathbb{Q}$). In general, the extension-by-zero of any character of Cl_K is a ray class character of conductor (1) and need not be principal (but is still necessarily primitive).

With the appropriate build-up we can generalize the L-function:

Definition 2.3.6: Weber L-function

Let K be a number field and χ a ray class character for K. Then the *Weber L-function* $L(s,\chi)$ is:

$$L(s,\chi) = \prod_{\mathfrak{p}} \left(1 - \chi(\mathfrak{p})\mathrm{Nm}(\mathfrak{p})^{-s}\right)^{-1} = \sum_{\mathfrak{a}} \chi(\mathfrak{a})\mathrm{Nm}(\mathfrak{a})^{-s}$$

If $K = \mathbb{Q}$, then Weber L-functions are Dirichlet L-functions. If K is any number field, then the Weber L-function for $\mathbb{1}$ is $L(s,\mathbb{1}) = \zeta_K(s)$.

Proposition 2.3.7: Meromorphicity and Holomorphicity of Weber L-function

Let χ be a ray class character of modulus $\mathfrak{m}$ for a number field K of degree n. Then $L(s,\chi)$ extends to a meromorphic function on $\mathrm{Re}(s) > 1 - \frac{1}{n}$ that has at most a simple pole at $s = 1$ and is holomorphic if χ is non-principal.

***Proof*:**
The same proof as for the analytic class number formula [2, Analytic Class Number Formula] works for $\mathrm{Cl}_K^{\mathfrak{m}}$, essentially identically (*mutatis mutandis*): replace $\mathrm{covol}(\mathcal{O}_K)$ with $\mathrm{covol}(\mathfrak{m}_0)$, replace the regulator $R_K = \mathrm{covol}(\pi(\mathrm{Log}(\mathcal{O}_K^\times)))$ with $R_K^{\mathfrak{m}} := \mathrm{covol}(\pi(\mathrm{Log}(\mathcal{O}_K^\times \cap K^{\mathfrak{m},1})))$, and replace $\omega_K = \#\mu_K$ with $\omega_K^{\mathfrak{m}} := \#(\mu_K \cap K^{\mathfrak{m},1})$. The exact value of ρ is not important here. The key point is that $\zeta_{K,\gamma}(s)$ has a meromorphic continuation to $\mathrm{Re}(s) > 1 - \frac{1}{n}$ with a simple pole at $s = 1$ whose residue ρ depends only on K and $\mathfrak{m}$ (not γ).

Associated to each ray class $\gamma \in \mathrm{Cl}_K^{\mathfrak{m}}$, we have a Dirichlet series

$$\zeta_{K,\gamma}(s) := \sum_{\mathfrak{a}\in\gamma} \mathrm{Nm}(\mathfrak{a})^{-s}$$

that is holomorphic on $\mathrm{Re}(s) > 1$. For the trivial modulus $\mathfrak{m}$, the analytic class number formula implies that $\zeta_{K,\gamma}(s)$ has a meromorphic continuation to $1 - \frac{1}{n}$ with a simple pole at $s = 1$ and residue

$$\rho = \frac{2^r(2\pi)^2 R_K}{\omega_K |D_K|^{1/2}}$$

independent of γ. Recall that in the ideal-counting estimate each $\gamma \in \mathrm{Cl}_K = \mathrm{cl}(\mathcal{O}_K)$ is treated separately and obtained the same value of ρ for each γ, leading to the residue $\rho_K = h_K \rho$ that appears in the analytic class number formula. So:

$$\begin{aligned} L(s,\chi) &= \sum_{\gamma \in \mathrm{Cl}_K^{\mathfrak{m}}} \chi(\gamma)\zeta_{K,\gamma}(s) \\ &= \sum_{\gamma \in \mathrm{Cl}_K^{\mathfrak{m}}} \chi(\gamma)(\zeta_{K,\gamma}(s) - \rho\zeta(s)) + \sum_{\gamma \in \mathrm{Cl}_K^{\mathfrak{m}}} \chi(\gamma)\rho\zeta(s), \end{aligned}$$

The first sum is a finite sum of functions holomorphic on $\mathrm{Re}(s) > 1 - \frac{1}{n}$ (since $\zeta(s)$ has a simple pole at $s = 1$ with residue 1), and the second sum vanishes whenever χ is non-principal (follows since the group character is non-trivial see [8, chapter 24.2]), as we sought to show.

Thus, it is natural to generalize Dirichlet's theorem on primes in arithmetic progressions to primes of a number field. Recall that the non-vanishing of $L(1,\chi)$ for non-principal χ was proven using the analytic class number formula for $\mathbb{Q}(\zeta_m)$, which is the Ray-class field of $\mathbb{Q}((m)\infty)$; the Dedekind-zeta function for $\mathbb{Q}(\zeta_m)$ can be written as the product of L-functions. Then, assuming the existence of the ray class field $K(\mathfrak{m})$, a similar approach works for Weber L-functions. However, not having proven the existence of $K(\mathfrak{m})$, an approach *not* depending on its existence will be taken. To that end, Let $\mathcal{C}$ be a congruence subgroup modulo $\mathfrak{m}$, $X(\mathcal{C})$ the set of primitive ray class characters whose kernels contain $\mathcal{C}$, so $X(\mathcal{C})$ is a subgroup of $X(\mathfrak{m})$ isomorphic to $\mathcal{I}_K^{\mathfrak{m}}/\mathcal{C}$; hence $X(\mathcal{C})$ can be viewed as a character group of $\mathcal{I}_K^{\mathfrak{m}}/\mathcal{C}$. Then:

Theorem 2.3.8: Dirichlet Density For Number Fields

Let $\mathcal{C}$ be a congruence subgroup of modulus $\mathfrak{m}$ for a number field K and let $n = [\mathcal{I}_K^{\mathfrak{m}} : \mathcal{C}]$. Then the set of primes $\{\mathfrak{p} \in \mathcal{C}\}$ has Dirichlet density:

$$d(\mathcal{C}) = \begin{cases} \frac{1}{n} & L(1,\chi) \neq 0 \text{ forall } \chi \neq \mathbb{1} \text{ in } X(\mathbb{C}) \\ 0 & \text{otherwise} \end{cases}$$

Corollary 2.3.10 shall show that in fact $d(\mathcal{C}) = 1/n$ always (no need for modulo $\mathfrak{m}$), which shall be easier to show after this result is proven.

***Proof*:**
This proof follows the same beats as section 18.4 in MIT. First, defin the indicator function:

$$\frac{1}{n} \sum_{\chi \in X(\mathcal{C})} \chi(\mathfrak{p}) = \begin{cases} 1 & \text{if } \mathfrak{p} \in \mathcal{C}, \\ 0 & \text{otherwise.} \end{cases}$$

Note that summing over $\chi \in X(\mathcal{C})$ is equivalent to summing over the character group of $\mathcal{I}_K^{\mathfrak{m}}/\mathcal{C}$, so Corollary 18.38 $\sum \chi(\mathfrak{p}) = 0$ unless the image of $\mathfrak{p}$ in $\mathcal{I}_K^{\mathfrak{m}}/\mathcal{C}$ is the identity, meaning that $\mathfrak{p} \in \mathcal{C}$, in which case $\sum \chi(\mathfrak{p}) = \#X(\mathcal{C}) = n$.

As $s \to 1^+$ we have

$$\log L(s, \chi) \sim \sum_{\mathfrak{p}} \chi(\mathfrak{p})\mathrm{Nm}(\mathfrak{p})^{-s},$$

and therefore

$$\begin{aligned}\sum_{\chi \in X(\mathcal{C})} \log L(s, \chi) &\sim \sum_{\chi \in X(\mathcal{C})} \sum_{\mathfrak{p}} \chi(\mathfrak{p})\mathrm{Nm}(\mathfrak{p})^{-s} \\ &\sim n \sum_{\mathfrak{p} \in \mathcal{C}} \mathrm{Nm}(\mathfrak{p})^{-s}.\end{aligned}$$

By Proposition 22.19, we may write

$$L(s, \chi) = (s-1)^{e(\chi)} g(s)$$

for some function $g(s)$ that is holomorphic and nonvanishing on a neighborhood of 1, where $e(\chi) := \mathrm{ord}_{s=1} L(s, \chi)$ is -1 when $\chi = \mathbb{1}$, and $e(\chi) \geq 0$ otherwise. We have

$$\log \frac{1}{s-1} - \sum_{\chi \neq \mathbb{1}} e(\chi) \log \frac{1}{s-1} \sim n \sum_{\mathfrak{p} \in \mathcal{C}} \mathrm{Nm}(\mathfrak{p})^{-s}.$$

Dividing both sides by $n \log \frac{1}{s-1}$ yields

$$\frac{1 - \sum_{\chi \neq \mathbb{1}} e(\chi)}{n} \sim \frac{\sum_{\mathfrak{p} \in \mathcal{C}} \mathrm{Nm}(\mathfrak{p})^{-s}}{\log \frac{1}{s-1}} \quad (\text{as } s \to 1^+),$$

thus

$$d(\mathcal{C}) = \lim_{s \to 1^+} \frac{\sum_{\mathfrak{p} \in \mathcal{C}} \mathrm{Nm}(\mathfrak{p})^{-s}}{\log \frac{1}{s-1}} = \frac{1 - \sum_{\chi \neq \mathbb{1}} e(\chi)}{n}.$$

The $e(\chi)$ are integers and the Dirichlet density is nonnegative, so either $e(\chi) = 0$ for all $\chi \neq \mathbb{1}$, in which case $L(1, \chi) \neq 0$ for all $\chi \neq \mathbb{1}$ and $d(\mathcal{C}) = \frac{1}{n}$, or $e(\chi) = 1$ for exactly one of the $\chi \neq \mathbb{1}$ and $d(\mathcal{C}) = 0$ (in fact this never happens, as noted above).

Corollary 2.3.9: Dirichlet Density when Multiplying by Ideal

Let $\mathcal{C}$ be a congruence subgroup of modulus $\mathfrak{m}$ for a number field K and let $n = [\mathcal{I}_K^{\mathfrak{m}} : \mathcal{C}]$. For every ideal $I \in \mathcal{I}_K^{\mathfrak{m}}$, the set $\{\mathfrak{p} \in I\mathcal{C}\}$ has Dirichlet density:

$$d(I\mathcal{C}) = \begin{cases} \frac{1}{n} & L(1, \chi) \neq 0 \text{ for all } \chi \neq \mathbb{1} \text{ in } X(\mathbb{C}) \\ 0 & \text{otherwise} \end{cases}$$

***Proof*:**
The proof is identical to theorem 2.3.8 with the indicator function:

$$\frac{1}{n}\sum_{\chi\in X(\mathcal{C})}\chi(I)^{-1}\chi(\mathfrak{p})=\begin{cases}1 & \mathfrak{p}\in I\mathcal{C}\\ 0 & \text{otherwise}\end{cases}$$

Since then:

$$\sum_{\chi\in X(\mathbb{C})}\chi(I)^{-1}\log L(s,\chi)\sim\sum_{\chi\in X(\mathbb{C})}\sum_{\mathfrak{p}}\chi(I)^{-1}\chi(\mathfrak{p})N(\mathfrak{p})^{-s}\sim n\sum_{\mathfrak{p}\in\mathbb{C}}N(\mathfrak{p})^{-s}.$$

Corollary 2.3.10: Dirichlet Density For Number Fields Always Exists

Let $\mathcal{C}$ be a congruence subgroup of modulus $\mathfrak{m}$ for a number field K and let $n=[\mathcal{I}_K^{\mathfrak{m}}:\mathcal{C}]$. Then for every ideal, $I\in\mathcal{I}_K^{\mathfrak{m}}$, the set $\{\mathfrak{p}\in I\mathcal{C}\}$ has Dirichlet density $\frac{1}{n}$, and for every $\chi\neq\mathbb{1}$ in $X(\mathcal{C})$, $L(1,\chi)\neq 0$

***Proof*:**
Let $I_1,...,I_n\in I_K^{\mathfrak{m}}$ be coset representative of $\mathcal{C}see\mathcal{I}K^{\mathfrak{m}}$. Then all but finitely many primes of K lie in each $I_j\mathcal{C}$, hence the density of all of these must add up to 1:

$$d(I_1\mathcal{C})+\cdots+d(I_n\mathcal{C})=1$$

By corollary 2.3.9 each $d(I_j\mathcal{C})$ must be either 0 or $1/n$, which as they sum to 1 must in fact all be $1/n$. But then $L(1,\chi)\neq 0$ and $\chi\neq\mathbb{1}$ in $X(\mathcal{C})$, as we sought to show.

Corollary 2.3.11: Comparing Dirichlet Density With Split Primes

Let L/K be an abelian extension of number fields and let $\mathcal{C}$ be a congruence subgroup for a modulus $\mathfrak{m}$ of K. Then if $\mathrm{Spl}(L)\preceq\{\mathfrak{p}\in\mathcal{C}\}$, then:

$$[\mathcal{I}_K^{\mathfrak{m}}:\mathcal{C}]\leq[L:K]$$

and $[\mathcal{I}_K^{\mathfrak{m}}:\mathcal{C}]=[L:K]$ if $\mathrm{Spl}(L)\sim\{\mathfrak{p}\in\mathcal{C}\}$

***Proof*:**
Recall that $\mathrm{Spl}(L)$ has polar density $1/[L:K]$, which by proposition 2.1.17 is also the Dirichlet density. We just showed the set $\{\mathfrak{p}\in\mathcal{C}\}$ has Dirichlet density $1/[\mathcal{I}_K^{\mathfrak{m}}:\mathcal{C}]$, and by assumption $\mathrm{Spl}(L)\preceq\{\mathfrak{p}\in\mathcal{C}\}$ hence:

$$\frac{1}{[L:K]}=d(\mathrm{Spl}(L))\leq d(\mathcal{C})=\frac{1}{[\mathcal{I}_K^{\mathfrak{m}}:\mathcal{C}]}$$

completing the proof.

2.4 Conductor and Takagi Group

We shall now show how to reduce the proof of the Artin reciprocity. The idea is that there is a way of characterizing the appropriate group by means of the norm (i.e. the norm group) which shall have the right properties to satisfy the Artin map kernel condition. We start by boxing the following:

Definition 2.4.1: Conductor of Abelian Extension

Let L/K be a finite abelian extension of local fields. Then the *conductor* of L/K, denoted $\mathfrak{c}(L/K)$ is given as follows:

1. If K is Archimedean, then if $K \cong \mathbb{R}$ we have $\mathfrak{c}(L/K) = 1$, and if $K \cong \mathbb{C}$ we have $\mathfrak{c}(L/K) = 0$
2. If K is nonarchimedean, and $\mathfrak{p}$ is a maximal ideal of its valuation ring $\mathcal{O}_K$, then:

$$\mathfrak{c}(L/K) = \min\left\{n \;\middle|\; 1+\mathfrak{p}^n \subseteq \mathrm{Nm}_{L/K}(L^\times)\right\}$$

   where we set $1+\mathfrak{p}^0 = \mathcal{O}_K^\times$. If L/K is a finite abelian extension of global fields, then its conductor is the modulus:

$$\mathfrak{c}(L/K) : M_K \to \mathbb{Z} \qquad \nu \mapsto \mathfrak{c}(L_w/K_\nu)$$

Like before, we may view the finite part of $\mathfrak{c}(L/K)$ as an $\mathcal{O}_K$-ideal, and the infinite part as the subset of ramified infinite places. Note too that conductors are defined even when the extension L/K is *not* abelian, which becomes important when looking at galois representation theory.

Proposition 2.4.2: Valuation of Conductor

Let L/K be a finite abelian extension of local or global fields. Then for each prime $\mathfrak{p} \subseteq K$,

$$\nu_{\mathfrak{p}}(\mathfrak{c}(L/K)) = \begin{cases} 0 & \text{iff } \mathfrak{p} \text{ is unramified} \\ 1 & \text{iff } \mathfrak{p} \text{ ramifies tamely} \\ \geq 2 & \text{iff } \mathfrak{p} \text{ is ramifies wildly} \end{cases}$$

***Proof*:**
The statement is local, and the Archimedean case is immediate. Fix a non-Archimedean prime $\mathfrak{p}$ of K and a prime $\mathfrak{q}|\mathfrak{p}$ of L. By local class field theory (theorem 3.1.2) the local reciprocity map identifies $K_\nu^\times/N_{L_\mathfrak{q}/K_\nu}(L_\mathfrak{q}^\times)$ with $\mathrm{Gal}(L_\mathfrak{q}/K_\nu)$, and the conductor exponent $\nu_\mathfrak{p}(\mathfrak{c}(L/K))$ is by definition the least $m \geq 0$ for which the higher unit group $1+\mathfrak{p}^m$ (with $1+\mathfrak{p}^0 := \mathcal{O}_{K_\nu}^\times$) is contained in the norm group $N := N_{L_\mathfrak{q}/K_\nu}(L_\mathfrak{q}^\times)$; by local reciprocity N is open of finite index, so such an m exists. Its position in the unit filtration is governed by the ramification of $L_\mathfrak{q}/K_\nu$, by the standard behaviour of local norm groups on the higher unit groups ([5], higher ramification):

- If $\mathfrak{p}$ is unramified, then $\mathcal{O}_{K_\nu}^\times = 1+\mathfrak{p}^0 \subseteq N$, so $\nu_\mathfrak{p}(\mathfrak{c}) = 0$.
- If $\mathfrak{p}$ is tamely ramified, then $N_{L_\mathfrak{q}/K_\nu}(1+\mathfrak{q}) = 1+\mathfrak{p}$, so $1+\mathfrak{p} \subseteq N$, while $\mathcal{O}_{K_\nu}^\times \not\subseteq N$ (else $\mathfrak{p}$ would be unramified); the least m with $1+\mathfrak{p}^m \subseteq N$ is therefore $m = 1$.

- If $\mathfrak{p}$ is wildly ramified, the tame identity fails, $N_{L_\mathfrak{q}/K_\nu}(1+\mathfrak{q}) \subsetneq 1+\mathfrak{p}$ and $1+\mathfrak{p} \not\subseteq N$; hence $m=0$ (as $\mathcal{O}_{K_\nu}^\times \supseteq 1+\mathfrak{p} \not\subseteq N$) and $m=1$ both fail, so $\nu_\mathfrak{p}(\mathfrak{c}) \geq 2$.

This is the stated trichotomy for the conductor exponent, as we sought to show.

Note that the finite part of the conductor of an abelian extension divides the discriminant ideal and is divisible by the same set of prime ideals! However, the valuation of the conductor at these primes is usually smaller than the discriminant

Example 2.4.3: Comparing Size of Conductor Vs Discriminant

Take $\mathbb{Q}(\zeta_p)/\mathbb{Q}$. It's discriminant is $(p)^{p-2}$, but it's conductor is $(p)\infty$.

Lemma 2.4.4: Conductor and Subfields

Let L_1/K and L_2/K be two finite abelian extensions of a local or global field K. Then if $L_1 \subseteq L_2$, $\mathfrak{c}(L_1/K)\big|\mathfrak{c}(L_2/K)$

***Proof*:**
If $K \cong \mathbb{R}, \mathbb{C}$, the result is immediate. In the non-Archimedean case recall:

$$\mathrm{Nm}_{L_2/K}(L_2^\times) = \mathrm{Nm}_{L_2/K}(\mathrm{Nm}_{L_2/L_1}(L_2^\times)) \subseteq N_{L_1/K}(L_1^\times)$$

We shall now find a good representation of the kernel of the Artin map $\psi_{L/K}^\mathfrak{m} : \mathcal{I}_K^\mathfrak{m} \to \mathrm{Gal}_K(L)$. First, recall from [1] that we can define the norm map $\mathrm{Nm}_{L/K} : \mathcal{I}_L \to \mathcal{I}_K$ via:

$$\prod_i \mathfrak{q}_i^{n_1} \mapsto \prod_i \mathfrak{p}_i^{n_i f_i}$$

Define from this the following:

Definition 2.4.5: Takagi Group

Let L/K be a finite abelian extension of number fields and let $\mathfrak{m}$ be a modulus for K divisible by the conductor of L/K. Then the *norm group* or *Takagi group* associated to $\mathfrak{m}$ is the congruence subgroup:

$$T_{L/K}^\mathfrak{m} = \mathcal{R}_K^\mathfrak{m}\mathrm{Nm}_{L/K}(\mathcal{I}_L^\mathfrak{m})$$

Proposition 2.4.6: Relating Norm Group and Kernel of Artin Map

Let L/K be a finite abelian extension of number fields and let $\mathfrak{m}$ be a modulus for K divisible by the conductor of L/K. Then

$$(\mathcal{R}_K^\mathfrak{m} \subseteq) \ker \psi_{L/K}^\mathfrak{m} \subseteq T_{L/K}^\mathfrak{m}$$

***Proof*:**
Take $\mathfrak{p} \subseteq K$ that is in $\ker \psi^{\mathfrak{m}}_{L/K}$. Then $\mathfrak{p}$ is coprime to $\mathfrak{m}$, and splits completely in L, so $e_{\mathfrak{p}} = f_{\mathfrak{p}} = 1$. Certainly there is at least one prime $\mathfrak{q} \subseteq L$ above $\mathfrak{p}$, and we know

$$\mathrm{Nm}_{L/K}(\mathfrak{q}) = \mathfrak{p}^{f_{\mathfrak{p}}} = \mathfrak{p}$$

But then:

$$\mathfrak{p} \in \mathrm{Nm}_{L/K}(\mathcal{I}^{\mathfrak{m}}_{L}) \subseteq T^{\mathfrak{m}}_{L/K}$$

completing the proof.

We next have the following very important result relating the sizes of field extensions and the index of the norm group in the ideal group:

Theorem 2.4.7: Fundamental Inequality I

Let L/K be a finite abelian extension of number fields and let $\mathfrak{m}$ be a modulus for K divisible by the conductor of L/K. Then:‘

$$[\mathcal{I}^{\mathfrak{m}}_{K} : T^{\mathfrak{m}}_{L/K}] \leq [L : K]$$

***Proof*:**
As $T^{\mathfrak{m}}_{L/K}$ is a congruence subgroup that contains all primes of K that split completely in L by the above proposition, this follows by corollary 2.3.11.

From this, we reduced the proof of Artin reciprocity to showing $T^{\mathfrak{m}}_{L/K} \subseteq \ker \psi^{\mathfrak{m}}_{L/K}$ for any modulus $\mathfrak{m}$ divisible by the conductor L/K (so that with proposition 2.4.6 we would get $T^{\mathfrak{m}}_{L/K} = \ker \psi^{\mathfrak{m}}_{L/K}$). Let us formulate what we're proving with this new insight:

Theorem 2.4.8: Fundamental Theorem of Class Field Theory (Ideal-Theoretic)

1. **Existence**: The ray class field $K(\mathfrak{m})$ exists
2. **Completeness**: If L/K i as finite abelian extension, then $L \subseteq K(\mathfrak{m})$ if and only if $\mathfrak{c}(L/K)\big|\mathfrak{m}$. In particular, every finite abelian L/K lies in a ray class field.
3. **Artin Reciprocity**: For each subextension L/K of $K(\mathfrak{m})$, $\ker \psi^{\mathfrak{m}}_{L/K} = T^{\mathfrak{m}}_{L/K}$ with conductor $\mathfrak{c}(L/K)\big|\mathfrak{m}$ and a canonical isomorphism:

$$\mathcal{I}^{\mathfrak{m}}_K / T_{L/K} \cong \mathrm{Gal}_K(L)$$

We thus get a commutative diagram of canonical bijections:

$$\begin{tikzcd}
\{\text{Abelian } L/K \text{ with } \mathfrak{c}(L/K\big|\mathfrak{m}\} \arrow[r, "L \mapsto T^{\mathfrak{m}}_{L/K}"] \arrow[d, "L \mapsto \mathrm{Gal}_K(L)"'] & \{\text{congruence subgroups } \mathcal{C} \subseteq \mathcal{I}^{\mathfrak{m}}_K\} \arrow[d, "\mathcal{C} \mapsto \mathcal{I}^{\mathfrak{m}}_K/\mathcal{C}"] \\
\{\text{quotients of } \mathrm{Gal}_K(K(\mathfrak{m}))\} & \{\text{quotients of } \mathrm{Cl}^{\,\mathfrak{m}}_K\} \arrow[l, "\psi^{\mathfrak{m}}_{L/K}"]
\end{tikzcd}$$

As this series of notes is following [9], we shall prove the specific case where the modulus is trivial. The build-up of the modulus seems to be there to help in understanding how this work would eventually help the reader to understand the general case. In Neukirch, he doesn't introduce the modulus and just assumes the trivial modulus. If $\mathfrak{m}$ is trivial, we get the *Hilbert class field*, which we define here:

Definition 2.4.9: Hilbert Class Field

Let K be a number field. The *Hilbert class field* H of K is the maximal *abelian* extension of K unramified at every finite prime; equivalently, it is the class field of K for the trivial modulus $\mathfrak{m} = 1$, so that $\mathrm{Gal}_K(H) \cong \mathrm{Cl}_K$.

Two refinements are worth flagging. Requiring H to be unramified at the infinite places as well gives the smaller *narrow* Hilbert class field, whose Galois group is the narrow class group. Dropping the abelian condition gives the larger maximal unramified extension K_{nr}/K (inside a fixed algebraic closure $\overline{K}$), which need not be abelian; the Hilbert class field is its maximal abelian subextension.

We shall spend the next few sections showing that the Hilbert Class field exists and satisfies the properties given in theorem 2.4.8. It shall be shown that in this case the Hilbert class field will be a finite extension of K. Before moving onto to his the poofs, a few interesting facts are pointed out but not proved:

1. Infinite unramified extensions exist (and are necessarily then nonabelian, as it corresponds to an infinite galois groups) Such a construction would come from a tower of Hilbert class fields. We would start with $K_0 = K$ and then construct the Hilbert class field K_{n+1} of K_n. This would give a tower of finite abelian extensions:

$$K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$$

Taking $L = \cup_n K_n$, either get that some K_n will have class number one and so $K_N = K_n$ for all $N \geq n$, stabilizing the tower, or this does not happen. In 1964, Golod and Shafarevich showed there are infinite towers of Hilbert class fields, an example being

$$K_0 = \mathbb{Q}(\sqrt{-30030}) = \mathbb{Q}(\sqrt{-2 \cdot 3 \cdot 5 \cdot 11 \cdot 13})$$

2. It was shown that imaginary quadratic field with discriminant $|D| \leq 420$ all have finite Hilbert class field towers that stabilize at either K_2 or K_3.

3. Extensions arising from Hilbert class fields are necessarily solvable, however there are infinite nonsolvable unramified extensions, it was shown that the bi-quadratic extension:

$$\mathbb{Q}(\sqrt{17601097}, \sqrt{17380678572169893})$$

has class number one and its maximal unramified extension is an infinite extension.

The following are also not proven but gives a strong reason to use HCF:

Proposition 2.4.10: Properties of Hilbert Class Fields

Let E be the Hilbert Class field. Then:

1. E is a finite Galois extension of K and $[E : K] = h_K$, where h_K is the class number of K.

2. The ideal class group of K is isomorphic to the Galois group of E over K.

3. Every ideal of O_K extends to a principal ideal of the ring extension O_E (principal ideal theorem).

4. Every prime ideal P of O_K decomposes into the product of h_K/f prime ideals in O_E, where f is the order of $[P]$ in the ideal class group of O_K.

***Proof*:**
not in this book.

2.5 The Herbrand Quotient in Class Field Theory

The Artin reciprocity proof of section 2.6 runs on Herbrand-quotient computations for the unit and ideal class groups of a cyclic extension. Their foundations are recalled in chapter 1: the Tate cohomology $\hat{H}^n(G, A)$ of a finite group (definition 1.1.1), its long exact sequence (theorem 1.1.2), the two-periodicity of the cyclic case (theorem 1.1.4), and the Herbrand quotient with its multiplicativity (definition 1.1.5, proposition 1.1.6). This section develops the cyclic periodicity in the explicit form the proof uses and records the quotient computations it invokes for units and ideals.

Let us now look at the Tate cohomology of cyclic groups, as it will be important for the Artin Reciprocity proof. Let $G = \langle g \rangle$ be a finite cyclic group. Then $I_G = \langle g - 1 \rangle$ (as an ideal of $\mathbb{Z}[G]$). Then for any G-module A, we may create the free resolution:

$$\cdots \to \mathbb{Z}[G] \xrightarrow{N_G} \mathbb{Z}[G] \xrightarrow{g-1} \mathbb{Z}[G] \xrightarrow{N_g} \mathbb{Z}[g] \xrightarrow{g-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

As I_G is principal, $\mathrm{im}_G = \ker(g-1)$, and so this sequence is exact. Next, as G is abelian, $\mathbb{Z}[G]$ is commutative, and so there is no difference between a left and right $\mathbb{Z}[G]$-module. In particular, we can view any G-module A the following as G-modules: $\mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} A$ and $\mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G], A)$ with the G-module action being:

$$g(h \otimes a) = gh \otimes a = h \otimes ga \qquad (g\varphi)(h) = \varphi(gh)$$

Note the importance of working with a commutative group G for these to be well-defined.

Theorem 2.5.1: Characterizing Tate Cohomology Groups with Cyclic Group

Let $G = \langle g \rangle$ be a finite cyclic group and A a G-module. Then for all $n \in \mathbb{Z}$,

$$\hat{H}^{2n}(G, A) \cong \hat{H}_{2n-1}(G, A) \cong \hat{H}^0(G, A)$$
$$\hat{H}_{2n}(G, A) \cong \hat{H}^{2n-1}(G, A) \cong \hat{H}_0(G, A)$$

***Proof*:**
First, recall that:

$$\mathrm{Hom}_{\mathbb{Z}[G]}(G, A) \cong A \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} A$$

via $\varphi \mapsto \varphi(1)$ and $a \mapsto 1 \otimes a$ which are isomorphisms which preserve "g-multiplication" give endomorphisms:

$$(g\varphi)(1) = g\varphi(1) \qquad 1 \otimes ga = g(1 \otimes a)$$

Now, take the free resolution

$$\cdots \to \mathbb{Z}[G] \xrightarrow{N_G} \mathbb{Z}[G] \xrightarrow{g-1} \mathbb{Z}[G] \xrightarrow{N_g} \mathbb{Z}[g] \xrightarrow{g-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

Then compute $H^n(G, A)$ using the cochain complex:

$$0 \to A \xrightarrow{g-1} A \xrightarrow{N_G} A \xrightarrow{g-1} A \xrightarrow{N_G} A \to \cdots$$

and the chain complex:

$$\cdots \to A \xrightarrow{N_G} A \xrightarrow{g-1} A \xrightarrow{N_G} A \xrightarrow{g-1} A \to 0$$

Next, recall that $A^G = \ker(g-1)$ for all $n \geq 1$, hence:

$$H^{2n}(G, A) = H_{2n-1}(G, A) = \ker(g-1)/\mathrm{im}(N_G) = \mathrm{coker}\hat{N}_G = \hat{H}^0(G, A)$$

Hence, $\hat{H}^{2n}(G, A) = \hat{H}_{2n-1}(G, A) = \hat{H}^0(G, A)$ for all $n \in \mathbb{Z}$. As $\hat{H}^{-2n}(G, A) = \hat{H}_{2n-1}(G, A)$. Thus, $\hat{H}_{-2n+1} = \hat{H}^{2n}$ for all $n \geq 0$.

Similarly, we have $\mathrm{im}(g-1) = I_G$ for all $n \geq 1$, and so:

$$H_{2n}(G, A) = H^{2n-1}(G, A) = \ker(N_G)/\mathrm{im}(g-1) = \ker \hat{N}_G = \hat{H}_0(G, A)$$

Hence, $\hat{H}_{2n}(G, A) = \hat{H}^{2n-1}(G, A) = \hat{H}_0(G, A)$ for all $n \in \mathbb{Z}$. As $\hat{H}_{-2n}(G, A) = \hat{H}^{2n-1}(G, A)$. Thus, $\hat{H}^{-2n+1} = \hat{H}_{2n}$ for all $n \geq 0$, completing the proof.

Thus, given G is a finite cyclic group, all the Tate (co)homology groups are determined by $\hat{H}^0(G,A)$ and $\hat{H}_0(G,A)$! This complete determination by two groups motivates the following definition:

Definition 2.5.2: Herbrand Quotient

Let G be a finite cyclic group and let A be a G-module. Then define:

$$h^n(A) = h^n(G,A) = \#\hat{H}^n(G,A)$$
$$h_n(A) = h_n(G,A) = \#\hat{H}_n(G,A)$$

When $h^0(A), h_0(A) < \infty$ are both finite, Then the *Herbrand quotient* is:

$$h(A) = \frac{h^0(A)}{h_0(A)} \in \mathbb{Q}$$

For a future proof, we put down the following corollary:

Corollary 2.5.3: Exact Hexagon

Let G be a finite cycle group. Then given an exact sequence of G-modules:

$$0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \to 0$$

There is a corresponding exact hexagon:

$$\begin{tikzcd}
 & \hat{H}^0(G,A) \arrow[r, "\hat{\alpha}^0"] & \hat{H}^0(G,B) \arrow[dr, "\hat{\beta}^0"] & \\
\hat{H}_0(G,C) \arrow[ur, "\hat{\delta}_0"] & & & \hat{H}^0(G,C) \arrow[dl, "\hat{\delta}_0"] \\
 & \hat{H}_0(G,B) \arrow[ul, "\hat{\beta}_0"] & \hat{H}_0(G,A) \arrow[l, "\hat{\alpha}_0"] &
\end{tikzcd}$$

***Proof*:**
put together the above results.

Corollary 2.5.4: Herbrand Quotient Additive Function

Let G be a finite cycle group. Then given an exact sequence of G-modules:

$$0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \to 0$$

If any two $h(A), h(B), h(C)$ are defined, so is the third, and

$$h(B) = h(A)h(C)$$

***Proof*:**
By using the exact hexagon, we can compute:

$$h^0(A) = \#\hat{H}^0(G,A) = \#\ker\hat{\alpha}^0\#\hat{\alpha}^0 = \#\ker\alpha^{\#}\ker\beta^0$$

And, repeating the same for $h^0(B)$ and $\hat{H}^1(G,B)$, we get:

$$h^0(A)h^0(C)h_0(B) = \#\ker\hat{\alpha}^0\#\ker\hat{\delta}^0\#\ker\hat{\alpha}_0\#\ker\hat{\beta}_0\#\ker\hat{\delta}_0\#\ker\hat{\alpha}^0 = h^0(A)h^0(C)h_0(B)$$

Similarly for $\hat{H}^0(G,B)$, $\hat{H}_0(G,A)$, $\hat{H}_0(G,C)$, we get:

$$h^0(B)h_0(A)h_0(C) = \#\ker\hat{\beta}^0\#\ker\hat{\delta}^0\#\ker\hat{\alpha}_0\#\ker\hat{\beta}_0\#\ker\hat{\delta}_0\#\ker\hat{\alpha}^0 = h^0(A)h^0(C)h_0(B).$$

Now, if any two $h(A), h(B), h(C)$ are defined, then we see that *at least 4* of the groups in the exact hexagon are finite, and the remaining two (non-adjacent) are forced to be finite. Then re-arranging, we get:

$$h(B) = h(A)h(C)$$

as we sought to show.

Corollary 2.5.5: Special Case of Herbrand Quotient Additive Function

Let G be a finite cyclic group and let A, B be G-modules. Then if $h(A)$ and $h(B)$ are defined,

$$h(A\oplus B) = h(A)h(B)$$

***Proof*:**
Take the short exact sequence

$$0\to A\to A\oplus B\to B\to 0$$

and apply corollary 2.5.4.

Corollary 2.5.6: Herbrand Quotient for Finite and induced Groups

Let G be a finite cyclic group. Then if A is an induced or finite G-module, then

$$h(A) = 1$$

***Proof*:**
Start with A being an induced G-module. Then, as induced and coinduced modules are Tate-acyclic (see [4]), we get $h_0(A) = h^0(A) = h(A) = 1$. Next, if A is finite, then we get the exact sequence:

$$0\to AG\to A\xrightarrow{g-1} A\to A_G\to 0$$

Which give s$\#A^G = \#\ker(g-1) = \#\mathrm{coker}(g-1) = \#A_G$. Hence:

$$h_0(A) = \#\ker\hat{N}_G = \#\mathrm{coker}\hat{N}_G = h^0(A)$$

giving $h(A) = 1$, completing the proof.

Corollary 2.5.7: Herbrand Quotient For Finitely Generated Groups

Let G be a finite cyclic group. Then if A is a G-module that is also a finitely generated abelian group, then:

$$h(A) = h(A/A_{\text{tor}})$$

if either is defined.

***Proof*:**
Take the short exact sequence:

$$0 \to A_{\text{tor}} \to A \to A/A_{\text{tor}} \to 0$$

Corollary 2.5.8: Herbrand Quotient For Trivial G-module With f.g. Group

Let G be a finite cyclic group. Then if A is a trivial G-module that is also a finitely generated abelian group, then:

$$h(A) = (\#G)^r$$

where r is the rank of A

***Proof*:**
As A is finitely generated, $A/A_{\text{tor}} \cong \mathbb{Z}^r$ for appropriate r where we take $\mathbb{Z}$ with the trivial G-module structure. Then $\mathbb{Z}_G = \mathbb{Z} = \mathbb{Z}^G$, and so $\hat{N}_G : \mathbb{Z}_G \to \mathbb{Z}^G$ is multiplication by $\#G$. Thus,

$$h(Z) = \#\text{coker}\hat{N}_G / \# \ker \hat{N}_G = \#G$$

Now use induction on corollary 2.5.5.

Corollary 2.5.9: Herbrand Quotient and G-module Morphism

Let G be a finite cyclic group and let $\alpha : A \to B$ be a morphism of G-modules with finite kernel and cokernel. Then if either $h(A)$ or $h(B)$ is defined,

$$h(A) = h(B)$$

***Proof*:**
Apply the additive function property to the exact seuqences:

$$0 \to \ker \alpha \to A \to \text{im}\alpha \to 0$$
$$0 \to \text{im}\alpha \to B \to \text{coker}\alpha \to 0$$

which, by using the finiteness so that $h(\ker \alpha) = h(\text{coker}\alpha) = 1$ we get:

$$h(A) = h(\ker \alpha)h(\text{im}\alpha) = h(\text{im}\alpha) = h(\text{im}\alpha)h(\text{coker}\alpha) = h(B)$$

completing the proof.

Corollary 2.5.10: Herbrand Quotient and Finite-index Submodules

Let G be a finite cyclic group and let $A \subseteq B$ be a sub G-modules with finite index. Then if either $h(A)$ or $h(B)$ is defined,

$$h(A) = h(B)$$

***Proof*:**
Use corollary 2.5.9 on the inclusion $A \to B$.

2.6 Artin Reciprocity Proof

We now return to proving Artin reciprocity. Recall that it was proven in theorem 2.4.7 that:

$$[\mathcal{I}_K^{\mathfrak{m}} : T_{L/K}^{\mathfrak{m}}] \leq [L : K] = \mathcal{I}_K^{\mathfrak{m}} : \ker \psi_{L/K}^{\mathfrak{m}}]$$

We must now prove that:

$$[\mathcal{I}_K^{\mathfrak{m}} : T_{L/K}^{\mathfrak{m}}] \geq [L : K]$$

which would given the isomorphism:

$$\mathcal{I}_K^{\mathfrak{m}} / T_{L/K}^{\mathfrak{m}} \xrightarrow{\sim} \mathrm{Gal}_K(L)$$

We shall first prove it in the special case of a cyclic extension L/K, and assume $\mathfrak{m}$ is trivial (so that L/K is unramified), and then generalize for all unramified abelian extensions.

Ambiguous Class Number Formula

Let L/G be a finite cyclic extension with galois group G. Then note that $L, L^\times, \mathcal{O}_L, \mathcal{O}_L^\times, \mathcal{I}_L, \mathcal{P}_L$ are all G-modules as we may define a natural G-action on them. Let us compute the Tate cohomology of these G-modules and the Herbrand quotients $h(A)$. The Herbrand quotient (if the quantities are finite) is given by:

$$\hat{H}^0(A) = \hat{H}^0(G, A) = \mathrm{coker}\hat{N}_G = A^G / \mathrm{im}\hat{N}_G = \frac{A[\sigma - 1]}{N_G(A)}$$

$$\hat{H}_0(A) = \hat{H}_0(G, A) = \ker \hat{N}_G = A_G[\hat{N}_G] = \frac{A[N_G}{(\sigma - 1)A}$$

Let us find the norm element. For the multiplicative groups $\mathcal{O}_L^\times$, $L^\times$, $\mathcal{I}_L$, $\mathcal{P}_L$, the norm element $N_G = \sum_i^n \sigma^i$ will correspond to the action of the field norm $N_{L/K}$ and ideal norm $N_{L/K}$ that we've defined previously, given we identify the codomain of the norm map with a subgroup of its domain (i.e. Identifying $L^\times$ and $\mathcal{O}_L^\times$ with $K^\times$ and $\mathcal{O}_K^\times$ via inclusion, and identifying $\mathcal{I}_K$ to $\mathcal{I}_L$ via the inclusion $\mathcal{I}_K \hookrightarrow \mathcal{I}_L$, which restricts to a map $\mathcal{P}_K \hookrightarrow \mathcal{P}_L$, as when extending ideals we may get more principal ideals[4]). We may thus think of the norm map on $\mathcal{I}_L$ as taking the field norm which takes an element of $\mathcal{I}_L$ and gives an element $\mathcal{I}_K$ which corresponds to the map $I \mapsto I^{\#G}$, in which case $\mathcal{I}_K$ is a subgroup of the G-invariants $\mathcal{I}_L^G$ (namely, as L/K is unramified!).

[4]Review the number theory notes in [1] if this is unfamiliar

To un-clutter notation, we shall use N to denote both the (relative) field norm $N_{L/K}$ and the ideal norm $N_{L/K}$. Let us recall the following from field theory

Lemma 2.6.1: Character Independence

Let L/K be a finite extension of fields. Then the set $\mathrm{Aut}_K(L)$ is a linearly independent subset of the set of L-vector space functions $L \to L$

***Proof*:**
see [8, chapter 18]

Theorem 2.6.2: Trivial Tate Cohomology for L/K

Let L/K be a finite galois extension with $G = \mathrm{Gal}_K(L)$, and for any G-module A, let $\hat{H}^n(A) = \hat{H}^n(G, A)$. Then:

1. $\hat{H}^0(L)$ and $\hat{H}^1(L)$ are trivial
2. $\hat{H}^0(L^\times) \cong K^\times/N(L^\times)$, while $\hat{H}^1(L^\times)$ is trivial

***Proof*:**

1. First, by definition $L^G = K$, the trace map $T : L \to K$ is not identically zero as L/K is separable and is furthermore surjective, and $N_G(L) = T(L) = K$. So $\hat{H}^0(L) = K/K = 0$

   For $\hat{H}^1(L)$, since T is surjective choose $\alpha \in L$ with $T(\alpha) = \sum_{\tau\in G} \tau(\alpha) = 1$. Consider a 1-cocycle $f : G \to L$ so that $f(\sigma\tau) = f(\sigma) + \sigma(f(\tau))$. Let $\beta = \sum_{\tau\in G} f(\tau)\tau(\alpha)$. Then for all $\sigma \in G$:

   $$\sigma(\beta) = \sum_{\tau\in G} \sigma(f(\tau))\sigma(\tau(\alpha)) = \sum_{\tau\in G} (f(\sigma\tau)-f(\sigma))(\sigma\tau)(\alpha) = \sum_{\tau\in G} (f(\tau)-f(\sigma))\tau(\alpha) = \beta - f(\sigma),$$

   so $f(\sigma) = \beta - \sigma(\beta) = (1-\sigma)(\beta)$, exhibiting f as a 1-coboundary; hence $\hat{H}^1(L) = H^1(G, L)$ is trivial.

2. For the multiplicative group, first note that $(L^\times)^G = K^\times$, and so $\hat{H}^0(L^\times) = K^\times/N_G L^\times = K^\times/N(L^\times)$. For $\hat{H}^1(L^\times)$, take $f : G \to L^\times$, in particular $f(\sigma\tau) = f(\sigma)\sigma(f(\tau))$. Then by the linear independence of characters (lemma 2.6.1), $\alpha \mapsto \sum_{\tau\in G} f(\tau)\tau(\alpha)$ is not the zero map. Now, let $\beta = \sum_{\tau\in G} f(\tau)\tau(\alpha) \in L^\times$ be a nonzero element in its image. Then for all $\sigma \in G$:

   $$\sigma(\beta) = \sum_{\tau\in G} \sigma(f(\tau))\sigma(\tau(\alpha)) = \sum_{\tau\in G} f(\sigma\tau)f(\sigma)^{-1}(\sigma\tau)(\alpha) = f(\sigma)^{-1} \sum_{\tau\in G} f(\tau)\tau(\alpha) = f(\sigma)^{-1}\beta,$$

   so $f(\sigma) = \beta/\sigma(\beta)$, exhibiting f as a 1-coboundary; hence $\hat{H}^1(L^\times) = H^1(G, L^\times)$ is trivial, as we sought to show.

This gives the famous Hilbert Theorem 90 in cohomological form

Theorem 2.6.3: Hilbert Theorem 90

Let L/K be a finite cyclic extension with Galois group $\mathrm{Gal}_K(L) = \langle\sigma\rangle$. Then $N(\alpha) = 1$ if and only if $\alpha = \beta/\sigma(\beta)$ for some $\beta \in L^\times$

***Proof*:**
By theorem 2.5.1

$$\hat{H}^1(L^\times) \cong \hat{H}^{-1}(L^\times) = \hat{H}_0(L^\times) = L^\times[N_G]/(\sigma - 1)(L^\times)$$

And so by theorem 2.6.2

$$L^\times[N_G] = (\sigma - 1)L^\times$$

giving us the result, as we sought to show.

Let us now compute the Herbrand quotient for $\mathcal{O}_L^\times$ in the case that L/K is a finite cyclic extension of number fields. For the following, recall the discussion of infinite primes in number fields in *Local Field Theory* [5]. We shall use that notation for the following:

Theorem 2.6.4: Herbrand Quotient For Units of Ring of Integers

Let L/K be a cyclic extension of number fields with cyclic galois group $G = \langle\sigma\rangle$. Then the Herbrand quotient of the G-module $\mathcal{O}_L^\times$ is:

$$h(\mathcal{O}_L^\times) = \frac{e_\infty(L/K)}{[L : K]}$$

***Proof*:**
By Dirichlet's unit theorem, we can simplify the proof by taking elements $\epsilon_1, ..., \epsilon_{r+s} \in \mathcal{O}_L^\times$ and taking the subgroup A generated by these elements and noting that by corollary 2.5.10:

$$h(A) = h(\mathcal{O}_L^\times)$$

For each field embedding $\varphi\colon K \hookrightarrow \mathbb{C}$, let E_φ be the free $\mathbb{Z}$-module with basis given by $\{\varphi|\varphi\}$ consisting of the $[L : K] = n$ embeddings $\varphi : L \hookrightarrow \mathbb{C}$ with G-action

$$\sigma(\varphi) = \varphi \circ \sigma$$

Let ν be the infinite place of K corresponding to φ, and let A_ν be the free $\mathbb{Z}$-module with basis $\{w|\nu\}$ consisting of the places of L that extend ν also equipped withthe G-action given by the G action on $\{w\big|\nu\}$. Next, let $\pi : E_\varphi \to A_\nu$ be the G-module homomorphism sending each embedding $\varphi\big|\varphi$ to the corresponding $w\big|\nu$. Let $m := \#\{w|v\}$ and define $\tau := \sigma^m$; then τ is either trivial or has order 2, and in either case generates the decomposition group D_w for all $w|v$ (since G is abelian). We have an exact sequence

$$0 \to \ker\pi \longrightarrow E_\varphi \xrightarrow{\pi} A_v \to 0$$

where $\ker\pi = (\tau - 1)E_\varphi$. If v is unramified, then $\ker\pi = 0$ and $h(A_v) = h(E_\varphi) = 1$, since $E_\varphi \simeq \mathbb{Z}[G] \simeq \mathrm{Ind}^G(\mathbb{Z})$.

Otherwise, order $\{w|v\} = \{w_0, \ldots, w_{m-1}\}$ so

$$\ker \pi = (\tau - 1)E_\varphi = \left\{ \sum_{0 \le i < m} a_i(w_i - w_{m+i}) : a_i \in \mathbb{Z} \right\}$$

and observe that $(\ker \pi)^G = 0$, since τ acts on π as negation, and $(\ker \pi)_G \simeq \mathbb{Z}/2\mathbb{Z}$, since $(\sigma - 1)\ker \pi = \{\sum a_i(w_i - w_{m+i}) : a_i \in \mathbb{Z} \text{ with } \sum a_i \equiv 0 \bmod 2\}$ (which is killed by N_G). So in this case $h(\ker \pi) = 1/2$, and therefore $h(A_v) = h(E_\varphi)/h(\ker \pi) = 2$, As the Herbrand quotient is an additive function (corollary 2.5.4), and in every case we have $h(A_v) = e_v$, where $e_v \in \{1, 2\}$ is the ramification index of v.

Now consider the exact sequence of G-modules

$$0 \to \mathbb{Z} \to \bigoplus_{v|\infty} A_v \xrightarrow{\psi} A \to 1$$

where ψ sends each infinite place $w_1, \ldots, w_{r+s}$ of L to the corresponding $\varepsilon_1, \ldots, \varepsilon_{r+s} \in A$ given by Dirichlet's unit theorem (each A_v contains either n or $n/2$ of the w_i in its $\mathbb{Z}$-basis). The kernel of ψ is the trivial G-module $(\sum_i w_i)\mathbb{Z} \simeq \mathbb{Z}$, since we have $\psi(\sum_i w_i) = \prod_i \varepsilon_i = 1$ and no other relations among the ε_i, by Dirichlet's unit theorem, by corollary 2.5.8 $h(\mathbb{Z}) = \#G = [L : K]$, and by corollary 2.5.5, $h(\bigoplus A_v) = \prod h(A_v) = \prod e_v$, so

$$h(A) = \frac{e_\infty(L/K)}{[L : K]}$$

which from our initial observation in the proof implies that

$$h(\mathcal{O}_L^\times) = \frac{e_\infty(L/K)}{[L : K]}$$

as we sought to show.

Lemma 2.6.5: Herbrand Quotient For Ideal Group is Trivial

Let L/K be a cyclic extension of number fields with galois group G. Then for the G-module $\mathcal{I}_L$,

$$h_0(\mathcal{I}_L) = 1 \qquad h^0(\mathcal{I}_L) = e_0(L/K)[\mathcal{I}_K : N(\mathcal{I}_L)]$$

***Proof*:**
This is essentially a lot of computation. It is clear that $I \in \mathcal{I}_L^G \Leftrightarrow v_{\sigma(\mathfrak{q})}(I) = v_\mathfrak{q}(I)$ for all primes $\mathfrak{q} \in \mathcal{I}_L$. If we put $\mathfrak{p} := \mathfrak{q} \cap \mathcal{O}_K$, then for $I \in \mathcal{I}_L^G$ the value of $v_\mathfrak{q}(I)$ is constant on $\{\mathfrak{q}|\mathfrak{p}\}$, since G acts transitively on this set. It follows that $\mathcal{I}_L^G$ consists of all products of ideals of the form $(\mathfrak{p}\mathcal{O}_L)^{1/e_\mathfrak{p}}$. Therefore $[\mathcal{I}_L^G : \mathcal{I}_K] = e_0(L/K)$ and $h^0(\mathcal{I}_L) = [\mathcal{I}_L^G : N(\mathcal{I}_L)] = e_0(L/K)[\mathcal{I}_K : N(\mathcal{I}_L)]$ as claimed.

Next, for each prime $\mathfrak{q}|\mathfrak{p}$ recall that $N(\mathfrak{q}) = \mathfrak{p}^{f_\mathfrak{p}}$. Thus if $N(I) = \mathcal{O}_K$ then $N(\prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{v_\mathfrak{q}(I)}) = \mathfrak{p}^{f_\mathfrak{p} \sum_{\mathfrak{q}|\mathfrak{p}} v_\mathfrak{q}(I)} = \mathcal{O}_K$, equivalently, $\sum_{\mathfrak{q}|\mathfrak{p}} v_\mathfrak{q}(I) = 0$, for every prime $\mathfrak{p}$ of K. Order $\{\mathfrak{q}|\mathfrak{p}\}$ as

$\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ so that $\mathfrak{q}_{i+1} = \sigma(\mathfrak{q}_i)$ and $\mathfrak{q}_1 = \sigma(\mathfrak{q}_r)$, let $n_i := v_{\mathfrak{q}_i}(I)$, and define

$$J_{\mathfrak{p}} := \mathfrak{q}_1^{n_1} \mathfrak{q}_2^{n_1-n_2} \mathfrak{q}_3^{n_1-n_2-n_3} \cdots \mathfrak{q}_r^{n_1-n_2-\cdots-n_r}.$$

Then

$$\begin{aligned}\sigma(J_{\mathfrak{p}})/J_{\mathfrak{p}} &= \mathfrak{q}_2^{n_1-(n_1-n_2)} \mathfrak{q}_3^{n_1-n_2-(n_1-n_2-n_3)} \cdots \mathfrak{q}_r^{n_1-\cdots-n_{r-1}-(n_1-\cdots-n_r)} \mathfrak{q}_1^{n_1-\cdots-n_r-n_1} \\ &= \mathfrak{q}_2^{n_2} \mathfrak{q}_3^{n_3} \cdots \mathfrak{q}_r^{n_r} \mathfrak{q}_1^{-n_2-\cdots-n_r} = \mathfrak{q}_2^{n_2} \mathfrak{q}_3^{n_3} \cdots \mathfrak{q}_r^{n_r} \mathfrak{q}_1^{n_1} = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{v_{\mathfrak{q}}(I)},\end{aligned}$$

since $n_1 + \cdots + n_r = 0$ implies $n_1 = -n_2 - \cdots - n_r$. It follows that $I = \sigma(J)/J$ where $J := \prod_{\mathfrak{p}|\mathfrak{m}} J_{\mathfrak{p}}$, thus $I_L[N_G] = (\sigma - 1)(I_L)$ and $h_0(\mathcal{I}_L) = 1$.

Theorem 2.6.6: Ambiguous Class Number Formula

Let L/K be a cyclic extension of number fields with galois group G. Then the G-invariant subgroup of the G-module Cl_L has cardinality:

$$\#\mathrm{Cl}_L^G = \frac{e(L/K)\#\mathrm{Cl}_K}{n(L/K)[L:K]}$$

where $n(L/K) = [\mathcal{O}_K^\times : N(L^\times) \cap \mathcal{O}_K^\times] \in \mathbb{N}_{>0}$

***Proof*:**
The ideal class group Cl_L is the quotient of $\mathcal{I}_L$ by its subgroup $\mathcal{P}_L$ of principal fractional ideals. We thus have a short exact sequence of G-modules

$$1 \to \mathcal{P}_L \to \mathcal{I}_L \to \mathrm{Cl}_L \to 1.$$

The corresponding exact sequence in (standard) cohomology begins

$$1 \to \mathcal{P}_L^G \to \mathcal{I}_L^G \to \mathrm{Cl}_L^G \to H^1(\mathcal{P}_L) \to 1,$$

By lemma 2.6.5, $H^1(\mathcal{I}_L) \simeq \hat{H}_0(\mathcal{I}_L)$ is trivial. Therefore

$$\#\mathrm{Cl}_L^G = [\mathcal{I}_L^G : \mathcal{P}_L^G] h^1(\mathcal{P}_L). \tag{2.1}$$

Using the inclusions $\mathcal{P}_K \subseteq \mathcal{P}_L^G \subseteq \mathcal{I}_L^G$ we can rewrite the first factor on the right hand side as

$$[\mathcal{I}_L^G : \mathcal{P}_L^G] = \frac{[\mathcal{I}_L^G : \mathcal{P}_K]}{[\mathcal{P}_L^G : \mathcal{P}_K]} = \frac{[\mathcal{I}_L^G : \mathcal{I}_K][\mathcal{I}_K : \mathcal{P}_K]}{[\mathcal{P}_L^G : \mathcal{P}_K]} = \frac{e_0(L/K)\#\mathrm{Cl}_K}{[\mathcal{P}_L^G : \mathcal{P}_K]}, \tag{2.2}$$

where $[\mathcal{I}_L^G : \mathcal{I}_K] = e_0(L/K)$ follows from the proof of lemma 2.6.5. Now, consider the short exact sequence

$$1 \to \mathcal{O}_L^\times \to L^\times \overset{\alpha \mapsto (\alpha)}{\to} \mathcal{P}_L \to 1.$$

The corresponding long exact sequence in cohomology begins

$$1 \to \mathcal{O}_K^\times \to K^\times \to \mathcal{P}_L^G \to H^1(\mathcal{O}_L^\times) \to 1 \to H^1(\mathcal{P}_L) \to H^2(\mathcal{O}_L^\times) \to H^2(L^\times), \tag{2.3}$$

where by Hilbert's 90th Theorem $H^1(L^\times)$ is trivial. We have $K^\times/\mathcal{O}_K^\times \simeq \mathcal{P}_K$, thus

$$[\mathcal{P}_L^G : \mathcal{P}_K] = h^1(\mathcal{O}_L^\times) = \frac{h^0(\mathcal{O}_L^\times)}{h(\mathcal{O}_L^\times)} = \frac{h^0(\mathcal{O}_L^\times)[L:K]}{e_\infty(L/K)},$$

by theorem 2.6.4. Combining this identity with (2.1) and (2.2) gives

$$\#\mathrm{Cl}_L^G = \frac{e(L/K)\#\mathrm{Cl}_K}{[L:K]} \cdot \frac{h^1(\mathcal{P}_L)}{h^0(\mathcal{O}_L^\times)}. \tag{2.4}$$

We can write the second factor on the right hand side using the second part of the long exact sequence in (2.3). Next, by theorem 2.5.1 we have $H^2(-) = \hat{H}^2(-) = \hat{H}^0(-)$. Thus

$$H^1(\mathcal{P}_L) \simeq \ker(\hat{H}^0(\mathcal{O}_L^\times) \to \hat{H}^0(L^\times)) \simeq \ker(\mathcal{O}_K^\times/N(\mathcal{O}_L^\times) \to K^\times/N(L^\times)),$$

so $h^1(\mathcal{P}_L) = [\mathcal{O}_K^\times \cap N(L^\times) : N(\mathcal{O}_L^\times)]$. We have $h^0(\mathcal{O}_L^\times) = [\mathcal{O}_K^\times : N(\mathcal{O}_L^\times)]$, thus

$$\frac{h^0(\mathcal{O}_L^\times)}{h^1(\mathcal{P}_L)} = [\mathcal{O}_K^\times : N(L^\times) \cap \mathcal{O}_K^\times] = n(L/K),$$

and plugging this into (2.4) yields the desired formula, completing the proof.

2.6.1 Artin Reciprocity: Unramified Cyclic Case Proof

Let us now prove the Artin Reciprocity, in particular the proof was reduced to showing the second fundamental inequality:

Theorem 2.6.7: Fundamental Inequality II

Let L/K be a totally unramified cyclic extension of number fields. Then:

$$[\mathcal{I}_K : N(\mathcal{I}_L)\mathcal{P}_K] \geq [L:K]$$

Note that the totally unramified condition applies to infinite primes as well.

***Proof*:**
First,

$$[\mathcal{I}_K : N(\mathcal{I}_L)\mathcal{P}_K] = \frac{[\mathcal{I}_K : \mathcal{P}_K]}{[N(\mathcal{I}_L)\mathcal{P}_K : \mathcal{P}_K]} = \frac{\#\mathrm{Cl}_K}{[N(\mathcal{I}_L)\mathcal{P}_K : \mathcal{P}_K]}$$

Re-writing the denominator on the right hand side, we get:

$$\begin{aligned}
[N(\mathcal{I}_L)\mathcal{P}_K : \mathcal{P}_K] &= [N(\mathcal{I}_L) : N(\mathcal{I}_L) \cap \mathcal{P}_K] && \text{2nd iso. thm} \\
&= [\mathcal{I}_L : N^{-1}(\mathcal{P}_K)] && \text{Snake Lem.} \\
&= [\mathcal{I}_L/\mathcal{P}_L : N^{-1}(\mathcal{P}_K)/\mathcal{P}_L] && \text{3rd iso. thm} \\
&= [\mathrm{Cl}_L : \mathrm{Cl}_L[N_G]] && \mathrm{Cl}_L[N_G] := N^{-1}(\mathcal{P}_K)/\mathcal{P}_L \\
&= \#N_G(\mathrm{Cl}_L)
\end{aligned}$$

Now, as $h^0(\mathrm{Cl}_L) = [\mathrm{Cl}_L^G : N_G(\mathrm{Cl}_L)]$, by theorem 2.6.6 we get:

$$[\mathcal{I}_K : N(\mathcal{I}_L)\mathcal{P}_K] = \frac{\#\mathrm{Cl}_K h^0(\mathrm{Cl}_L)}{\#\mathrm{Cl}_L^G} = \frac{h^0(\mathrm{Cl}_L)n(L/K)[L:K]}{e(L/K)} \overset{!}{\geq} [L:K]$$

where the $\overset{!}{\geq}$ comes from the fact that the extension is totally unramified, so $e(L/K) = 1$, and that $h^0(\mathrm{Cl}_L), n(L/K) \geq 1$, which gives the desired inequality as we sought to show.

Theorem 2.6.8: Artin Reciprocity Law: Unramified Cyclic Case

Let L/K be a totally unramified cyclic extension of number fields. Then:

$$[\mathcal{I}_K : T_{L/K}] = [L:K]$$

which implies the Artin map induces an isomorphism:

$$\mathcal{I}_K / T_{L/K} \cong \mathrm{Gal}_K(L)$$

***Proof*:**
We combine the inequalities from theorem 2.4.7 and theorem 2.6.7 to get $[\mathcal{I}_K : T_{L/K}] = [L:K]$. For the isomorphism, as $\ker \psi_{L/K} \subseteq T_{L/K}$ and $\psi_{L/K}$ is surjective, we get by counting the index that $\ker \psi_{L/K} = T_{L/K}$, implying $\mathcal{I}_K/T_{L/K} \cong \mathrm{Gal}_K(L)$, as we sought to show.

Corollary 2.6.9: Class Number Away From Galois Group

Let L/K be a totally unramified cyclic extension of number fields. Then

$$\#\mathrm{Cl}_L^G = \frac{\#\mathrm{Cl}_K}{[L:K]}$$

and the Tate cohomology groups of Cl_L are all trivial

***Proof*:**
By the Artin Reciprocity Law in the unramified case and the Fundamental Inequality II, we get

$$n(L/K) = 1 \qquad h^0(\mathrm{Cl}_K) = 1 \qquad e(L/K) = 1$$

and so by the Ambiguous Class Number Formula:

$$\#\mathrm{Cl}_L^G = \frac{\#\mathrm{Cl}_L}{[L:K]}$$

Furthermore as Cl_L is finite by corollary 2.5.6 and

$$h(\mathrm{Cl}_K) = \frac{h^-(\mathrm{Cl}_L)}{h_0(\mathrm{Cl}_L)} = 1 \qquad \Rightarrow \qquad h_0(\mathrm{Cl}_L) = 1$$

And so $\hat{H}^{-1}(\mathrm{Cl}_L)$ and $\hat{H}^0(\mathrm{Cl}_L)$ are trivial, which by theorem 2.5.1 implies all the Tate cohomology groups are trivial, as we sought to show.

Corollary 2.6.10: Norm map is Surjective

Let L/K be a totally unramified cyclic extension of number fields. Then every unit in $\mathcal{O}_K^\times$ is the norm of an element of L, i.e. There is a surjection $\mathrm{Nm}_{L/K} : L^\times \to \mathcal{O}_K^\times$

***Proof*:**

$$n(L/K) = [\mathcal{O}_K^\times : N(L^\times) \cap \mathcal{O}_K^\times] = 1 \qquad \Rightarrow \qquad \mathcal{O}_K^\times = N(L^\times) \cap \mathcal{O}_K^\times$$

2.6.2 Artin Reciprocity: Abelian Case Proof

We now generalize to the case of an unramified abelian extension. We shall prove it for the case where the modulus is trivial (and hence that he Hilbert class field exists), however we can slightly generalize our prove to show that that $K(\mathfrak{m})$ exists given that a finite cyclic extension K/L with modulus $\mathfrak{m}$ exists. We do now show Artin reciprocity in the cyclic case with the trivial modulus, but this build-up shows that we have in fact done a good deal of the work towards it:

Definition 2.6.11: Class Field For Modulus

Let $\mathfrak{m}$ be a modulus for a number field K and let L/K be a finite abelian extension ramified only at primes $\mathfrak{p}\big|\mathfrak{m}$. Then we say that L is a *class field* for $\mathfrak{m}$ is

$$\ker \psi^{\mathfrak{m}}_{L/K} = T^{\mathfrak{m}}_{L/K}$$

where $\psi^{\mathfrak{m}}_{L/K} : \mathcal{I}^{\mathfrak{m}}_K \to \mathrm{Gal}_K(L)$ is the Artin map

Note that we can make this definition stronger than some books due to our build-up, namely we have proven the surjectivity of the Artin map which we factor into the definition.

Lemma 2.6.12: Compositum of Class Fields

Let $\mathfrak{m}$ be a modulus for a number field K. Then if L_1, L_2 are class fields for $\mathfrak{m}$, then so is their compositum

$$L = L_1L_2$$

***Proof*:**

Certainly, $L = L_1L_2$ is ramified only at primes ramified in either L_1, L_2, so L is ramified only at primes $\mathfrak{p}\big|\mathfrak{m}$. Then similarly to theorem 2.1.24 a prime $\mathfrak{p} \nmid fm$ splits completely in L if it splits completely in L_1 and L_2, implying

$$\ker \psi^{\mathfrak{m}}_{L/K} = \ker \psi^{\mathfrak{m}}_{L_1/K} \cap \ker^{\mathfrak{m}}_{L_2/K}$$

As the norm map is transitive on towers, if $I = N_{L?K}(J)$, then $I = N_{L_1/K}(N_{L/L_1}(J))$ and $I = N_{L_2/K}(N_{L/L_2}(J))$, and so:

$$N(\mathcal{I}^{\mathfrak{m}}_L) \subseteq N(\mathcal{I}^{\mathfrak{m}}_{L_1}) \cap N(\mathcal{I}^{\mathfrak{m}}_{L_2}) \qquad \Rightarrow \qquad T^{\mathfrak{m}}_{L/K} \subseteq T^{\mathfrak{m}}_{L_1/K} \cap T^{\mathfrak{m}}_{L_2/K}$$

Finally, if L_1 and L_2 are class fields for $\mathfrak{m}$, then

$$T^{\mathfrak{m}}_{L/K} \subseteq T^{\mathfrak{m}}_{L_1/K} \cap T^{\mathfrak{m}}_{L_2/K} = \ker \psi^{\mathfrak{m}}_{L_1/K} \cap \ker^{\mathfrak{m}}_{L_2/K} = \ker \psi^{\mathfrak{m}}_{L/K}$$

Then as $\ker^{\mathfrak{m}}_{L/K} \subseteq T_{L/K}\mathfrak{m}$, we have equality $T_{L/K} = \ker \psi^{\mathfrak{m}}_{L/K}$, completing the proof.

Corollary 2.6.13: Extending to Abelian Case

Let $\mathfrak{m}$ be a modulus for a number field K. Then if every finite cyclic extension of K with conductor dividing $\mathfrak{m}$ is a class field for $\mathfrak{m}$, then so is every abelian extension of K with conductor dividing $\mathfrak{m}$

***Proof*:**
Let L/K be a finite abelian extension with conductor $\mathfrak{c}\big|\mathfrak{m}$. Then the conductor of any subextension of L divides $\mathfrak{c}$ by lemma 2.4.4, and hence $\mathfrak{m}$.

Now, write $G = \mathrm{Gal}_K(L) \cong H_1 \times \cdots \times H_r$ as a product of cyclic groups, and take

$$L_i = L^{\overline{H}_i} \qquad \overline{H}_i = \prod_{j \neq i} H_J \subseteq G$$

Then the galois groups for these fields are cyclic

$$\mathrm{Gal}_K(L_i) = G/\overline{H_i} \cong H_i$$

Noting that $L = L_1 \cdots L_r$, which is a compositum of linearly disjoint cyclic extensions of K, we have by lemma 2.6.12 that if each L_i are class fields for $\mathfrak{m}$, so is L, completing the proof.

We now use this to finish off by showing the proof of the Fundamental Theorem of Class Field Theory given a trivial modulus:

Theorem 2.6.14: Fundamental Theorem of C.F.T. (Ideal-theoretic): Unramified

Let K be a number field with Hilbert class field H. Then:

1. H/K is a finite extension with $\mathrm{Gal}_K(H)$ isomorphic to a quotient of Cl_K
2. *Partial Existence*: $K(1)$ exists if and only if $\mathrm{Gal}_K(H) \cong \mathrm{Cl}_K$ in which case $K(1) = H$
3. *Completeness*: every unramified abelian extension of K is a subfield of H
4. *Artin Reciprocity*: for every unramified abelian extension of K, we have $\ker \psi_{L/K} = T_{L/K}$ along with a canonical isomorphism:

$$\mathcal{I}_K/T_{L/K} \cong \mathrm{Gal}_K(L)$$

***Proof*:**
Artin Reciprocity (4). Theorem 2.6.8 and corollary 2.6.13 give $\ker \psi_{L/K} = T_{L/K}$ and the canonical isomorphism $\mathcal{I}_K/T_{L/K} \cong \mathrm{Gal}_K(L)$ for every unramified abelian extension L/K. Distinct such extensions correspond to distinct quotients of Cl_K: a prime splits completely in L precisely when it lies in the kernel of the Artin map, and an unramified abelian extension is determined by its set of completely split primes.

Existence and Completeness (1)-(3). That the Hilbert class field H exists, is finite over K with $\mathrm{Gal}_K(H) \cong \mathrm{Cl}_K$, and contains every unramified abelian extension of K, is the *existence* half of the theory and is not established by reciprocity alone. Its proof is deferred to the Existence Theorem (theorem 3.3.4), established via the idelic approach of the next chapter.

2.7 Application: Solving Diophantine Equations

Throughout this chapter, we have built a massive machine. We have moved from the geometry of lattices to the analysis of zeta functions, and finally the Global Artin Reciprocity Law (theorem 3.3.3), which established a the isomorphism between the ideal class group (an algebraic structure of "internal" arithmetic) and the Galois group (the algebraic structure of "external" symmetry).

However, it is reasonable to pause and ask: *Why?* The motivating problem of Number Theory is not necessarily to understand the structure of infinite abelian extensions for their own sake, but to solve equations involving integers. Specifically, we want to know when a polynomial $f(x) \in \mathbb{Z}[x]$ has roots modulo p, or when a prime p can be represented by a specific quadratic form $x^2 + ny^2$.

It turns out that Artin Reciprocity is not merely a structural result; it is a computational engine. It allows us to convert difficult questions about factorization in high-degree field extensions into simple questions about modular arithmetic in the base field.

2.7.1 Splitting of Primes as Congruence Conditions

The most direct application of Class Field Theory is determining the factorization behavior of primes in an extension. Recall the Dedekind-Kummer theorem [1]: if $L = K(\theta)$ is defined by a polynomial $f(x)$, the factorization of a prime $\mathfrak{p}$ in $\mathcal{O}_L$ mirrors the factorization of $f(x)$ modulo $\mathfrak{p}$ given it is coprime to the conductor (which we now see is a "trivial" blocker that originates by working with global fields rather than adeles).

Specifically, $f(x)$ splits completely into linear factors modulo $\mathfrak{p}$ if and only if $\mathfrak{p}$ splits completely in L. Artin Reciprocity shows exactly when this happens.

Theorem 2.7.1: The Splitting Theorem

Let L/K be a finite abelian extension of number fields with conductor $\mathfrak{m}$. Let $\psi^{\mathfrak{m}}_{L/K} : \mathcal{I}^{\mathfrak{m}}_K \to \mathrm{Gal}(L/K)$ be the Artin map. Then a prime $\mathfrak{p}$ of K (coprime to $\mathfrak{m}$) splits completely in L if and only if:

$$\mathfrak{p} \in \ker \psi^{\mathfrak{m}}_{L/K}$$

Since $\ker \psi^{\mathfrak{m}}_{L/K} = T^{\mathfrak{m}}_{L/K}$ (the Takagi group), this is equivalent to saying that $\mathfrak{p}$ splits completely if and only if $\mathfrak{p} = (\alpha)$ for some $\alpha \in K^{\mathfrak{m},1}$ (up to norms). In simple terms:

splitting is determined by congruence conditions modulo $\mathfrak{m}$

***Proof*:**

Recall that for an unramified prime $\mathfrak{p}$, the Artin symbol $\left(\frac{L/K}{\mathfrak{p}}\right)$ is the Frobenius automorphism $\sigma_{\mathfrak{p}}$. By definition, the decomposition group $D_{\mathfrak{p}}$ is generated by $\sigma_{\mathfrak{p}}$. The prime $\mathfrak{p}$ splits completely if and only if $e = f = 1$, which for an unramified prime implies $[D_{\mathfrak{p}} : 1] = 1$, i.e., $\sigma_{\mathfrak{p}} = \mathrm{id}$.

Thus, $\mathfrak{p}$ splits completely if and only if $\left(\frac{L/K}{\mathfrak{p}}\right) = 1$, which is exactly the statement that $\mathfrak{p} \in \ker \psi^{\mathfrak{m}}_{L/K}$.

This theorem is powerful because it shows that for abelian equations, the question "Does $f(x) \equiv 0 \pmod{p}$ have a solution?" reduces to "Is $p \equiv a \pmod{m}$?"

Example 2.7.2: Splitting in a Cyclic Cubic Field

Consider the polynomial $f(x) = x^3 - 3x + 1 \in \mathbb{Z}[x]$. We wish to know for which primes p this polynomial splits completely (has 3 roots modulo p).

Checking manually is tedious:

- $p = 5$: $x^3 - 3x + 1$ is irreducible (no roots).
- $p = 7$: Irreducible.
- $p = 17$: $17 \equiv 8 \pmod{9}$. $f(-3) = -27 + 9 + 1 = -17 \equiv 0$. It splits!
- $p = 19$: $19 \equiv 1 \pmod{9}$. $f(x)$ splits.

Is there a pattern? The discriminant of $f(x)$ is $D = 81 = 9^2$. Since the discriminant is a square, the Galois group is a subgroup of $A_3 \cong C_3$. Thus, the splitting field L is a cyclic cubic extension of $\mathbb{Q}$.

By the Kronecker-Weber theorem, $L \subseteq \mathbb{Q}(\zeta_m)$ for some m. The conductor of L turns out to be 9. The Galois group of $\mathbb{Q}(\zeta_9)$ is $(\mathbb{Z}/9\mathbb{Z})^\times \cong \{1, 2, 4, 5, 7, 8\}$. This group has order 6.

Our field L corresponds to the unique subgroup of index 3 (order 2) in $(\mathbb{Z}/9\mathbb{Z})^\times$. This subgroup is $H = \{1, 8\} \pmod{9}$.

By theorem 2.7.1, a prime p splits completely in L (and thus $f(x)$ has 3 roots) if and only if the Artin symbol acts trivially, which means the Frobenius element lies in the fixing subgroup. In the context of cyclotomic fields, the Artin map sends $p \to (p \pmod{m})$.

Thus, $f(x) \equiv 0 \pmod{p}$ has 3 solutions if and only if:

$$p \equiv 1 \text{ or } 8 \pmod{9}$$

We have turned a hard algebra problem into checking the last digit of a prime in base 9.

2.7.2 Primes of the form $x^2 + ny^2$

Fermat famously proved that an odd prime p can be written as $p = x^2 + y^2$ if and only if $p \equiv 1 \pmod{4}$. Euler conjectured, and Class Field Theory finally proved, generalizations of this for $x^2 + ny^2$.

This problem is intimately tied to the *Hilbert Class Field* and its generalization, the *Ring Class Field.*

Let $K = \mathbb{Q}(\sqrt{-n})$. If $h_K = 1$ (the class number is 1), then $\mathcal{O}_K$ is a PID. In this case, $p = x^2 + ny^2$ is equivalent to p splitting in K (and thus being the norm of a principal element). This happens if $\left(\frac{-n}{p}\right) = 1$.

However, if $h_K > 1$, simply splitting in K is not enough; the prime ideals lying above p must be *principal.* Class Field Theory characterizes principal ideals as those that split completely in the Hilbert Class Field H.

Theorem 2.7.3: Representation by Quadratic Forms

Let $n > 0$ be a square-free integer. Let $K = \mathbb{Q}(\sqrt{-n})$. Let H be the Hilbert Class Field of K. A prime p (not dividing $2n$) is of the form $x^2 + ny^2$ if and only if p splits completely in H.

***Proof*:**
$p = x^2 + ny^2 = (x + y\sqrt{-n})(x - y\sqrt{-n})$ implies that p is the norm of an element $\alpha \in \mathcal{O}_K$ (assuming $\mathcal{O}_K = \mathbb{Z}[\sqrt{-n}]$, which holds if $-n \equiv 2, 3 \pmod{4}$). This is equivalent to saying $p\mathcal{O}_K = (\alpha)(\bar{\alpha})$ splits into principal prime ideals.

By the properties of the Hilbert Class Field (proposition 2.4.10), the Artin map $\psi_{H/K}$ induces an isomorphism $\mathrm{Cl}_K \cong \mathrm{Gal}(H/K)$. The kernel of this map is the set of principal ideals. Therefore, a prime ideal $\mathfrak{p}$ of K is principal if and only if $\psi_{H/K}(\mathfrak{p}) = 1$, which means $\mathfrak{p}$ splits completely in H.

Consequently, p is of the form $x^2 + ny^2$ iff p splits in K into $\mathfrak{p}\bar{\mathfrak{p}}$ AND $\mathfrak{p}$ is principal iff p splits in K and $\mathfrak{p}$ splits in H iff p splits completely in H.

Example 2.7.4: Primes of the form $x^2 + 5y^2$

Let $n = 5$. Consider $K = \mathbb{Q}(\sqrt{-5})$. The class number is $h_K = 2$. The Hilbert Class Field H must be an unramified abelian extension of degree 2 over K. Consider $H = K(i) = \mathbb{Q}(\sqrt{-5}, i)$.

- The discriminant of K is -20.
- H is generated over K by i, a root of x^2+1. The discriminant of this polynomial is -4. Since -4 and -20 share factors, we must be careful checking unramified-ness, but computation shows H/K is indeed everywhere unramified.

Thus $H = \mathbb{Q}(\sqrt{-5}, \sqrt{-1}) = \mathbb{Q}(\sqrt{5}, \sqrt{-1})$. By the theorem, $p = x^2 + 5y^2$ iff p splits completely in H. For a rational prime to split completely in a biquadratic field $\mathbb{Q}(\sqrt{5}, \sqrt{-1})$, it must split in the quadratic subfields $\mathbb{Q}(\sqrt{-1})$ and $\mathbb{Q}(\sqrt{5})$.

- Splits in $\mathbb{Q}(\sqrt{-1}) \iff p \equiv 1 \pmod 4$.
- Splits in $\mathbb{Q}(\sqrt{5}) \iff p \equiv 1, 4 \pmod 5$ (Quadratic reciprocity).

Combining these via Chinese Remainder Theorem:

$$p \equiv 1, 9 \pmod{20}$$

Thus, we have derived a static congruence condition for a representation problem that eluded mathematicians for centuries.

2.7.3 Explicit Class Field Theory

The previous examples illustrate a satisfying workflow:

1. Start with a base field K.
2. Identify a specific abelian extension L (like the Hilbert Class Field).
3. Use Artin Reciprocity to describe splitting in L via congruences in K.

However, step 2 contains a gap. We know H *exists* by the Existence Theorem, but can we write it down? Can we generate it?

For $K = \mathbb{Q}$, the Kronecker-Weber theorem shows exactly how to generate abelian extensions: use roots of unity ζ_n. The values of the analytic function $e^{2\pi i z}$ at rational points $z \in \mathbb{Q}$ generate the abelian extensions.

For $K = \mathbb{Q}(\sqrt{-n})$ (imaginary quadratic), the theory of *Complex Multiplication* provides the answer. The abelian extensions are generated by the values of the j-invariant function (a modular function) at quadratic irrationalities.

For example, the Hilbert Class Field of $K = \mathbb{Q}(\sqrt{-23})$ ($h_K = 3$) is generated by the real root of $x^3 - x - 1$. This polynomial comes from the modular equation related to the j-invariant.

This brings us to the boundary of the "Hilbert's 12th Problem": explicitly constructing class fields for arbitrary number fields. While we have solved it for $\mathbb{Q}$ and imaginary quadratic fields, the general case remains one of the great open problems in number theory.

Exercise 2.7.1

1. Show that $p = x^2 + 27y^2$ if and only if $p \equiv 1 \pmod 3$ and 2 is a cubic residue mod p. (Hint: The ring of integers of $\mathbb{Q}(\sqrt{-27})$ is not $\mathbb{Z}[\sqrt{-27}]$, so you must work with the *ring class field* of the order $\mathbb{Z}[\sqrt{-27}]$. The polynomial $x^3 - 2$ defines this extension).
2. Determine the congruence conditions on p for the polynomial $x^4 - 2 \pmod p$ to split completely. (Hint: The Galois group is D_8, which is non-abelian. You will need to look at the lattice of

subfields and apply CFT to the abelian pieces, or use the Chebotarev Density Theorem to find the density of such primes).

3

Class Field Theory: Adèle Approach

In this chapter, we shall study all completions of a number ring at once. Recall that each $\mathbb{Q}_p$ is locally compact. The product topology $X = \prod_{p \leq \infty} \mathbb{Q}_p$ is *not* necessarily locally compact: if $x \in X$ has a compact neighborhood $C \subseteq X$, then as each open subset must be $\mathbb{Q}_p$ for all but finitely many positions, we see that $\pi_i(C)$ only compact iff almost all $\mathbb{Q}_p$ are compact, which we know is not the case.

3.1 build-up: Local Class Field Theory

The goal of this section is to classify all finite abelian extensions of a given local field K. Instead of doing this for each individual finite abelian extension L/K, we shall instead to it all together with our developed theory of Adele rings by using infinite galois theory (see [8, chapter 18.4]).

Definition 3.1.1: Maximal Abelian Extension

Let K be a field with separable closure K^{sep}. Then the field:

$$K^{\text{ab}} = \bigcup_{\substack{L \subseteq K^{\text{sep}} \\ L/K \text{ finite Ab.}}} L$$

is the *maximal abelian extension* of K.

As the categories of separable over residue fields correspond to unramified extensions ([5]) $K^{nr} \subseteq K^{\text{ab}}$ (recall that in the Archimedean case $K = K^{nr}$ and $K^{\text{ab}} = K^{\text{sep}} = \mathbb{C}$, as $\mathbb{C}/\mathbb{R}$ is ramified). We thus have:

$$K \subseteq K^{nr} \subseteq K^{\text{ab}} \subseteq K^{\text{sep}}$$

By infinite galois theory, we have that:

$$\mathrm{Gal}_K(K^{\text{ab}}) \cong \varprojlim_{L} \mathrm{Gal}_K(L)$$

where L ranges over all finite extensions of K in Y^{ab} ordered by inclusion. We then have the galois correspondence:

$$\left\{\begin{matrix}\text{Finite Extension } E/K \\ K \subseteq E \subseteq K^{\text{ab}}\end{matrix}\right\} \underset{(K^{\text{ab}})^H}{\overset{\mathrm{Gal}_E(K^{\text{ab}})}{\rightleftarrows}} \left\{\begin{matrix}\text{closed subgroups } H \\ H \subseteq \mathrm{Aut}_K(K^{\text{ab}})\end{matrix}\right\}$$

Then finite abelian extensions L/K correspond to pen subgroups of $\mathrm{Gal}_K(K^{\text{ab}})$ (which have finite index as $\mathrm{Gal}_K(K^{\text{ab}})$ is compact). If K is Archimedean so that $K \in \{\mathbb{R}, \mathbb{C}\}$, then this correspondence is immediate: $\mathbb{R}^{\text{ab}} = \mathbb{C}$, $\mathbb{C}^{\text{ab}} = \mathbb{C}$. So let's look at the nonarchimedean case.

Let K is a nonarchimedean local field with ring of integers $\mathcal{O}_K$, maximal ideal $\mathfrak{p}$, and residue field $\mathbb{F}_\mathfrak{p} = \mathcal{O}_K/\mathfrak{p}$. Then if L/K is finite unramified with residue field $\mathbb{F}_\mathfrak{q} = \mathcal{O}_L/\mathfrak{q}$. By [5], we have the natural isomorphism:

$$\langle \mathrm{Frob}_{L/K} \rangle = \mathrm{Gal}_K(L) \cong \mathrm{Gal}_{\mathbb{F}_\mathfrak{p}}(\mathbb{F}_\mathfrak{q}) = \langle x \mapsto x^{\#\mathbb{F}_\mathfrak{p}} \rangle$$

where $\mathrm{Frob}_{L/K} \in \mathrm{Gal}_K(L)$ generates it. Then as shown in section 2.1, we can naturally define the Artin map:

$$\psi_{L/K} : \mathcal{I}_K \to \mathrm{Gal}_K(L) \qquad \mathfrak{p} \mapsto \mathrm{Frob}_{L/K}$$

which corresponds to the quotient map $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ with $n = [L : K]$. Now, notice that each $I \in \mathcal{I}_K$ is principal when working over a local field as $\mathcal{O}_K$ is a DVR (and hence a PID), and so every I is of the form (x) for $x \in K^\times$. We may thus naturally extend the Artin map to $K^\times$ via:

$$\psi_{L/K}(x) := \psi_{L/K}((x))$$

which sends every uniformizer π to the Frobenius element.

We would like to extend this map to $\theta_K : K^\times \to \mathrm{Gal}_K(K^{\text{ab}})$, which is known as the *local Artin homomorphism*.

Theorem 3.1.2: Local Artin Reciprocity

Let K be a local field. Then there is a unique continuous homomorphism

$$\theta_K : K^\times \to \mathrm{Gal}_K(K^{\mathrm{ab}})$$

such that for each finite intermediary extension $K \subseteq L \subseteq K^{\mathrm{ab}}$ the homomorphism:

$$\theta_{L/K} : K^\times \to \mathrm{Gal}_K(L)$$

which is given by $\varphi_{L/K}\mathrm{Res}_{L/K} \circ \theta_K$ (where Res is the natural restriction map $\mathrm{Res}_{L/K} : \mathrm{Gal}_K(K^{\mathrm{ab}}) \twoheadrightarrow \mathrm{Gal}_K(L)$) has the properties that:

1. if K is nonarchimedean, L/K unramified, then $\theta_{L/K}(\pi) = \mathrm{Frob}_{L/K}$ for every uniformizer π of $\mathcal{O}_K$
2. $\theta_{L/K}$ is surjective with kernel $N_{L/K}(L^\times)$, which induces:

$$\frac{K^\times}{N_{L/K}(L^\times)} \cong \mathrm{Gal}_K(L)$$

The restriction map can be defined, and hence interpreted, in many different ways:

- the map $\sigma \mapsto \sigma|_L$
- the quotient map $\mathrm{Gal}_K(K^{\mathrm{ab}}) \twoheadrightarrow \frac{\mathrm{Gal}_K(K^{\mathrm{ab}})}{\mathrm{Gal}_L(K^{\mathrm{ab}})}$
- The projection from $\mathrm{Gal}_K(K^{\mathrm{ab}}) = \varprojlim_L \mathrm{Gal}(L/K) \subseteq \prod_L \mathrm{Gal}_K(L)$

By the last point, these are exactly the maps that appear in the inverse system defining $\mathrm{Gal}_K(K^{\mathrm{ab}})$.

***Proof*:**
This is the central theorem of local class field theory; in the cohomological approach it is a consequence of the local fundamental class. Take L/K finite abelian with $G = \mathrm{Gal}_K(L)$. The group $H^2(G, L^\times) \cong \mathrm{Br}(L/K)$ carries a canonical generator, the *local fundamental class*, whose image under the invariant map $\mathrm{inv}_K \colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ is $1/[L : K]$. Cup product with this class gives, by Tate–Nakayama for the local class formation, an isomorphism

$$\frac{K^\times}{N_{L/K}(L^\times)} \cong \hat{H}^0(G, L^\times) \cong \hat{H}^{-2}(G, \mathbb{Z}) = G^{\mathrm{ab}} = G$$

which is property (2); passing to the limit over L produces the continuous θ_K, unique subject to (1). The invariant map, local Tate duality, and the Tate–Nakayama theorem underlying this are developed in *Group and Galois Cohomology* [4, Local Reciprocity]; the displayed isomorphism is recalled as corollary 1.4.2. Property (1), that a uniformizer maps to the Frobenius in the unramified case, identifies $\theta_{L/K}$ on $\mathcal{I}_K$ with the ideal-theoretic Artin map $\psi_{L/K}$ defined above, so the local and ideal-theoretic constructions agree. It remains to assemble these local maps into the global Artin homomorphism, carried out in section 3.3.

There are some interesting contrasts between the local case using Adeles and the global cases using places. Notice that :

1. There is no modulus! K^{ab} contains all abelian extension of K, with non being omitted
2. The ray class group $\mathrm{Cl}_K^{\mathfrak{m}}$ is now replaced by the quotient $K^\times$
3. The Takagi group $N_{L/K}(\mathcal{I}_L^{\mathfrak{m}})\mathcal{R}_K^{\mathfrak{m}} \subseteq \mathcal{I}_K^{\mathfrak{m}}$ is replaced by the (simpler!) $N_{L/K}(L^\times) \subseteq K^\times$

One thing we don't have is that $K^\times \to \mathrm{Gal}_K(K^{\text{ab}})$ is an isomorphism. We shall need to take the profinite completion of $K^\times$. To build up to this, let us better understand the group that shows up in this reciprocity:

Definition 3.1.3: Local Norm Group

Let K be a local field. Then the *local norm group* with respect to L, or just *norm group*, is the subgroup of the form:

$$N_{L/K}(L^\times)$$

Holding theorem 3.1.2 to be true, it suffices to study norm groups! The following shows how the results translate:

Theorem 3.1.4: Galois Correspondence Via Norm Groups

The map $L \mapsto N_{L/K}(L^\times)$ define an inclusion reversing bijection between the finite abelian extensions L/K in K^{ab} and the norm groups in $K^\times$ where:

1. $N_{L/K}(L_1L_2)^\times = N(L_1^\times) \cap N(L_2^\times)$
2. $N((L_1 \cap L_2)^\times) = N(L_1^\times)N(L_2^\times)$

Furthermore, every norm group of K has finite index in $K^\times$, and every subgroup of $K^\times$ that contains a norm group is a norm group

***Proof*:**
First, if $L_1 \subseteq L_2$ are two extensions of K then by transitivity of the field norm:

$$\mathrm{N}_{L_2/K} = \mathrm{N}_{L_1/K} \circ \mathrm{N}_{L_2/L_1},$$

and therefore $\mathrm{N}(L_2^\times) \subseteq \mathrm{N}(L_1^\times)$; the map $L \mapsto \mathrm{N}(L^\times)$ thus reverses inclusions.

This immediately implies $\mathrm{N}((L_1L_2)^\times) \subseteq \mathrm{N}(L_1^\times) \cap \mathrm{N}(L_2^\times)$, since $L_1, L_2 \subseteq L_1L_2$. For the reverse inclusion, let us consider the commutative diagram

$$\begin{tikzcd}
K^\times \arrow[r, "\theta_{L_1L_2/K}"] \arrow[dr, "\theta_{L_1/K}\times\theta_{L_2/K}"'] & \mathrm{Gal}(L_1L_2/K) \arrow[d, "\mathrm{res}\times\mathrm{res}"] \\
& \mathrm{Gal}(L_1/K) \times \mathrm{Gal}(L_2/K)
\end{tikzcd}$$

By Local Artin Reciprocity (theorem 3.1.2), each $x \in \mathrm{N}(L_1^\times) \cap \mathrm{N}(L_2^\times) \subseteq K^\times$ lies in the kernel of $\theta_{L_1/K}$ and $\theta_{L_2/K}$, hence in the kernel of $\theta_{L_1L_2/K}$ (by the diagram), and therefore in $\mathrm{N}((L_1L_2)^\times)$, again by theorem 3.1.2. This proves (1).

We now show that $L \mapsto \mathrm{N}(L^\times)$ is a bijection; it is surjective by definition, so we just need to show it is injective. If $\mathrm{N}(L_2^\times) = \mathrm{N}(L_1^\times)$ then by (1) we have

$$\mathrm{N}((L_1L_2)^\times) = \mathrm{N}(L_1^\times) \cap \mathrm{N}(L_2^\times) = \mathrm{N}(L_1^\times) = \mathrm{N}(L_2^\times),$$

and theorem 3.1.2 implies $\mathrm{Gal}(L_1L_2/K) \cong \mathrm{Gal}(L_1/K) \cong \mathrm{Gal}(L_2/K)$, which forces $L_1 = L_2$; thus $L \mapsto \mathrm{N}(L^\times)$ is injective.

We now prove (2). The field $L_1 \cap L_2$ is the largest extension of K that lies in both L_1 and L_2, while $\mathrm{N}(L_1^\times)\mathrm{N}(L_2^\times)$ is the smallest subgroup of $K^\times$ containing both $\mathrm{N}(L_1^\times)$ and $\mathrm{N}(L_2^\times)$; they therefore correspond under the inclusion reversing bijection $L \mapsto \mathrm{N}(L^\times)$ and we have $\mathrm{N}((L_1 \cap L_2)^\times) = \mathrm{N}(L_1^\times)\mathrm{N}(L_2^\times)$ as desired.

The fact that every norm group has finite index in $K^\times$ follows immediately from the isomorphism $\mathrm{Gal}(L/K) \cong K^\times/\mathrm{N}_{L/K}(L^\times)$ given by theorem 3.1.2, since $\mathrm{Gal}(L/K)$ is finite.

Finally, let us prove that every subgroup of $K^\times$ that contains a norm group is a norm group. Suppose $\mathrm{N}(L^\times) \subseteq H \subseteq K^\times$, for some finite abelian L/K, and subgroup H of $K^\times$, and put $F := L^{\theta_{L/K}(H)}$. We have a commutative diagram

$$\begin{tikzcd} K^\times \arrow[r, "\theta_{L/K}"] \arrow[dr, "\theta_{F/K}"'] & \mathrm{Gal}(L/K) \arrow[d, "\mathrm{res}"] \\ & \mathrm{Gal}(F/K) \end{tikzcd}$$

in which $\mathrm{Gal}(L/F) = \theta_{L/K}(H)$ is precisely the kernel of the map $\mathrm{Gal}(L/K) \to \mathrm{Gal}(F/K)$ induced by restriction. It follows from theorem 3.1.2 that

$$H = \ker \theta_{F/K} = \mathrm{N}(F^\times)$$

is a norm group, as we sought to show.

Lemma 3.1.5: Norm Groups Finite Index Subgroup of Units

Let L/K be any extension of local fields. Then if $N(L^\times)$ has finite index in $K^\times$, it is open

***Proof*:**
Proof. The lemma is clear if K is Archimedean (either $L = K$ and $\mathrm{N}(L^\times) = K^\times$, or $L \cong \mathbb{C}$, $K \cong \mathbb{R}$, and $[K^\times : \mathrm{N}(L^\times)] = [\mathbb{R}^\times : \mathbb{R}_{>0}] = 2$), so assume K is nonarchimedean. Suppose $[K^\times : \mathrm{N}(L^\times)] < \infty$. The unit group $\mathcal{O}_L^\times$ is compact, so $\mathrm{N}(\mathcal{O}_L^\times)$ is compact (since $\mathrm{N} : L^\times \to K^\times$ is continuous), thus closed in the Hausdorff space $K^\times$. For any $\alpha \in L$,

$$\alpha \in \mathcal{O}_L^\times \iff |\alpha| = 1 \iff |\mathrm{N}_{L/K}(\alpha)| = 1 \iff \mathrm{N}_{L/K}(\alpha) \in \mathcal{O}_K^\times,$$

and therefore

$$\mathrm{N}(\mathcal{O}_L^\times) = \mathrm{N}(L^\times) \cap \mathcal{O}_K^\times.$$

It follows that $\mathrm{N}(\mathcal{O}_L^\times)$ is the kernel of the homomorphism $\mathcal{O}_K^\times \hookrightarrow K^\times \to K^\times/\mathrm{N}(L^\times)$ and therefore $[\mathcal{O}_K^\times : \mathrm{N}(\mathcal{O}_L^\times)] \le [K^\times : \mathrm{N}(L^\times)] < \infty$. Thus $\mathrm{N}(\mathcal{O}_L^\times)$ is a closed subgroup of finite index in $\mathcal{O}_K^\times$, hence open (its complement is a finite union of closed cosets, hence closed), and $\mathcal{O}_K^\times$ is open[1] in $K^\times$, so $\mathrm{N}(\mathcal{O}_L^\times)$ is open in $K^\times$, and therefore $\mathrm{N}(L^\times)$ is open in $K^\times$, since $\mathrm{N}(L^\times)$ is a union of cosets of the open subgroup $\mathrm{N}(\mathcal{O}_L^\times)$.

The existence part of the Local Artin Reciprocity shows the converse holds. For reference purposes, we shall box this result:

Theorem 3.1.6: Local Artin Reciprocity Norm Group

K is a local field and H is a finite index open subgroup of $K^\times$, then there is a unique extension L/K in K^{ab} where $N_{L/K}(L^\times) = H$.

Now, let us focus back on $\theta_K : K^\times \to \mathrm{Gal}_K(K^{\mathrm{ab}})$. This cannot be an isomorphism as $\mathrm{Gal}_K(K^{\mathrm{ab}})$ is compact, while $K^\times$ is not. However, the local existence portion of Local Artin Reciprocity that after taking the profinite completion, the local Artin homomorphism becomes an isomorphism

Theorem 3.1.7: Main Theorem of Local Class Field Theory

Let K be a lcoal field. Then the local Artin homomorphism induces a natural isomorphism:

$$\widehat{\theta}_K : \widehat{K^\times} \xrightarrow{\cong} \mathrm{Gal}_K(K^{\mathrm{ab}})$$

of profinite groups

***Proof*:**
By infinite galois theory, the Galois group $\mathrm{Gal}(K^{\mathrm{ab}}/K)$ is a profinite group, isomorphic to the inverse limit

$$\mathrm{Gal}(K^{\mathrm{ab}}/K) \cong \varprojlim_L \mathrm{Gal}(L/K), \tag{3.1}$$

where L ranges over the finite extensions of K in K^{ab} ordered by inclusion.

It follows from lemma 3.1.5, the comment under it, and the definition of the profinite completion, that

$$\widehat{K^\times} \cong \varprojlim_L K^\times/\mathrm{N}(L^\times), \tag{3.2}$$

where L ranges over finite abelian extensions of K (in K^{sep}). By local Artin reciprocity, for each finite abelian extension L/K we have an isomorphism

$$\theta_{L/K} : K^\times/\mathrm{N}(L^\times) \xrightarrow{\sim} \mathrm{Gal}(L/K),$$

and these isomorphisms commute with inclusion maps between finite abelian extensions of K. We thus have an isomorphism of the inverse systems appearing in (3.1) and (3.2). The isomorphism is canonical because the Artin homomorphism θ_K is unique and the isomorphisms in (3.1) and (3.2) are both canonical, completing the proof.

Proposition 3.1.8: Local Artin Homomorphism is Unique

Let K be a local field and assume every finite index open subgroup of $K^\times$ is a norm group. Then there is at most one homomorphism $\theta : K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$ of topological groups that has the properties given in Local Artin Reciprocity Law.

***Proof*:**
The proposition is clear when K is Archimedean, so assume it is nonarchimedean. Let $\mathfrak{p} = (\pi)$ be the maximal ideal of $\mathcal{O}_K$, and for each integer $n \geq 0$ let $K_{\pi,n}/K$ be the finite abelian extension given by theorem 3.1.6 corresponding to the finite index subgroup $(1+\mathfrak{p}^n)\langle\pi\rangle$ of $K^\times$; here $1+\mathfrak{p}^n$ and $\langle\pi\rangle$ denote subgroups of $K^\times$, with $1+\mathfrak{p}^0 := \mathcal{O}_K^\times$, and we note that $K^\times \cong \mathcal{O}_K^\times\langle\pi\rangle$.

Suppose $\theta : K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$ is a continuous homomorphism as given by Local Artin Reciprocity. Then $\theta(\pi)$ fixes $K_\pi := \bigcup_n K_{\pi,n}$, since $\pi \in \mathrm{N}(K_{\pi,n}) = \ker\theta_{K_{\pi,n}/K}$. We also know that $\theta_{L/K}(\pi) = \mathrm{Frob}_{L/K}$ for all finite unramified extensions L/K, which uniquely determines the action of $\theta(\pi)$ on K^{nr}, and hence on $K^{\mathrm{ab}} = K_\pi K^{nr}$.

Now suppose $\theta' : K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$ is another continuous homomorphism as given by Local Artin Reciprocity. By the argument above we must have $\theta'(\pi) = \theta(\pi)$ for every uniformizer π of $\mathcal{O}_K$, and $K^\times$ is generated by its subset of uniformizers: if we fix one uniformizer π, every $x \in K^\times$ can be written as $u\pi^n = (u\pi)\pi^{n-1}$ for some $u \in \mathcal{O}_K^\times$ and $n \in \mathbb{Z}$, and $u\pi$ is another uniformizer). It follows that $\theta(x) = \theta'(x)$ for all $x \in K^\times$ and therefore $\theta = \theta'$ is unique, as we sought to show.

Hence, it is worth-while understanding the group $\widehat{K}$. If K is Archimedean $\widehat{K^\times}$ is trivial or cyclic of order 2. If K is nonarchimedean, let us first pick a uniformizer π for the maximal ideal $\mathfrak{p}$ of $\mathcal{O}_K$. Then each $x \in K^\times$ can be uniquely represented as $u\pi^{\nu(x)}$ where $u \in \mathcal{O}_K^\times$ and $\nu(x) \in \mathbb{Z}$. Then we get:

$$K^\times \cong \mathcal{O}_K^\times \times \mathbb{Z} \qquad x \mapsto \left(\frac{x}{\pi^{\nu(x)}}, \nu(x)\right)$$

Taking now the profinite completion we get:

$$\widehat{K^\times} \cong \mathcal{O}_K^\times \times \widehat{\mathbb{Z}}$$

recalling that $\mathcal{O}_K$ is already a profinite completion (see *Local Field Theory* [5]):

$$\mathcal{O}_K^\times \cong \mathbb{F}_\mathfrak{p}^\times \times (1+\mathfrak{p}) \cong \mathbb{F}_\mathfrak{p}^\times \times \varprojlim_n \mathcal{O}_K(1+\mathfrak{p}^n)$$

This isomorphism is dependent on the choice of π (of which are uncountably many), and hence is not a canonical isomorphism. Overall, we now get the commutative diagram of exact sequences:

$$\begin{tikzcd}
1 \arrow[r] & \mathcal{O}_K^\times \arrow[r] \arrow[d, "\cong"] & K^\times \arrow[r, "\nu"] \arrow[d, "\theta_K"] & \mathbb{Z} \arrow[r] \arrow[d, hook, "\varphi"] & 0 \\
1 \arrow[r] & \mathrm{Gal}_{K^{nr}}(K^{\mathrm{ab}}) \arrow[r] & \mathrm{Gal}_K(K^{\mathrm{ab}}) \arrow[r, "\mathrm{res}"] & \mathrm{Gal}_K(K^{nr}) \arrow[r] & 1
\end{tikzcd}$$

where the bottom row is the profinite completion of the top row! The right-most map is given by:

$$\mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}} \cong \mathrm{Gal}_{\mathbb{F}_\mathfrak{p}}(\overline{\mathbb{F}}_\mathfrak{p}) \cong \mathrm{Gal}_K(K^{nr})$$

sending 1 to the sequence of Frobenius elements. The Frobenius $\varphi(1)$ is a topological generator, making a dense subset. Note that $\varphi(-1)$ is also a generator; sometimes $\varphi(1)$ is called the arithmetic generator while $\varphi(-1)$ is called the geometric generators for reason beyond this book.

The group $\mathrm{Gal}_{K^{nr}}(K^{\mathrm{ab}}) \cong \mathcal{O}_K^\times$ corresponds to the inertia subgroup of $\mathrm{Gal}_K(K^{\mathrm{ab}})$. As the top sequence splits, so does the bottom so that for each choice of uniformizer:

$$\mathrm{Gal}_K(K^{\mathrm{ab}}) \cong \mathrm{Gal}_{K^{nr}}(K^{\mathrm{ab}}) \times \mathrm{Gal}_K(K^{nr}) \cong \mathcal{O}_K^\times \times \widehat{\mathbb{Z}}$$

Furthermore, for each choice of uniformizer $\pi \in \mathcal{O}_K$, we get

$$K^{\mathrm{ab}} = K_\pi K^{nr}$$

which corresponds to $K^\times = \mathcal{O}_K^\times \pi^{\mathbb{Z}}$, where the field $K_\pi \subseteq K^{\mathrm{ab}}$ is the subfield fixed by $\theta_K(\pi) \in \mathrm{Gal}_K(K^{\mathrm{ab}})$. Equivalently, K_π si the compositum of all totally ramified finite extensions L/K in K^{ab} where $\pi \in N(L^\times)$.

The simplest such decomposition is given by taking $K = \mathbb{Q}_p$ and $\pi = p$ in which case $K^{\mathrm{ab}} = K_\pi K^{nr}$ is

$$\mathbb{Q}_p^{\mathrm{ab}} = \bigcup_n \mathbb{Q}_p(\zeta_{p^n}) \cdot \bigcup_{m \perp p} \mathbb{Q}_p(\zeta_m)$$

where the first union is fixed by $\theta_K(p)$ and the second union is fixed by $\theta_K(\mathcal{O}_K^\times)$.

Summary

Overall we get a bijection between these three sets:

1. finite-index open subgroups of $K^\times$
2. open subgroups of $\mathrm{Gal}_K(K^{\mathrm{ab}})$
3. finite extensions of K in K^{ab}

which fully classifies abelian extension using *only* information about $K^\times$!

3.2 Ring of Adeles and Strong Approximation

Recall that local fields are locally compact (see *Local Field Theory* [5]). Many of the results we have shown use the property of locally compact fields. We require local compactness "in the limit"; that is when considering all local fields. We thus take the following particular topology express invented for solving this problem:

Definition 3.2.1: Restricted Product

Let $(X_i)_{i\in I}$ be a family of topological spaces indexed by I, and let $U_i \subseteq X_i$ be a family of open subsets. Then the *restricted product* is the set:

$$\prod_{i\in I}{}' (X_i, U_i) = \{(x_i)_{i\in I} \mid x_i \in U_i \text{ for a.e. } i \in I\} \subseteq \prod X_i$$

with topology given by the basis:

$$\mathcal{B} = \left\{ \prod_{i\in I} V_i \;\middle|\; V_i \subseteq X_i \text{ is open for all } i \in I \text{ and } V_i = U_i \text{ for a.e. } i \in I \right\}$$

where almost all means all but finitely many.

Essentially, instead of having X_i at a.e. position, we shall have U_i. Note that this does not mean that that the sets X_i for a specific index i does not exist: the set

$$X_i \times \prod_{i\neq j\in I} U_j$$

is within the restricted product for each index i. This shows that in the finite product cases, the two topologies and sets are identical. Notice that each projection map is still continuous, As $\prod X_i$ is supposed to be the coarsest space with these maps being continuous, we may expect more open sets in the restricted product, and indeed we do: $\prod_i U_i$ is open in the restricted product but not the product (unless, $U_i = X_i$ for a.e. indexes). This also shows that the restricted product is not in the subspace topology, as there are no open sets of $\prod_i X_i$ that can be intersected to get $\prod_i U_i$.

Note that the topology and set of the restricted product does not depend on any particular U_i. Indeed, if $U_i = U_i'$ for a.e. indices, then:

$$\prod_{i\in I}{}' (X_i, U_i) = \prod_{i\in I}{}' (X_i, U_i')$$

Let us show that this topology is the colimit of appropriate subspaces of the product space. Given any $x \in X \prod_{i\in I}' (X_i, U_i)$ define the (possibly empty) finite set:

$$S(x) = \{i \in I \mid x_i \notin U_i\}$$

From this, define:

$$X_s = \{x \in X \mid S(x) \subseteq S\} = \prod_{i\in S} X_i \times \prod_{i\notin S} U_i$$

Notice that each X_s is an open subset and can be viewed as a subspace of X or as a direct product of appropriate X_i and U_i (the basis $\mathcal{B}$ can be restricted to a basis $\mathcal{B}_S$ that generated both topologies). Notice that if $S \subseteq T$ then $X_S \subseteq X_T$, and hence we may partially order the family of topological space X_S via the inclusion and consider the colimit:

$$\varinjlim_{S} X_S = \coprod X_S / \sim$$

with the equivalence being given by $x \sim \iota_{ST}(x)$ where $\iota_{ST} : X_S \to X_T$ is the inclusion map. Then it is a good exercise in topology to show that the following are homeomorphic:

$$X \cong_{\mathbf{Top}} \varinjlim X_S$$

Due to this characterization, we can conclude that the restricted product of locally compact spaces is locally compact when the family $(U_i)_{i\in I}$ are a.e. compact (can you see why this fixes the problem of the continuous image of compact sets is compact?)

Let us now work with topological spaces that are relevant to number theory. Let K be a global field, M_K denote the set of places of K, K_ν the corresponding local field, and $\mathcal{O}_\nu$ the valuation ring of K_ν where if ν is Archimedean then $\mathcal{O}_\nu = K_\nu$. Then

Definition 3.2.2: Adèle Ring

Let K be a global field. Then the *adèle ring* or *adele ring* is the restricted product:

$$\mathbb{A}_K = \prod_{\nu\in M_K}' (K_\nu, \mathcal{O}_\nu)$$

which is a subset (not subspace) of $\prod_\nu K_\nu$. Define addition and multiplication component-wise.

Recall that each $\mathcal{O}_\nu$ is compact when ν is non-Archimedean, and that there are only finitely many Archimedean valuations, hence $\mathbb{A}_K$ is locally compact. As K_ν is Hausdroff and the restricted topology is finer, $\mathbb{A}_K$ is also Hausdroff. For each $a \in \mathbb{A}_K$, denote by a_ν the projection into K_ν. For each finite set of place S, we can define the subring of S-adeles:

$$\mathbb{A}_{K,S} = \prod_{\nu\in S} K_\nu \times \prod_{\nu\notin S} \mathcal{O}_\nu$$

so that $\mathbb{A}_K \cong \varinjlim_S \mathbb{A}_{K,S}$, showing that $\mathbb{A}_K$ is also a topological ring. A simple example would be when $K = \mathbb{Q}$, in which case $\mathbb{A}_\mathbb{Q}$ is the union of the rings:

$$\mathbb{R} \times \prod_{p\in S} \mathbb{Q}_p \times \prod_{p\notin S} \mathbb{Z}_p$$

where S varies over all finite sets of primes. Taking the canonical embeddings $K \hookrightarrow K_\nu$, we may induce a canonical embedding:

$$K \hookrightarrow \mathbb{A}_K \qquad x \mapsto (x, x, ...)$$

which is well-defined as $x \in \mathcal{O}_\nu$ for a.e. ν. For reasons that should be illuminating to the reader to ponder, the image K in $\mathbb{A}_K$ is called the *principal adele* subring (which is also a field).

The normalized absoltue value $\| - \|_\nu$ of K_ν can naturally be extended to $\mathbb{A}_K$ for each ν via:

$$\|(a)\|_\nu = \|a_\nu\|_\nu$$

And by the product formula for absolute values [5] we may define the *adelic absolute value* (or *adelic norm*) to be:

$$\|(a)\| = \prod_{\nu\in M_K} \|a\|_\nu \in \mathbb{R}_{\geq 0}$$

Note that unless $\|a\|_\nu = 1$ for all but finitely many ν, this product will naturally converge to 0. If$\|a\| \neq 0$, this is equal to the *Arakelov divisor*. For any nonzero principal adele a, we may consider it as $a \in K^\times$, and by the product formula $\|a\| = 1$ (showing they form the group of units).

The Haar measure on $\mathbb{A}_K$ (the product of the local normalized Haar measures μ_ν, compatible via the product formula) is developed in *Local Field Theory* [5]; it is used in the strong-approximation argument below.

Next, let's consider a base change. Let us see that the embedding $K \hookrightarrow \mathbb{A}_K$ makes $\mathbb{A}_K$ into a K-vector space. If L/K is any finite separable extension of K, we can take:

$$\mathbb{A}_K \otimes_K L$$

giving an L-vector space. As a topological K-vector space, $\mathbb{A}_K \otimes_K L$ is the product of $[L : K]$ copies of $\mathbb{A}_K$. This product does indeed act how we may expect it to:

Lemma 3.2.3: Adele Ring Change of Basis

Let L/K be a finite separable extension of a global field K. Then there is a natural isomorphism of topological rings:

$$\mathbb{A}_L \cong \mathbb{A}_K \otimes L$$

such that the following diagram commutes:

$$\begin{tikzcd} L \arrow[r, "\cong"] \arrow[d, hook] & K \otimes_K L \arrow[d, hook] \\ \mathbb{A}_L \arrow[r, "\cong"] & \mathbb{A}_K \otimes_K L \end{tikzcd}$$

***Proof*:**
The statement is local. For each place v of K the finite separable K_v-algebra $K_v \otimes_K L$ decomposes as the product of the completions of L at the places above v,

$$K_v \otimes_K L \cong \prod_{w|v} L_w$$

and at all but finitely many v (those unramified in L) this carries $\mathcal{O}_{K_v} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ isomorphically onto $\prod_{w|v} \mathcal{O}_{L_w}$. Since L is a finite K-module, tensoring the restricted product $\mathbb{A}_K$ by L commutes with the product, and the integrality condition defining the restricted product is preserved at almost every place by the displayed isomorphism. Hence

$$\mathbb{A}_K \otimes_K L \cong \prod_v{}'(K_v \otimes_K L) \cong \prod_v{}' \prod_{w|v} L_w \cong \prod_w{}' L_w = \mathbb{A}_L$$

as topological rings, the primes denoting restricted products. The principal embeddings match up since $a \in L$ maps to $1 \otimes a$ on one side and to its diagonal image $(a)_w$ on the other, as we sought to show.

Lemma 3.2.4: Adele Ring Change of Basis As Vector Space

Let L/K be a degree n separable extension of a global field K. Then there is a natural isomorphism of topological K-vector spaces and local compact groups:

$$\mathbb{A}_L \cong \mathbb{A}_K \oplus \cdots \oplus \mathbb{A}_K$$

which restricts to an isomorphism $L \cong K \oplus \cdots \oplus K$ of the principal adeles of $\mathbb{A}_L$

***Proof*:**
immediate from lemma 3.2.3.

More importantly, we can understand the topology of L within $\mathbb{A}_L$. Recall that a group $H \leqslant G$ is *cocompact* if G/H is compact.

Theorem 3.2.5: Principal Adeles Cocompact

Let L be a global field considered as the principal adele ring $L \subseteq \mathbb{A}_L$. Then it is a discrete cocompact subgroup of $\mathbb{A}_L$ seen as an additive group.

***Proof*:**
If $K = \mathbb{Q}$ or $K = \mathbb{F}_q(t)$ (i.e. K is a rational subfield), then it follows from lemma 3.2.4, that if we show it for K, then it follows for L, so assume K is a rational subfield, and identify it with its image in $\mathbb{A}_K$. To show it's discrete, we can show 0 is an isolated point. Take:

$$U = \{a \in \mathbb{A}_K \mid \|a\|_\infty < 1,\ \|a\|_\nu \leq 1,\ \forall \nu < \infty\}$$

where ∞ denotes the unique infinite place of K. By the product formula, $\|a\| = 1$ for all $a \in K^\times \subseteq \mathbb{A}_K$, and so

$$U \cap K = \{0\}$$

showing it is discrete.

To show K is cocompact, i.e. $\mathbb{A}_K/K$ is compact, take the sets

$$U_\infty = \{x \in K_\infty \mid \|x\|_\infty \leq 1\} \qquad W = \{a \in \mathbb{A}_K \mid \|a\|_\nu \leq 1,\ \forall \nu\}$$

Then we may decompose W as:

$$W = U_\infty \times \prod_{\nu < \infty} \mathcal{O}_\nu \subseteq \mathbb{A}_{K,\{\infty\}} \subseteq \mathbb{A}_K$$

which is a product of compact sets, hence compact. If we show that W contains a set of coset representatives for K in $\mathbb{A}_K$, we're done, as then $\mathbb{A}_K/K$ is the image of the comapct set W under the continuous quotient $\mathbb{A}_K \to \mathbb{A}_K/K$.

Let $a = (a_\nu)$ be any element of $\mathbb{A}_K$. To show it has a representative in W, we need that $a = b + c$ for some $b \in W$ and $c \in K$, which we will do by constructing $c \in K$ so that $b = a - c \in W$.

For each $\nu < \infty$ define $x_\nu \in K$ as follows: put $x_\nu := 0$ if $\|a_\nu\|_\nu \leq 1$ (almost all ν), and otherwise choose $x_\nu \in K$ so that $\|a_\nu - x_\nu\|_\nu \leq 1$ and $\|x_\nu\|_w \leq 1$ for $w \neq \nu$. To show that such an x_ν

exists, let us first suppose $a_\nu = r/s \in K$ with $r, s \in \mathcal{O}_K$ coprime (note that $\mathcal{O}_K$ is a PID), and let $\mathfrak{p}$ be the maximal ideal of $\mathcal{O}_\nu$. The ideals $\mathfrak{p}^{\nu(s)}$ and $\mathfrak{p}^{-\nu(s)}(s)$ are coprime, so we can write $r = r_1 + r_2$ with $r_1 \in \mathfrak{p}^{\nu(s)}$ and $r_2 \in \mathfrak{p}^{-\nu(s)}(s) \subseteq \mathcal{O}_K$, so that $a_\nu = r_1/s + r_2/s$ with $\nu(r_1/s) \geq 0$ and $w(r_2/s) \geq 0$ for all $w \neq \nu$. If we now put $x_\nu := r_2/s$, then $\|a_\nu - x_\nu\|_\nu = \|r_1/s\|_\nu \leq 1$ and $\|x_\nu\|_w = \|r_2/s\|_w \leq 1$ for all $w \neq \nu$ as desired. We can approximate any $a'_\nu \in K_\nu$ by such an $a_\nu \in K$ with $\|a'_\nu - a_\nu\|_\nu < \epsilon$ and construct x_ν as above so that $\|a_\nu - x_\nu\|_\nu \leq 1$ and $\|a'_\nu - x_\nu\|_\nu \leq 1 + \epsilon$; but for sufficiently small ϵ this implies $\|a'_\nu - x_\nu\|_\nu \leq 1$, since the nonarchimedean absolute value $\| - \|_\nu$ is discrete.

Finally, let $x := \sum_{\nu < \infty} x_\nu \in K$ and choose $x_\infty \in \mathcal{O}_K$ so that

$$\|a_\infty - x - x_\infty\|_\infty \leq 1.$$

For $a_\infty - x \in \mathbb{Q}_\infty \cong \mathbb{R}$, we can take $x_\infty \in \mathbb{Z}$ in the real interval $[a_\infty - x - 1, a_\infty - x + 1)$. For $a_\infty - x \in \mathbb{F}_q(t)_\infty \cong \mathbb{F}_q((t^{-1}))$ we can take $x_\infty \in \mathbb{F}_q[t]$ to be the polynomial of least degree for which $a_\infty - x - x_\infty \in \mathbb{F}_q[[t^{-1}]]$.

Now let $c := \sum_{\nu \leq \infty} x_\nu \in K \subseteq \mathbb{A}_K$, and let $b := a - c$. Then $a = b + c$, with $c \in K$, and we claim that $b \in W$. For each $\nu < \infty$ we have $x_w \in \mathcal{O}_\nu$ for all $w \neq \nu$ and

$$\|b\|_\nu = \|a - c\|_\nu = \left\| a_\nu - \sum_{w \leq \infty} x_w \right\|_\nu \leq \max(\|a_\nu - x_\nu\|_\nu, \max(\{\|x_w\|_\nu : w \neq \nu\})) \leq 1,$$

by the nonarchimedean triangle inequality. For $\nu = \infty$ we have $\|b\|_\infty = \|a_\infty - c\|_\infty \leq 1$ by our choice of x_∞, and $\|b\|_\nu \leq 1$ for all ν, so $b \in W$ as claimed and the theorem follows.

Let us put this result to good use by showing the strong approximation Theorem for Adele rings. To show this result, let us generalize the Minkowski Lattice Point Theorem to bound the size a prime via a principal prime:

Theorem 3.2.6: Adelic Blichfeldt-Minkowski Lemma

Let K be a global field. Then there exists a positive constant B_K such that for any $a \in \mathbb{A}_K$ with $\|a\| > B_K$, there exists a nonzero principal adele $x \in K \subseteq \mathbb{A}_K$ such that

$$\|x\|_\nu \leq \|a\|_\nu \qquad \forall \nu \in M_K$$

***Proof*:**
Let $b_0 := \operatorname{covol}(K)$ be the measure of a fundamental region for K in $\mathbb{A}_K$ under the normalized Haar measure μ on $\mathbb{A}_K$ from [5] (as K is cocompact, b_0 is finite). Now define

$$b_1 := \mu(\{z \in \mathbb{A}_K : \|z\|_\nu \leq 1 \text{ for all } \nu \text{ and } \|z\|_\nu \leq \frac{1}{4} \text{ if } \nu \text{ is Archimedean}\}).$$

Then $b_1 \neq 0$, since K has only finitely many Archimedean places. Now let $B_K := b_0/b_1$.

Suppose $a \in \mathbb{A}_K$ satisfies $\|a\| > B_K$. We know that $\|a\|_\nu \leq 1$ for all almost all ν, so $\|a\| \neq 0$ implies that $\|a\|_\nu = 1$ for almost all ν. Let us now consider the set

$$T := \{t \in \mathbb{A}_K : \|t\|_\nu \leq \|a\|_\nu \text{ for all } \nu \text{ and } \|t\|_\nu \leq \frac{1}{4}\|a\|_\nu \text{ if } \nu \text{ is Archimedean}\}.$$

From the definition of b_1 we have

$$\mu(T) = b_1\|a\| > b_1 B_K = b_0;$$

this follows from the fact that the Haar measure on $\mathbb{A}_K$ is the product of the normalized Haar measures μ_ν on each of the K_ν. Since $\mu(T) > b_0$, the set T is not contained in any fundamental region for K, so there must be distinct $t_1, t_2 \in T$ with the same image in $\mathbb{A}_K/K$, equivalently, whose difference $x = t_1 - t_2$ is a nonzero element of $K \subseteq \mathbb{A}_K$. We have

$$\|t_1 - t_2\|_\nu \le \begin{cases} \max(\|t_1\|_\nu, \|t_2\|_\nu) \le \|a\|_\nu & \text{finite place} \\ \|t_1\|_\nu + \|t_2\|_\nu \le 2 \cdot \frac{1}{4}\|a\|_\nu \le \|a\|_\nu & \text{real place} \\ (\|t_1 - t_2\|_\nu^{1/2})^2 \le (\|t_1\|_\nu^{1/2} + \|t_2\|_\nu^{1/2})^2 \le (2 \cdot \frac{1}{2}\|a\|_\nu^{1/2})^2 \le \|a\|_\nu & \text{complex place} \end{cases}$$

Here we have used the fact that the normalized absolute value $\|-|_\nu$ satisfies the nonarchimedean triangle inequality when ν is nonarchimedean, $\|-\|_\nu$ satisfies the Archimedean triangle inequality when ν is real, and $\|-\|_\nu^{1/2}$ satisfies the Archimedean triangle inequality when ν is complex. Thus $\|x\|_\nu = \|t_1 - t_2\|_\nu \le \|a\|_\nu$ for all places $\nu \in M_K$ as desired.

Theorem 3.2.7: Strong Approximations For Places

Let $M_K = S \sqcup T \sqcup \{w\}$ be a partition of the places of a global field K, where S is finite. Then fixing a $a_\nu \in K$ and $\epsilon_\nu \in \mathbb{R}_{>0}$ for each S, there exists a $x \in K$ such that

$$\|x - a_\nu\|_\nu \le \epsilon_\nu, \forall \nu \in S$$
$$\|x_\nu\|_\nu \le 1, \forall \nu \in T$$

with no constraint on $\|x\|_w$

***Proof*:**
Let $W = \{z \in \mathbb{A}_K : \|z\|_v \le 1 \text{ for all } v \in M_K\}$ as in the proof of theorem 3.2.5. Then W contains a complete set of coset representatives for $K \subseteq \mathbb{A}_K$, so $\mathbb{A}_K = K + W$. For any nonzero $u \in K \subseteq \mathbb{A}_K$ we also have $\mathbb{A}_K = K + uW$: given $c \in \mathbb{A}_K$ write $u^{-1}c \in \mathbb{A}_K$ as $u^{-1}c = a + b$ with $a \in K$ and $b \in W$ and then $c = ua + ub$ with $ua \in K$ and $ub \in uW$. Now choose $z \in \mathbb{A}_K$ such that

$$0 < \|z\|_v \le \epsilon_v \text{ for } v \in S, \quad 0 < \|z\|_v \le 1 \text{ for } v \in T, \quad \|z\|_w > B \prod_{v \ne w} \|z\|_v^{-1},$$

where B is the constant in the Blichfeldt-Minkowski Lemma (this is clearly possible: every $z = (z_v)$ with $\|z_v\|_v \le 1$ is an element of $\mathbb{A}_K$). We have $\|z\| > B$, so there is a nonzero $u \in K \subseteq \mathbb{A}_K$ with $\|u\|_v \le \|z\|_v$ for all $v \in M_K$.

Now let $a = (a_v) \in \mathbb{A}_K$ be the adele with a_v given by the hypothesis of the theorem for $v \in S$ and $a_v = 0$ for $v \notin S$. We have $\mathbb{A}_K = K + uW$, so $a = x + y$ for some $x \in K$ and $y \in uW$. Therefore

$$\|x - a_v\|_v = \|y\|_v \le \|u\|_v \le \|z\|_v \le \begin{cases} \epsilon_v & \text{for } v \in S, \\ 1 & \text{for } v \in T, \end{cases}$$

as we sought to show.

As a direct consequence

Corollary 3.2.8: Density of Principal Adeles

Let K be a global field and let w be any place of K. Then K is dense in the restricted product $\prod'_{\nu\neq w}(K_\nu, \mathcal{O}_\nu)$.

***Proof*:**
immediate consequence.

It was remarked that these results can be extended for algebraic groups (algebraic variety with compatible group structure), showing there is something deeper going on in this approximation.

3.2.1 Units of Adèle Ring

Let us now try to find:

$$\mathbb{A}_K^\times = \left\{(a) \in \mathbb{A}_K \;\middle|\; a_\nu \in K_\nu^\times,\ \forall \nu \in M_k \text{ and } a_\nu \in \mathcal{O}_\nu^\times \text{ for a.e. } \nu \in M_K\right\}$$

where if ν is Archimedean then $\mathcal{O}_K^\times$ is either $\mathbb{R}^\times$ or $\mathbb{C}^\times$. Unfortunately, $\mathbb{A}_K^\times$ is not a topological subgroup of $\mathbb{A}_K$, as the inversion map $a \mapsto a^{-1}$ is not continuous. To see this, take the case where $K = \mathbb{Q}$. Then for each prime p, consider $a(p) = (1, ..., 1, p, 1, ...) \in \mathbb{A}_\mathbb{Q}$ so that $a(p)_p = p$ and $a(p)_q = 1$ for $q \neq p$. Ten every basic open set $U \subseteq \mathbb{A}_\mathbb{Q}$ around 1 must have the form (for some finite $S \subseteq M_\mathbb{Q}$):

$$\prod_{\nu\in S} U_\nu \times \prod_{\nu\notin S} \mathcal{O}_\nu$$

where $1_\nu \in U_\nu$. Then certainly U contains $a(p)$ for sufficiently large p. Thus, we get that in $\mathbb{A}_\mathbb{Q}$ the following limit is:

$$\lim_{p\to\infty} a(p) = 1$$

However, U does not contain $a(p)^{-1}$ for any sufficiently large p, hence we also have:

$$\lim_{p\to\infty} a(p)^{-1} \neq 1^{-1} = 1$$

Thus $a \mapsto a^{-1}$ is *not* a continuous map in the subspace $\mathbb{A}_K^\times$. The problem is more general than our number-theoretic context, as a topological ring has no continuous inverse condition, there is no good reason that $R^\times \subseteq R$ be a topological group in the subspace topology (unless, R is a subring of a topological field which explains why this problem did not occur for $\mathcal{O}_K^\times$ and why this cannot happen for the ring of adeles as it is not an integral domain).

To resolve this, we impose the weakest topology that will make $R^\times$ a topological group, namely, we take the embedding:

$$\varphi : R^\times \to R \times R \qquad r \mapsto (r, r^{-1})$$

and put the weakest topology on $R^\times$ that makes this map $\varphi : R^\times \to \varphi(R^\times)$ a homeomorphism. This is the subspace topology of $\varphi(R^\times) \subseteq R \times R$. The map $r \mapsto r^{-1}$ is naturally continuous as it is the composition of φ with the projection onto the second component.

With this in mind, let us return to $\mathbb{A}_K^\times$. Then we have $\varphi : \mathbb{A}_K^\times \to \mathbb{A}_K \times \mathbb{A}_K$, and each $\varphi(a) = (a, a^{-1})$ lies in a product $U \times V$ of basic open sets $U, V \subseteq \mathbb{A}_K$, which forces both a, a^{-1} to lie in $\mathcal{O}_\nu$, and hence in $\mathcal{O}_\nu^\times$ for almost all ν. From this, we see that the induced topology on $\mathbb{A}_K^\times$ is the restricted product with open sets that we would have hopped to have:

Definition 3.2.9: Idele Group

Let K be a global field. Then the *idele group* of K is the topological group:

$$\mathbb{I}_K = \prod_\nu{}' (K_\nu^\times, \mathcal{O}_\nu^\times)$$

with multiplication defined pointwise. Then $\mathbb{I}_K$ is the set $\mathbb{A}_K^\times$ with the restricted product topology instead of the subspace topology. The canonical embedding $K \hookrightarrow \mathbb{A}_K$ restricts to a canonical embedding $K^\times \hookrightarrow \mathbb{I}_K$. The *idele class group* is the topological group:

$$C_K = \frac{\mathbb{I}_K}{K^\times}$$

From here on out, we may use $\mathbb{A}_K^\times$ and $\mathbb{I}_K$ interchangeably, understanding that $\mathbb{A}_K^\times$ has the restricted product topology instead of the subspace topology. Furthermore, $\mathbb{I}_K$ is a locally compact group (it is certainly Hausdroff as it is finer than $\mathbb{A}_K^\times$ with the subspace topology which is Hausdroff, and each $\mathcal{O}_K^\times$ is compact with $K_\nu^\times$ being locally compact). Next, $K^\times$ is a discrete subgroup of $\mathbb{I}_K$: as K is a discrete subset of $\mathbb{A}_K$ (by theorem 3.2.5, $K \times K$ is a discrete subset of $\mathbb{A}_K \times \mathbb{A}_K$, and as $K^\times$ lies in the image of $\mathbb{A}_K^\times \hookrightarrow \mathbb{A}_K \times \mathbb{A}_K$ and $K \hookrightarrow \mathbb{A}_K \times \mathbb{A}_K$, $K^\times$ is discrete in $\mathbb{A}_K^\times$, and hence in $\mathbb{I}_K$ (as the topology is finer). We shall return to cocompactness shortly.

Let us see how this new topology fixes the counter-example presented earlier for the lack of continuity. Take again the sequence $(a(p))_p$. Then this sequence no longer converges to 1 in $\mathbb{I}_\mathbb{Q}$. Notice that for the basic open neighborhood of 1:

$$\prod_\nu \mathcal{O}_\nu^\times = \prod_p \mathbb{Z}_p^\times \times \mathbb{R}^\times \subseteq \mathbb{I}_\mathbb{Q}$$

none of the $a(p)$ lie in this set, and hence this sequence cannot converge to 1. In fact, this implies it cannot converge at all for if it converges to some $1 \neq x \in \mathbb{I}_\mathbb{Q}$, then it would converge to $1 \neq x \in \mathbb{A}_\mathbb{Q}^\times \subseteq \mathbb{A}_\mathbb{Q}$ which we know is not the case. The high-level idea is that we added more open sets to this topology, increasing the requirement for sequences to converge (and at the same time, make it easier for the inverse map to be continuous).

With this, we may now associate the elements of $\mathbb{I}_K$ to the ideals in $\mathcal{I}_K$. Consider the surjective homomorphism:

$$\mathbb{I}_K \to \mathcal{I}_K \qquad a \mapsto \prod \mathfrak{p}^{\nu_\mathfrak{p}(a)}$$

Then the image of the composition

$$K^\times \hookrightarrow \mathbb{I}_K \twoheadrightarrow \mathcal{I}_K$$

is $\mathcal{P}_K$, and so we have a surjective homomorphism from $\mathbb{I}_K$ to $C_K = \mathbb{I}_K/K^\times$ onto $\mathrm{Cl}_K = \mathcal{I}_K/\mathcal{P}_K$

which makes the following diagram of exact sequences commute:

$$\begin{tikzcd} 1 \arrow[r] & K^\times \arrow[r] \arrow[d, two heads] & \mathbb{I}_K \arrow[r] \arrow[d, two heads] & C_K \arrow[r] \arrow[d, two heads] & 1 \\ 1 \arrow[r] & \mathcal{P}_K \arrow[r] & \mathcal{I}_K \arrow[r] & \operatorname{Cl}_K \arrow[r] & 1 \end{tikzcd}$$

When working with K and $\mathbb{A}_K$, it was also shown that K is cocompact, however $K^\times$ is not cocompact, and so the idele class group C_K is locally compact but not compact. This can be seen more if we recall that the group of units $\mathcal{O}_K^\times$ is not a cocompact subgroup of $K_\mathbb{R}^\times$ as $\log(\mathcal{O}_K^\times)$ is not a (full) sub-lattice in $\mathbb{R}^{r+s} \cong \log(K_\mathbb{R}^\times)$ (recall it lies in the trace-zero hyperplane, see [1]). We must thus get the corresponding trace zero hyperplane for $\mathbb{I}_K$. Take the continuous homomorphism between topological groups:

$$\| - \| : \mathbb{I}_K \to \mathbb{R}_{>0}^\times \qquad a \mapsto \|a\| = \prod_\nu \|a\|_\nu$$

where $\| - \|$ is the adelic norm. Then $\|a\| > 0$ for $a \in \mathbb{I}_K$ as $a_\nu \in \mathcal{O}_\nu^\times$ for a.e. ν, which implies $\|a\|_\nu = 1$ for a.e. ν, making the product effectively a finite product, and it is nonzero as $a_\nu \in K_\nu^\times$ is nonzero for all $\nu \in M_K$. This motivates the following natural topological group:

Definition 3.2.10: 1-Ideles Group

Let K be a global field. Then the group if *1-ideles* is the topological group

$$\mathbb{I}_K^1 = \ker \| - \| = \{a \in \mathbb{I}_K \mid \|a\| = 1\}$$

An important result that we shall need is that the topology of $\mathbb{I}_K^1$ is the same as the subspace topology induced by $\mathbb{A}_K$

Lemma 3.2.11: Topology of 1-Idele Group

The group of 1-ideles $\mathbb{I}_K^1$ is a closed subset of $\mathbb{A}_K$ and $\mathbb{I}_K$, and the two induced subspace topologies on $\mathbb{I}_K^1$ coincide.

Proof:
We first show that $\mathbb{I}_K^1$ is closed in $\mathbb{A}_K$, and therefore also in $\mathbb{I}_K$, since it has a finer topology. Consider any $x \in \mathbb{A}_K - \mathbb{I}_K^1$. We will construct an open neighborhood U_x of x that is disjoint from $\mathbb{I}_K^1$. The union of the U_x is then the open complement of the closed set $\mathbb{I}_K^1$. For each $\epsilon > 0$, finite $S \subseteq M_K$, and $x \in \mathbb{A}_K$ we define

$$U_\epsilon(x, S) := \{u \in \mathbb{A}_K : \|u - x\|_v < \epsilon \text{ for } v \in S \text{ and } \|u\|_v \le 1 \text{ for } v \notin S\},$$

which is a basic open set of $\mathbb{A}_K$ (a product of open sets U_v for $v \in S$ and $\mathcal{O}_v$ for $v \notin S$).

The case $\|x\| < 1$. Let S be a finite set containing the Archimedean places $v \in M_K$ and all v for which $\|x\|_v > 1$, such that $\prod_{v \in S} \|x\|_v < 1$: such an S exists since $\|x\| < 1$ and $\|x\|_v \le 1$ for almost all v. For all sufficiently small $\epsilon > 0$ the set $U_x := U_\epsilon(x, S)$ is an open neighborhood of x disjoint from $\mathbb{I}_K^1$ because every $y \in U_x$ must satisfy $\|y\| < 1$.

The case $\|x\| > 1$. Let B be twice the product of all the $\|x\|_v$ greater than 1. Let S be the finite set containing the Archimedean places $v \in M_K$, all nonarchimedean v with residue field cardinality less than $2B$, and all v for which $\|x\|_v > 1$. For all sufficiently small $\epsilon > 0$ the set $U_x := U_\epsilon(x, S)$ is an open neighborhood of x disjoint from $\mathbb{I}_K^1$ because for every $y \in U_x$, either $\|y\|_v = 1$ for all $v \notin S$, in which case $\|y\| > 1$, or $\|y\|_v < 1$ for some $v \notin S$, in which case $\|y\|_v < 1/(2B)$ and $\|y\| < 1$.

This proves that $\mathbb{I}_K^1$ is closed in $\mathbb{A}_K$, and therefore also in $\mathbb{I}_K$. To prove that the subspace topologies coincide, it suffices to show that for every $x \in \mathbb{I}_K^1$ and open $U \subseteq \mathbb{I}_K$ containing x there exists open sets $V \subseteq \mathbb{I}_K$ and $W \subseteq \mathbb{A}_K$ such that $x \in V \subseteq U$ and $V \cap \mathbb{I}_K^1 = W \cap \mathbb{I}_K^1$; this implies that every neighborhood basis in the subspace topology of $\mathbb{I}_K^1 \subseteq \mathbb{I}_K$ is a neighborhood basis in the subspace topology of $\mathbb{I}_K^1 \subseteq \mathbb{A}_K$ (the latter is a priori coarser than the former).

So consider any $x \in \mathbb{I}_K^1$ and open neighborhood $U \subseteq \mathbb{I}_K$ of x. Then U contains a basic open set

$$V = \{u \in \mathbb{A}_K : \|u - x\|_v < \epsilon \text{ for } v \in S \text{ and } \|u\|_v \leq 1 \text{ for } v \notin S\},$$

for some $\epsilon > 0$ and finite $S \subseteq M_K$ (take $S = \{v \in M_K : \|x\| \neq 1 \text{ or } \pi_v(U) \neq \mathcal{O}_v\}$ and $\epsilon > 0$ small enough). If we now put $W := U_\epsilon(x, S)$ then $x \in V \subseteq U$ and $V \cap \mathbb{I}_K^1 = W \cap \mathbb{I}_K^1$ as desired.

Theorem 3.2.12: Fujisaki's Lemma

Let K be any global field. Then the principal ideles $K^\times \subseteq \mathbb{I}_K$ are a discrete cocompact subgroup of the group of 1-ideles $\mathbb{I}_K^1$.

***Proof*:**
As $K^\times$ is discrete in $\mathbb{I}_K$, it is discrete in the subspace $\mathbb{I}_K^1$. As in the proof of theorem 3.2.5, to prove that $K^\times$ is cocompact in $\mathbb{I}_K^1$ it suffices to exhibit a compact set $W \subseteq \mathbb{A}_K$ for which $W \cap \mathbb{I}_K^1$ surjects onto $\mathbb{I}_K^1/K^\times$ under the quotient map (here we are using Lemma 26.8: $\mathbb{I}_K^1$ is closed so $W \cap \mathbb{I}_K^1$ is compact).

To construct W we first choose $a \in \mathbb{A}_K$ such that $\|a\| > B_K$, where B_K is the Blichfeldt-Minkowski constant in theorem 3.2.6, and let

$$W := L(a) = \{x \in \mathbb{A}_K : \|x\|_v \leq \|a\|_v \text{ for all } v \in M_K\}.$$

Now consider any $u \in \mathbb{I}_K^1$. We have $\|u\| = 1$, so $\|\frac{a}{u}\| = \|a\| > B_K$, and by theorem 3.2.6 there is a $z \in K^\times$ for which $\|z\|_v \leq \|\frac{a}{u}\|_v$ for all $v \in M_K$. Therefore $zu \in W$. Thus every $u \in \mathbb{I}_K^1$ can be written as $u = z^{-1} \cdot zu$ with $z^{-1} \in K^\times$ and $zu \in W \cap \mathbb{I}_K^1$. Thus $W \cap \mathbb{I}_K^1$ surjects onto $\mathbb{I}_K^1/K^\times$ under the quotient map $\mathbb{I}_K^1 \to \mathbb{I}_K^1/K^\times$, which is continuous, and it follows that $\mathbb{I}_K^1/K^\times$ is compact.

Fujisaki's lemma is a generalization of many previous results, and can be used to prove many of the finiteness or bounding results seen in [1], including the proof of Dirichlet's Unit Theorem.

Definition 3.2.13: Norm-1 Idele Class Group

Let K be a global field. Then the [compact] group

$$C_K^1 = \mathbb{I}_K^1/K^\times$$

is the *norm-1 idele class group.*

Let us now generalize the idele norm to map to idele groups:

Definition 3.2.14: Idele Norm

Let L/K be a finite separable extension of global field.s Then the *idele norm* $N_{L/K} : \mathbb{I}_L \to \mathbb{I}_K$ is given by $N_{L/K}(b_w) = (a_\nu)$ for each

$$a_\nu = \prod_{w|\nu} N_{L_w/K_\nu}(b_w)$$

with $N_{L_w/K_\nu} : L_w \to K_\nu$ is the field norm.

By the above exact sequence, the Idele norm extends the field norm on the subgroup of principal ideles $L^\times \subseteq \mathbb{I}_L$, and we have the commutative diagram relating the three norms:

$$\begin{tikzcd}
L^\times \arrow[r] \arrow[d, "\mathrm{N}_{L/K}"] & \mathbb{I}_L \arrow[r] \arrow[d, "\mathrm{N}_{L/K}"] & \mathcal{I}_L \arrow[d, "\mathrm{N}_{L/K}"] \\
K^\times \arrow[r] & \mathbb{I}_K \arrow[r] & \mathcal{I}_K
\end{tikzcd}$$

which gives:

$$\begin{tikzcd}
C_L \arrow[r, two heads] \arrow[d, "\mathrm{N}_{L/K}"] & \mathrm{Cl}_L \arrow[d, "\mathrm{N}_{L/K}"] \\
C_K \arrow[r, two heads] & \mathrm{Cl}_K
\end{tikzcd}$$

3.3 Artin Homomorphism

Let us now combine the local Artin homomorphisms to define a global Artin homomorphism. To setup, let K be a global field, choose a separable closure K^{sep}, for each ν of K choose a separable closure K_ν^{sep}, and let $K^{\mathrm{ab}}, K_\nu^{\mathrm{ab}}$ denote the maximal abelian extensions within these separable closures. All abelian extension for the remainder of this chapter shall be within these maximal abelian extensions. Then recall that for each K_ν we have:

$$\theta_{K_\nu} : K_\nu^\times \to \mathrm{Gal}_{K_\nu}(K_\nu^{\mathrm{ab}})$$

where for each finite abelian extension L/K and each place $w|\nu$ of L, composing θ_{K_ν} with the natural map $\mathrm{Gal}_{K_\nu}(K^{\mathrm{ab}}) \twoheadrightarrow \mathrm{Gal}_{K_\nu}(L_w)$ gives the surjective homomorphism:

$$\theta_{L_w/K_\nu} : K_\nu^\times \to \mathrm{Gal}_{K_\nu}(L_w)$$

which has kernel $N_{L_w/K_\nu}(L_w^\times)$. In the case where K_ν is nonarchimedean and L_w/K_ν is unramified, for all uniformizers π_ν of K_ν:

$$\theta_{L_w/K_\nu}(\pi_\nu) = \mathrm{Frob}_{L_w/K_\nu}$$

Next, define:

$$\begin{aligned}\varphi_w : \mathrm{Gal}_{K_\nu}(L_w) &\to \mathrm{Gal}_K(L)\\ \sigma &\mapsto \sigma|_L\end{aligned}$$

which is certainly well-defined and injective (recall each $y \in L_w$ can be written as lx, $l \in L$ and $x \in K_\nu$). In particular, if ν is Archimedean, $\varphi_w(\mathrm{Gal}_{K_\nu}(L_w))$ is trivial or generated by the involution given by complex conjugation in $L_w \cong \mathbb{C}$, and if ν is finite and $\mathfrak{q}$ is the prime of L corresponding go $w|\nu$, then $\varphi_w(\mathrm{Gal}_{K_\nu}(L_w))$ is the familiar decomposition group $D_\mathfrak{q} \subseteq \mathrm{Gal}_K(L)$. The image is the stabilizer of w under the action of $\mathrm{Gal}_K(L)$ on $\{w|\nu\}$. The composition $\varphi_w \circ \theta_{L_w/K_\nu} : K_\nu^\times \to \mathrm{Gal}_K(L)$ gives a map independence of choice of $w|\nu$ (namely since $\varphi_w(\theta_{L_w/K_\nu}(\pi_\nu) = \mathrm{Frob}_\nu$.

Let us now bring in $\mathbb{I}_K$. For each place ν of K, we can embed $K_\nu^\times$ into the idele group $\mathbb{I}_K$ via:

$$\begin{aligned}\iota_\nu : K_\nu^\times &\hookrightarrow \mathbb{I}_K\\ \alpha &\mapsto (1, 1, ..., 1, \alpha, 1, ...)\end{aligned}$$

Note that the image intersects $K^\times \subseteq \mathbb{I}_K$ trivially. The embedding is compatible with the idele norm in that if L/K is any finite separable extension and w is a place of L that extends the place ν of K, then the diagram commutes

$$\begin{tikzcd} L_w^\times \arrow[r, "N_{L_w/K_\nu}"] \arrow[d, hook, "\iota_w"] & K_\nu^\times \arrow[d, hook, "\iota_\nu"] \\ \mathbb{I}_L \arrow[r, "N_{L/K}"] & \mathbb{I}_K \end{tikzcd}$$

Finally, we define a system of compatible homomorphisms to take the limit of. For each finite abelian extension L/K, for each place ν of K, we can pick a place w of L st $w|\nu$ and define

$$\begin{aligned}\varphi_{L/K} : \mathbb{I}_K &\to \mathrm{Gal}_K(L)\\ (a_\nu) &\mapsto \prod_\nu \varphi_w(\theta_{L_w/K_\nu}(a_\nu))\end{aligned}$$

This product is well-defined as $a_\nu \in \mathcal{O}_\nu^\times$ and ν is unramified in L for a.e. ν, which implies:

$$\varphi_w(\theta_{L_w/K_\nu}(a_\nu)) = \mathrm{Frob}_\nu^{\nu(a_\nu)} = 1 \qquad \text{a.e.}$$

It is furthermore a continuous homomorphism, with kernel being the union of open sets of the form

$$U_a = U_S \times \prod_{\nu \notin S} \mathcal{O}_\nu^\times \subseteq \ker \theta_{L/K}$$

where $a = (a_\nu) \in \ker \theta_{L/K}$, S contains all ramified ν and all ν such that $a_\nu \notin \mathcal{O}_\nu^\times$, and U_S is the kernel of $(a_\nu)_{\nu \in S} \mapsto \prod_{\nu \in S} \varphi_w(\theta_{L_w/K_\nu}))$.

These maps work well with finite abelian extensions: if $L_1 \subseteq L_2$ is a finite abelian extension of K then

$$\theta_{L_1/K}(a) = \theta_{L_2/K}(a)|_{L_1} \qquad \forall a \in \mathbb{I}_K$$

Thus, the collection $\theta_{L/K}$ where L ranges over finite abelian exensions of K in K^{ab} ordered by inclusion form a compatible system of homomorphism from $\mathbb{I}_K$ to $\varprojlim_L \text{Gal}_K(L) \cong \text{Gal}_K(K^{\text{ab}})$. By the universal property of profinite completion, they uniquely determine a continuous homomorphism. From this, we have built-up to the map:

Definition 3.3.1: Global Artin Homomorphism

Let K be a global field. Then the *global Artin homomorphism* is the continuous homomorphism:

$$\theta_K : \mathbb{I}_K \to \text{Gal}_K(K^{\text{ab}}) \cong \varprojlim_L \text{Gal}_K(L)$$

given by $\theta_{L/K} : \mathbb{I}_K \to \text{Gal}_K(L)$ where L ranges over all finite abelian extension of K in K^{ab}.

This map is the natural map that holds all the local artin homomorphism

Proposition 3.3.2: Universal Property Global Artin Map

Let K be a global field. Then the global Artin homomorphism θ_K is the unique continuous homomorphism $\mathbb{I}_K \to \text{Gal}_K(K^{\text{ab}})$ with the property that for every finite abelian extension L/K in K^{ab} and every place w of L lying over a place ν of K, the following diagram commutes:

$$\begin{tikzcd} K_v^\times \arrow[r, "\theta_{L_w/K_v}"] \arrow[d, hook, "\iota_v"] & \text{Gal}(L_w/K_v) \arrow[d, hook, "\varphi_w"] \\ \mathbb{I}_K \arrow[r, "\theta_{L/K}"] & \text{Gal}(L/K) \end{tikzcd}$$

where the homomorphism $\theta_{L/K}$ is defined by $\theta_{L/K}(a) = \theta_K(a)|_L$.

***Proof*:**
We just showed that θ_K satisfies the property, what's left to show is uniqueness.

For the sake of contradiction, assume that there is another continuous homomorphism $\theta'_K : \mathbb{I}_K \to \text{Gal}(K^{\text{ab}}/K)$ with the same property. We may view elements of $\text{Gal}(K^{\text{ab}}/K) \cong \varprojlim \text{Gal}(L/K)$ as elements of $\prod_{L/K} \text{Gal}(L/K)$, where L varies over finite abelian extensions of K in K^{ab}. If θ_K and θ'_K are not identical, then there must be an $a \in \mathbb{I}_K$ and a finite abelian extension L/K for which $\theta_{L/K}(a) \neq \theta'_{L/K}(a)$.

Let S be a finite set of places of K that includes all places v for which $a_v \notin \mathcal{O}_v^\times$ and all ramified places of L/K. Define $b \in \mathbb{I}_K$ by $b_v := 1$ for $v \in S$ and $b_v := a_v$ for $v \notin S$, so that $a = b \prod_{v\in S} \iota_v(a_v)$. Then $\theta_{L_w/K_v}(b_v) = 1$ for all places v, so we must have $\theta_{L/K}(b) = 1 = \theta'_{L/K}(b)$, and for $v \in S$ we have

$$\theta_{L/K}(\iota_v(a_v)) = \varphi_w(\theta_{L_w/K_v}(a_v)) = \theta'_{L/K}(\iota_v(a_v)),$$

by the commutativity of the diagram in the proposition. But then

$$\theta_{L/K}(a) = \theta_{L/K}(b) \prod_{v\in S} \theta_{L/K}(\iota_v(a_v)) = \theta'_{L/K}(b) \prod_{v\in S} \theta'_{L/K}(\iota_v(a_v)) = \theta'_{L/K}(a),$$

forcing $\theta'_K = \theta_K$, as we sought to show.

With this build-up, we can now state the global Artin reciprocity law

Theorem 3.3.3: Global Artin Reciprocity

Let K be a global field. Then the kernel of the global Artin homomorphism θ_K contains $K^\times$, and we get a continuous homomorphism:

$$\theta_K : C_K \to \mathrm{Gal}_K(K^{\mathrm{ab}})$$

where $C_K = \mathbb{I}_K/K^\times$, where every finite abelian extension $K \subseteq L \subseteq K^{\mathrm{ab}}$ we get the homomorphism

$$\theta_{L/K} : C_K \to \mathrm{Gal}_K(L)$$

by composing θ_K with the natural surjection $\mathrm{Gal}_K(K^{\mathrm{ab}}) \twoheadrightarrow \mathrm{Gal}_K(L)$ which has kernel $N_{L/K}(C_L)$, hence:

$$\frac{C_K}{N_{L/K}(C_L)} \cong \mathrm{Gal}_K(L)$$

***Proof*:**
This is just taking the quotient of the above map and following through the consequences, with the well-definedness needing to be the only non-immediate result to check.

Theorem 3.3.4: Global Existence Theorem

Let K be a global field. Then for every finite index open subgroup H of C_K, there exists a finite abelian extension $K \subseteq L \subseteq K^{\mathrm{ab}}$ where

$$N_{L/K}(C_L) = H$$

***Proof*:**
This is the existence half of global class field theory, and once reciprocity is in hand it follows from the isomorphism of theorem 3.3.5. Under $\widehat{\theta}_K : \widehat{C_K} \xrightarrow{\sim} \mathrm{Gal}_K(K^{\mathrm{ab}})$, an open finite-index subgroup $H \subseteq C_K$ has closure of finite index in $\widehat{C_K}$, carried to a closed finite-index subgroup of $\mathrm{Gal}_K(K^{\mathrm{ab}})$. By the Galois correspondence its fixed field L is a finite abelian extension of K, and by the reciprocity law $N_{L/K}(C_L) = \ker\theta_{L/K} = \theta_K^{-1}(\mathrm{Gal}_L(K^{\mathrm{ab}}))$, which is exactly H by construction. Thus $N_{L/K}(C_L) = H$, as sought. The reciprocity machinery this rests on is cited to [4].

Note that we have not specified whether θ_K is injective or surjective. It may in fact be one or the other:

1. When K is a number field, θ_K is surjective but not injective
2. When K is a global function field, θ_K is injective but not surjective, the image consisting on automorphisms $\sigma \in \mathrm{Gal}_K(K^{\mathrm{ab}})$ corresponding to integer powers of the Frobenius automorphism of $\mathrm{Gal}_k(k^{\mathrm{sep}})$ where k is the constant field of K

If we take the profinite completion will give an isomorphism:

Theorem 3.3.5: Main Theorem of Global Class Field Theory

Let K be a global field. Then the global Artin homomorphism θ_K induces a natural isomorphism

$$\widehat{\theta}_K : \widehat{C_K} \xrightarrow{\cong} \mathrm{Gal}_K(K^{\mathrm{ab}})$$

of profinite groups

***Proof*:**
This is the global reciprocity law, and as in the local case it comes from a *fundamental class*. The idèle class group C_K is a class formation for the absolute Galois group of K: for each finite abelian L/K the group $H^2(\mathrm{Gal}_K(L), C_L)$ is cyclic of order $[L : K]$ with a canonical generator, the global fundamental class, whose invariant is $1/[L : K]$ under the global invariant map $\mathrm{inv}_K \colon \mathrm{Br}(K) \hookrightarrow \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z}$ furnished by the Albert–Brauer–Hasse–Noether sequence. Cup product with this class (Tate–Nakayama for the global class formation) gives $C_K/N_{L/K}(C_L) \cong \mathrm{Gal}_K(L)$ for each L, compatibly under restriction; passing to the limit over L and completing yields $\widehat{\theta}_K \colon \widehat{C_K} \xrightarrow{\sim} \mathrm{Gal}_K(K^{\mathrm{ab}})$. The idèle-class class formation, the global fundamental class, and the ABHN sequence are established in *Group and Galois Cohomology* [4, Idèle Cohomology and the Global Fundamental Class].

We thus get the corerspondence

$$\left\{\begin{matrix}\text{Finite Extension } E/K \\ K \subseteq E \subseteq K^{\mathrm{ab}}\end{matrix}\right\} \underset{(K^{\mathrm{ab}})^{\theta_K(H)}}{\overset{N_{E/K}(C_E)}{\rightleftarrows}} \left\{\begin{matrix}\text{closed subgroups } H \\ H \subseteq \mathrm{Aut}_K(K^{\mathrm{ab}})\end{matrix}\right\}$$

with a corresponding isomorphism $C_K/H \cong \mathrm{Gal}_K(E)$ where $H = N_{E/K}(C_E)$. The global Artin homomorphism also has the following functorial property that can be thought of as the first part of Langland's program

Theorem 3.3.6: Functoriality of Artin Homomorphism

Let L/K be a finite separable extension of global fields (not necessarily abelian). Writing Res for the restriction $\mathrm{Gal}_L(L^{\mathrm{ab}}) \to \mathrm{Gal}_K(K^{\mathrm{ab}})$, $\sigma \mapsto \sigma|_{K^{\mathrm{ab}}}$ (well-defined since $K^{\mathrm{ab}} \subseteq L^{\mathrm{ab}}$), the diagram

$$\begin{tikzcd} C_L \arrow[r, "\theta_L"] \arrow[d, "N_{L/K}"'] & \mathrm{Gal}_L(L^{\mathrm{ab}}) \arrow[d, "\mathrm{Res}"] \\ C_K \arrow[r, "\theta_K"'] & \mathrm{Gal}_K(K^{\mathrm{ab}}) \end{tikzcd}$$

commutes; that is, $\theta_K \circ N_{L/K} = \mathrm{Res} \circ \theta_L$.

***Proof*:**
The global Artin map is assembled from the local ones (theorem 3.1.2): for an idèle $a = (a_w) \in \mathbb{I}_L$ one has $\theta_L(a) = \prod_w \theta_{L_w}(a_w)$, and similarly for K, so it suffices to check commutativity place by place. Fix a place v of K and a place $w|v$ of L. The local Artin maps satisfy $\theta_{K_v} \circ N_{L_w/K_v} = \mathrm{Res} \circ \theta_{L_w}$, a functoriality property of the local reciprocity map [4, Local

Reciprocity]. Since the global norm $N_{L/K}$ is the product of the local norms N_{L_w/K_v}, taking the product over all places gives $\theta_K \circ N_{L/K} = \mathrm{Res} \circ \theta_L$ on $\mathbb{I}_L$, which descends to the idèle class groups because θ_K is trivial on $K^\times$.

3.3.1 Relating to Ideal-Theoretic Version

In the ideal-theoretic case (the places approach), we directly worked with global fields, and hence needed to use a modulus to isolate the ramifying primes. This shall could be "resolved" instead with the right choice of open subgroup to mod C_K. Let K be a number field and let $\mathfrak{m} : M_K \to \mathbb{Z}_{\geq 0}$ be a modulus for K, which we view as a formal product $\mathfrak{m} = \prod_v v^{e_v}$ over the places v of K with $e_v \leq 1$ when v is Archimedean and $e_v = 0$ when v is complex. For each place v we define the open subgroup

$$U_K^{\mathfrak{m}}(v) := \begin{cases} \mathcal{O}_v^\times & \text{if } v \nmid \mathfrak{m}, \text{ where } \mathcal{O}_v^\times := K_v^\times \text{ when } v \text{ is infinite}), \\ \mathbb{R}_{>0} & \text{if } v | \mathfrak{m} \text{ is real, where } \mathbb{R}_{>0} \subseteq \mathbb{R}^\times \cong \mathcal{O}_v^\times := K_v^\times, \\ 1 + \mathfrak{p}^{e_v} & \text{if } v|\mathfrak{m} \text{ is finite, where } \mathfrak{p} = \{x \in \mathcal{O}_v : |x|_v < 1\}, \end{cases}$$

and let $U_K^{\mathfrak{m}} := \prod_v U_K^{\mathfrak{m}}(v) \subseteq \mathbb{I}_K$ denote the corresponding open subgroup of $\mathbb{I}_K$. The image $\overline{U}_K^{\mathfrak{m}}$ of $U_K^{\mathfrak{m}}$ in the idele class group $C_K = \mathbb{I}_K / K^\times$ is a finite index open subgroup. The idelic version of a ray class group is the quotient

$$C_K^{\mathfrak{m}} := \mathbb{I}_K / (U_K^{\mathfrak{m}} K^\times) = C_K / \overline{U}_K^{\mathfrak{m}},$$

and we have isomorphisms

$$C_K^{\mathfrak{m}} \cong \mathrm{Cl}_K^{\mathfrak{m}} \cong \mathrm{Gal}(K(\mathfrak{m})/K),$$

where $\mathrm{Cl}_K^{\mathfrak{m}}$ is the ray class group for the modulus $\mathfrak{m}$, and $K(\mathfrak{m})$ is the corresponding ray class field, which we can now define as the finite abelian extension L/K for which

$$\mathrm{N}_{L/K}(C_L) = \overline{U}_K^{\mathfrak{m}}$$

which we know exists by the Global Existence Theorem.

If L/K is any finite abelian extension, then $\mathrm{N}_{L/K}(C_L)$ contains $\overline{U}_K^{\mathfrak{m}}$ for some modulus $\mathfrak{m}$; this follows from the fact that the groups $\overline{U}_K^{\mathfrak{m}}$ form a fundamental system of open neighborhoods of the identity. Indeed, the conductor of the extension L/K is precisely the minimal modulus $\mathfrak{m}$ for which this is true. It follows that every finite abelian extension L/K lies in a ray class field $K(\mathfrak{m})$, with $\mathrm{Gal}(L/K)$ isomorphic to a quotient of a ray class group $C_K^{\mathfrak{m}}$.

3.4 The Kronecker-Weber Theorem

The archetype of the whole theory is the case $K = \mathbb{Q}$, where the abelian extensions are named explicitly: they are the subfields of the cyclotomic fields. Class field theory over $\mathbb{Q}$ (the reciprocity and existence theorems already established) turns this into a short and structural argument, since the Galois groups of the $\mathbb{Q}(\zeta_n)$ assemble into precisely the group realized by the idèle class group of $\mathbb{Q}$.

Theorem 3.4.1: Cyclotomic Reciprocity Over $\mathbb{Q}$

Fix $n \geq 1$ and set the modulus $\mathfrak{m} = (n)\infty$. The ideal-theoretic Artin map of section 2.1 descends to a canonical isomorphism from the ray *class* group

$$\mathrm{Cl}_{\mathbb{Q}}^{(n)\infty} = \mathcal{I}_{\mathbb{Q}}^{\mathfrak{m}}/\mathcal{R}_{\mathbb{Q}}^{(n)\infty} \xrightarrow{\cong} \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \xrightarrow[\omega]{\cong} (\mathbb{Z}/n\mathbb{Z})^{\times},$$

sending the class of (p) (for $p \nmid n$) to $\mathrm{Frob}_p \mapsto (p \bmod n)$, where ω is characterized by $\sigma(\zeta_n) = \zeta_n^{\omega(\sigma)}$. These isomorphisms are compatible with the transition maps $(\mathbb{Z}/nm\mathbb{Z})^{\times} \twoheadrightarrow (\mathbb{Z}/n\mathbb{Z})^{\times}$ and $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_{nm})) \twoheadrightarrow \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$.

***Proof*:**
That $\psi^{\mathfrak{m}}_{\mathbb{Q}(\zeta_n)/\mathbb{Q}}$ is surjective with $\mathrm{Frob}_p \mapsto (p \bmod n)$ under ω is the computation of section 2.1 for the cyclotomic extension (the ramified primes are exactly those dividing n, and $\mathbb{Q}(\zeta_n)$ is a ray class field for $\mathfrak{m} = (n)\infty$). Reciprocity in the form of theorem 2.4.8 identifies the kernel with the Takagi group; for the cyclotomic field this is the ray group $\mathcal{R}_{\mathbb{Q}}^{(n)\infty}$, so the induced map on $\mathrm{Cl}_{\mathbb{Q}}^{(n)\infty}$ is an isomorphism onto $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$, and composing with ω gives $(\mathbb{Z}/n\mathbb{Z})^{\times}$. Compatibility across n is the functoriality of the Artin map under the inclusions $\mathbb{Q}(\zeta_n) \subseteq \mathbb{Q}(\zeta_{nm})$, which restricts Frob_p compatibly and matches the reduction $(\mathbb{Z}/nm\mathbb{Z})^{\times} \twoheadrightarrow (\mathbb{Z}/n\mathbb{Z})^{\times}$.

Theorem 3.4.2: Global Kronecker-Weber Theorem

Every finite abelian extension $L/\mathbb{Q}$ is contained in a cyclotomic field $\mathbb{Q}(\zeta_n)$ for some $n \geq 1$. Equivalently, $\mathbb{Q}^{\mathrm{cyc}} := \bigcup_{n\geq 1} \mathbb{Q}(\zeta_n) = \mathbb{Q}^{\mathrm{ab}}$, and the reciprocity map induces $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}^{\mathrm{ab}}) \cong \varprojlim_n (\mathbb{Z}/n\mathbb{Z})^{\times} = \widehat{\mathbb{Z}}^{\times}$.

***Proof*:**
Taking the inverse limit of the compatible isomorphisms of theorem 3.4.1 gives $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}^{\mathrm{cyc}}) = \varprojlim_n \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong \varprojlim_n (\mathbb{Z}/n\mathbb{Z})^{\times} = \widehat{\mathbb{Z}}^{\times}$. On the arithmetic side, the idèle class group of $\mathbb{Q}$ decomposes as $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times}$, because $\mathbb{Q}$ has class number 1 and $\mathbb{Z}^{\times} = \{\pm 1\}$: every idèle is a rational multiple of a unique idèle in $\mathbb{R}_{>0} \times \prod_p \mathbb{Z}_p^{\times}$, the sign being absorbed by $\{\pm 1\} \subseteq \mathbb{Q}^{\times}$. The connected component of the identity is the archimedean factor $\mathbb{R}_{>0}$, so the profinite completion (the maximal totally disconnected quotient, quotient by the connected component) is $\widehat{C_{\mathbb{Q}}} \cong \widehat{\mathbb{Z}}^{\times}$. By theorem 3.3.5 the completed reciprocity map is an isomorphism $\widehat{\theta}_{\mathbb{Q}} \colon \widehat{C_{\mathbb{Q}}} \xrightarrow{\cong} \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}^{\mathrm{ab}})$, so $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}^{\mathrm{ab}}) \cong \widehat{\mathbb{Z}}^{\times}$.

It remains to promote this equality of profinite groups to the equality of fields $\mathbb{Q}^{\mathrm{cyc}} = \mathbb{Q}^{\mathrm{ab}}$; matching Galois groups abstractly does not by itself force the fields to coincide, and one must rule out a proper containment $\mathbb{Q}^{\mathrm{cyc}} \subsetneq \mathbb{Q}^{\mathrm{ab}}$. Under $\widehat{\theta}_{\mathbb{Q}}$ the field $\mathbb{Q}^{\mathrm{cyc}}$ corresponds to the norm subgroup $N := \bigcap_n N_{\mathbb{Q}(\zeta_n)/\mathbb{Q}}(C_{\mathbb{Q}(\zeta_n)}) = \bigcap_n \overline{U}_{\mathbb{Q}}^{(n)\infty}$ of $C_{\mathbb{Q}}$. The open subgroups $\overline{U}_{\mathbb{Q}}^{(n)\infty}$, running over all n, form a fundamental system of open neighborhoods of the identity in $\widehat{C_{\mathbb{Q}}}$, so their intersection is trivial; by the Global Existence Theorem (theorem 3.3.4) the trivial subgroup corresponds to $\mathbb{Q}^{\mathrm{ab}}$. Hence $\mathbb{Q}^{\mathrm{cyc}}$ and $\mathbb{Q}^{\mathrm{ab}}$ have equal norm subgroups and therefore coincide. Consequently any finite abelian $L/\mathbb{Q}$ lies in $\mathbb{Q}^{\mathrm{ab}} = \mathbb{Q}^{\mathrm{cyc}}$, and being finite, in $\mathbb{Q}(\zeta_n)$ for some n;

concretely one may take n divisible by the finite part of the conductor $\mathfrak{c}(L/\mathbb{Q})$ (theorem 2.4.8).

Theorem 3.4.3: Local Kronecker-Weber Theorem

Every finite abelian extension of $\mathbb{Q}_p$ is contained in a cyclotomic extension $\mathbb{Q}_p(\zeta_m)$ of $\mathbb{Q}_p$ for some $m \geq 1$.

***Proof*:**
This is the local statement, established in *Local Field Theory* [5, Local Kronecker-Weber Theorem]. It assembles to the global theorem prime by prime: for finite abelian $L/\mathbb{Q}$ and each ramified p, the completion $L_{\mathfrak{p}}/\mathbb{Q}_p$ embeds in some $\mathbb{Q}_p(\zeta_{m_p})$; writing $m_p = p^{n_p} m_p'$ with $p \nmid m_p'$, the prime-to-p factor is unramified and the p-power factor $\mathbb{Q}_p(\zeta_{p^{n_p}})$ carries the ramification, and $n = \prod_p p^{n_p}$ gives $L \subseteq \mathbb{Q}(\zeta_n)$. This agrees with the class-field-theoretic argument above.

3.5 Hilbert Symbols and Explicit Reciprocity

Local Artin reciprocity attaches to every element of $K^\times$ an automorphism of K^{ab}; pairing this automorphism against an n-th root packages the whole of local class field theory into a single computable symbol. The resulting *Hilbert symbol* converts the abstract reciprocity isomorphism into an arithmetic invariant, and its product formula across all places recovers, and vastly generalizes, Gauss's quadratic reciprocity.

Definition 3.5.1: Local Hilbert Symbol

Let K be a local field containing the group μ_n of n-th roots of unity (so $\operatorname{char} K \nmid n$), and let $\theta = \theta_K \colon K^\times \to \mathrm{Gal}_K(K^{\mathrm{ab}})$ be the local Artin map of theorem 3.1.2. For $a, b \in K^\times$ the *local Hilbert symbol* is

$$(a,b)_K := \frac{\theta_a\left(\sqrt[n]{b}\right)}{\sqrt[n]{b}} \in \mu_n,$$

where $\theta_a := \theta(a)$ acts on the Kummer extension $K(\sqrt[n]{b})/K$ and $\sqrt[n]{b}$ is any fixed n-th root of b.

The quotient lies in μ_n and is independent of the chosen root: any Galois conjugate of $\sqrt[n]{b}$ differs from it by an element of $\mu_n \subseteq K$, on which θ_a acts trivially. Thus $(a,b)_K$ measures precisely how far a is from being a norm out of $K(\sqrt[n]{b})$.

Theorem 3.5.2: Properties of the Local Hilbert Symbol

Let K be a local field containing μ_n. The pairing $(\,\cdot\,,\cdot\,)_K : K^\times \times K^\times \to \mu_n$ satisfies, for all $a, b \in K^\times$:

1. *(Bilinearity)* $(aa', b)_K = (a, b)_K\,(a', b)_K$ and $(a, bb')_K = (a, b)_K\,(a, b')_K$;
2. *(Skew-symmetry)* $(a, b)_K = (b, a)_K^{-1}$;
3. *(Steinberg and $-a$ relations)* $(a, 1-a)_K = 1$ whenever $a, 1-a \in K^\times$, and $(a, -a)_K = 1$;
4. *(Non-degeneracy)* if $(a, b)_K = 1$ for all $b \in K^\times$, then $a \in (K^\times)^n$.

***Proof*:**
Bilinearity is immediate from definition 3.5.1: θ is a homomorphism in a, and the choice $\sqrt[n]{bb'} = \sqrt[n]{b}\,\sqrt[n]{b'}$ (up to μ_n, which is fixed) gives the second identity. The remaining structure is the cup-product description of the symbol. Under the Kummer isomorphism $K^\times/(K^\times)^n \cong H^1(K, \mu_n)$ (using $\mu_n \subseteq K$ to identify $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$), the class of a is a 1-cocycle χ_a, and

$$(a, b)_K = \mathrm{inv}_K\big(\chi_a \cup \chi_b\big),$$

where $\cup : H^1(K, \mu_n) \times H^1(K, \mu_n) \to H^2(K, \mu_n^{\otimes 2})$ is the cup product and $\mathrm{inv}_K : H^2(K, \mu_n^{\otimes 2}) \cong \frac{1}{n}\mathbb{Z}/\mathbb{Z}$ the invariant map (the identity requires matching the uniformizer-to-Frobenius normalization of theorem 3.1.2 with the fundamental-class normalization of the invariant map). Skew-symmetry follows from graded-commutativity of the cup product on degree-one classes; the relations $(a, 1-a)_K = 1$ and $(a, -a)_K = 1$ are the Steinberg relations for this cup product; and non-degeneracy is local Tate duality. The cup product, the invariant map, and the duality are developed in *Group and Galois Cohomology* [4, cup product, invariant map].

Non-degeneracy is what makes the symbol a genuine *pairing*: it recovers the norm-residue isomorphism, the Kummer-theoretic incarnation of local reciprocity. Summing these local symbols over all places of a global field yields a single global constraint, the source of every classical reciprocity law.

Theorem 3.5.3: Hilbert Reciprocity (Product Formula)

Let K be a global field containing μ_n, and let $a, b \in K^\times$. For each place v write $(a, b)_{K_v}$ for the local Hilbert symbol of the completion K_v. Then $(a, b)_{K_v} = 1$ for all but finitely many v, and

$$\prod_v (a, b)_{K_v} = 1$$

as an identity in μ_n.

***Proof*:**
Fix b and set $L = K(\sqrt[n]{b})$, abelian of exponent dividing n. For each place v the local symbol $(a, b)_{K_v}$ records the action of $\theta_{K_v}(a)$ on $\sqrt[n]{b}$, and the local maps are compatible with the global Artin map θ_K of theorem 3.3.5 under $\mathrm{Gal}_K(K^{\mathrm{ab}}) \twoheadrightarrow \mathrm{Gal}_K(L)$. Assembling the local components, $\prod_v (a, b)_{K_v}$ is the action on $\sqrt[n]{b}$ of θ_K applied to the diagonal (principal) idèle of $a \in K^\times$. But

θ_K factors through $C_K = \mathbb{I}_K/K^\times$, so it kills principal idèles; hence the global action is trivial, which is exactly $\prod_v (a,b)_{K_v} = 1$. For finiteness: at a place $v \nmid n$ where a, b are both units, L/K is unramified and the tame symbol vanishes, so $(a,b)_{K_v} = 1$; only finitely many places fail this.

The product formula says the local symbols are not independent: knowing all but one determines the last. Specializing to $K = \mathbb{Q}$ and $n = 2$ turns this single relation into Gauss's law.

Corollary 3.5.4: Quadratic Reciprocity over $\mathbb{Q}$

For distinct odd primes p and q, $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$.

***Proof*:**
Take $K = \mathbb{Q}$, $n = 2$, $a = p$, $b = q$; here $\mu_2 = \{\pm 1\} \subseteq \mathbb{Q}^\times$. Hilbert reciprocity (theorem 3.5.3) gives $\prod_v (p,q)_v = 1$. Evaluating the local factors: at $v = \infty$, $q > 0$ is a square in $\mathbb{R}^\times$ so $(p,q)_\infty = 1$; at odd $\ell \notin \{p,q\}$ both are units and $\mathbb{Q}_\ell(\sqrt{q})/\mathbb{Q}_\ell$ is unramified, so $(p,q)_\ell = 1$ (tame vanishing); the tame evaluation at the odd places p, q gives $(p,q)_p = \left(\frac{q}{p}\right)$ and $(p,q)_q = \left(\frac{p}{q}\right)$; and the dyadic computation gives $(p,q)_2 = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$. Setting the product of these to 1 and rearranging (each factor being ± 1) yields the stated identity.

Quadratic reciprocity is thus the $n = 2$, $K = \mathbb{Q}$ case of a single adelic statement: the global Artin map annihilates principal idèles, so the local obstructions to being a norm must cancel place by place.

3.6 The Hasse Norm Theorem

Reciprocity has a Diophantine payoff. Whether $x \in K^\times$ is a norm from L is, a priori, a hard global question; over each completion it is elementary, decided by a Hilbert symbol. For *cyclic* extensions the two levels coincide exactly, the passage from local data to a global solution being furnished by the cohomology of the idèle class group. For a place v of K and $w|v$ of L, since L/K is Galois the local norm subgroup $N_{L_w/K_v}(L_w^\times) \subseteq K_v^\times$ is independent of $w|v$, so "x is a local norm at v" is unambiguous.

Theorem 3.6.1: Hasse Norm Theorem

Let L/K be a finite cyclic extension of global fields and $x \in K^\times$. Then x is a *global norm* ($x = N_{L/K}(y)$ for some $y \in L^\times$) if and only if x is a *local norm everywhere*: $x \in N_{L_w/K_v}(L_w^\times)$ for every place v of K.

The forward direction is formal: a global norm is a local norm at each v. The content is the converse, where cyclicity enters.

***Proof*:**
Let $G = \mathrm{Gal}_K(L)$, cyclic of order n, and let $\mathbb{I}_L$ be the idèle group of L. The short exact sequence of G-modules $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ has a long exact sequence in Tate cohomology, part of which reads

$$\widehat{H}^{-1}(G, C_L) \longrightarrow \widehat{H}^0(G, L^\times) \xrightarrow{\alpha} \widehat{H}^0(G, \mathbb{I}_L).$$

Here $\widehat{H}^0(G, L^\times) = K^\times/N_{L/K}(L^\times)$ is the group of global norm classes, and $\widehat{H}^0(G, \mathbb{I}_L) = \prod_v{}' K_v^\times/N_{L_w/K_v}(L_w^\times)$ is the restricted product of local norm classes, with α sending $[x]$ to its local components. Thus x is a local norm at every place precisely when $\alpha([x]) = 0$, that is, when $[x] \in \ker\alpha = \operatorname{im}\big(\widehat{H}^{-1}(G, C_L) \to \widehat{H}^0(G, L^\times)\big)$.

It therefore suffices to show $\widehat{H}^{-1}(G, C_L) = 0$. Since G is cyclic, cyclic periodicity ([4, Cyclic Tate Periodicity]) gives $\widehat{H}^{-1}(G, C_L) \cong \widehat{H}^1(G, C_L)$, and $\widehat{H}^1(G, C_L) = 0$ is an axiom of the global class formation, established via the idèle-class cohomology of [4, Idèle Cohomology and the Global Fundamental Class]. Hence $\ker\alpha = 0$, and every everywhere-local norm is a global norm, as we sought to show.

Corollary 3.6.2: Hasse Symbol Criterion for Quadratic Extensions

Let $L = K(\sqrt{d})/K$ be a quadratic extension of number fields. Then $x \in K^\times$ is a global norm from L if and only if the Hilbert symbol $(x, d)_v = 1$ at *every* place v of K.

***Proof*:**
A quadratic extension is cyclic, so theorem 3.6.1 applies. For each v the local norm group $N_{L_w/K_v}(L_w^\times)$ is the kernel of $(-, d)_v$, so "x is a local norm at v" is exactly "$(x, d)_v = 1$"; the theorem equates being a global norm with the vanishing of every local symbol.

The emphasis on *each* symbol is the whole force of the theorem. The product formula $\prod_v (x, d)_v = 1$ holds for every x by theorem 3.5.3, independently of whether x is a norm; a product of 1 is consistent with two symbols equal to -1 that cancel. Hasse's content is that being a global norm forces, and is forced by, the vanishing of each factor separately.

3.6.3 Warning: Cyclicity is Essential The Hasse norm theorem can fail once G is non-cyclic. The smallest failures are biquadratic: for $L = \mathbb{Q}(\sqrt{13}, \sqrt{17})$ with $\mathrm{Gal}_{\mathbb{Q}}(L) \cong (\mathbb{Z}/2\mathbb{Z})^2$, there exist elements of $\mathbb{Q}^\times$ that are local norms at every place yet are not global norms. Cohomologically the obstruction is the *knot group* $\widehat{H}^{-1}(G, C_L)$, which for this field is $\mathbb{Z}/2\mathbb{Z}$; the cyclic periodicity $\widehat{H}^{-1} \cong \widehat{H}^1$ used in the proof, which forces this group to vanish, is available only for cyclic G, and that is exactly where the hypothesis is spent.

3.7 The Conductor-Discriminant Formula

The conductor of a single ray class character records how deeply the associated cyclic character ramifies at each prime, a purely local datum read off from the higher ramification filtration of [5]. Assembling these data over the whole character group of $\mathrm{Gal}_K(L)$ reconstructs a global invariant: the discriminant of L/K. Throughout, L/K is finite abelian with $G = \mathrm{Gal}_K(L)$ and character group $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$; fixing a modulus $\mathfrak{m}$ divisible by $\mathfrak{c}(L/K)$, the Artin map $\psi_{L/K}^{\mathfrak{m}}$ (theorem 2.1.25) realizes each $\chi \in \widehat{G}$ as a ray class character $\chi \circ \psi_{L/K}^{\mathfrak{m}}$, whose conductor $\mathfrak{c}(\chi)$ (definition 2.3.3) is computed prime by prime by the local theory.

Proposition 3.7.1: Character Conductor via the Ramification Filtration

Let $\chi \in \widehat{G}$ and let $K_\chi = L^{\ker\chi}$ be the cyclic extension it cuts out. Then $\mathfrak{c}(\chi) = \mathfrak{c}(K_\chi/K)$, and for each nonarchimedean prime $\mathfrak{p}$ the exponent $\nu_\mathfrak{p}(\mathfrak{c}(\chi))$ is the last upper-numbering ramification break of $(K_\chi)_u/K_\nu$ (for $u|\mathfrak{p}$), computed from the higher ramification groups of [5].

***Proof*:**
Since χ factors through $\mathrm{Gal}_K(K_\chi) = G/\ker\chi$ as a faithful character, $\ker(\chi \circ \psi^{\mathfrak{m}}_{L/K}) = (\psi^{\mathfrak{m}}_{L/K})^{-1}(\ker\chi)$ is the congruence subgroup that is the kernel of $\psi^{\mathfrak{m}}_{K_\chi/K}$; by definition 2.3.3 and theorem 2.4.8 its conductor is $\mathfrak{c}(K_\chi/K)$. Locally, upper numbering is compatible with the quotient $G_v \to \mathrm{Gal}_{K_\nu}((K_\chi)_u)$ ([5, upper numbering passes to quotients]), so $\chi|_{G^v} = 1$ iff $G^v \subseteq \ker\chi$, i.e. iff the ramification group of $(K_\chi)_u/K_\nu$ in upper numbering v is trivial. Hence $\nu_\mathfrak{p}(\mathfrak{c}(\chi))$ is the last break of $(K_\chi)_u/K_\nu$, which by the local conductor computation (definition 2.4.1, theorem 3.1.2) is $\nu_\mathfrak{p}(\mathfrak{c}((K_\chi)_u/K_\nu))$, as claimed.

Theorem 3.7.2: Conductor-Discriminant Formula

Let L/K be a finite abelian extension of number fields with Galois group G and character group $\widehat{G}$. Then the relative discriminant factors as

$$\mathrm{disc}_{L/K} = \prod_{\chi\in\widehat{G}} \mathfrak{c}(\chi),$$

the product over all characters of their conductors, where $\mathfrak{c}(\chi)$ is the conductor of the ray class character $\chi \circ \psi^{\mathfrak{m}}_{L/K}$ (equivalently the Artin conductor of the one-dimensional representation χ).

***Proof*:**
Both sides are ideals of $\mathcal{O}_K$, so it suffices to fix a nonarchimedean prime $\mathfrak{p}$ and match the exponents. Fix $w|\mathfrak{p}$ of L and set $D = \mathrm{Gal}_{K_\nu}(L_w)$, the decomposition group, with lower-numbering ramification groups D_i of orders $g_i = |D_i|$ and residue degree $f = [D : D_0]$; since L/K is abelian these are independent of $w|\mathfrak{p}$, and there are $r = [G : D]$ primes above $\mathfrak{p}$.

Left side. The discriminant is the norm of the different, so summing over the r primes above $\mathfrak{p}$ (each with residue degree f and different exponent $\sum_{i\geq 0}(g_i - 1)$),

$$\nu_\mathfrak{p}\big(\mathrm{disc}_{L/K}\big) = r \cdot f \cdot \sum_{i\geq 0}(g_i - 1) = [G : D]\, f \sum_{i\geq 0}(g_i - 1),$$

using the different-ramification formula $v_w(\mathfrak{d}_{L_w/K_\nu}) = \sum_{i\geq 0}(g_i - 1)$ of [5, the different and the ramification filtration].

Right side. By proposition 3.7.1, $\nu_\mathfrak{p}(\mathfrak{c}(\chi))$ depends only on $\chi|_D$ and equals $\sum_{i\geq 0} \frac{1}{[D_0:D_i]}\big(1 - [\chi|_{D_i} = 1]\big)$. Summing over $\chi \in \widehat{G}$ and grouping by the restriction $\psi = \chi|_D \in \widehat{D}$ (each hit exactly $[G:D]$ times),

$$\sum_{\chi\in\widehat{G}} \nu_{\mathfrak{p}}\big(\mathfrak{c}(\chi)\big) = [G:D]\sum_{i\geq 0}\frac{1}{[D_0:D_i]}\sum_{\psi\in\widehat{D}}\big(1-[\,\psi|_{D_i}=1\,]\big).$$

By orthogonality the number of $\psi\in\widehat{D}$ trivial on D_i is $|D|/g_i$, so the inner sum is $|D|-|D|/g_i = |D|(g_i-1)/g_i$. With $[D_0:D_i]=g_0/g_i$ and $|D|=fg_0$, the summand becomes $\frac{g_i}{g_0}\cdot fg_0\cdot\frac{g_i-1}{g_i} = f(g_i-1)$, hence

$$\sum_{\chi\in\widehat{G}} \nu_{\mathfrak{p}}\big(\mathfrak{c}(\chi)\big) = [G:D]\,f\sum_{i\geq 0}(g_i-1) = \nu_{\mathfrak{p}}\big(\operatorname{disc}_{L/K}\big),$$

matching the left side. As $\mathfrak{p}$ was arbitrary, $\operatorname{disc}_{L/K}=\prod_\chi \mathfrak{c}(\chi)$, completing the proof.

The formula repackages the wildness of ramification, spread across the different by the filtration (D_i), as a sum of one-dimensional conductors, one per irreducible character of the abelian group G.

Corollary 3.7.3: Discriminant of a Cyclic Extension of Prime Degree

Let L/K be cyclic of prime degree ℓ with conductor $\mathfrak{c}(L/K)$. Then $\operatorname{disc}_{L/K}=\mathfrak{c}(L/K)^{\ell-1}$.

***Proof*:**
The ℓ characters of $G\cong\mathbb{Z}/\ell\mathbb{Z}$ are the trivial character (conductor (1)) and $\ell-1$ faithful characters, each cutting out $K_\chi = L$, so with conductor $\mathfrak{c}(L/K)$ by proposition 3.7.1. The product $\prod_\chi \mathfrak{c}(\chi)=\mathfrak{c}(L/K)^{\ell-1}$ equals $\operatorname{disc}_{L/K}$ by theorem 3.7.2, as we sought to show.

Applied to $\mathbb{Q}(\zeta_p)/\mathbb{Q}$ (p odd), the $p-2$ nontrivial characters each have p-conductor (p), recovering $\operatorname{disc}(\mathbb{Q}(\zeta_p))=\pm p^{p-2}$ (compare example 2.4.3).

3.8 Application: Chebotarev Density Theorem

We need some build-up

Proposition 3.8.1: Dirichlet Density to Ray Class Element

Let $\mathfrak{m}$ be a modulus for a number field K, and let $\mathrm{Cl}_K^{\mathfrak{m}}$ be the corresponding ray class group. Then for every ray class $c\in\mathrm{Cl}_K^{\mathfrak{m}}$, the Dirichlet density of the set of primes $\mathfrak{p}$ of K lie in c is

$$\frac{1}{\#\mathrm{Cl}_K^{\mathfrak{m}}}$$

***Proof*:**
Apply corollary 2.3.10 to $\mathcal{C}=\mathcal{R}_K^{\mathfrak{m}}$

As the ray class fields exists (theorem 2.4.8), we get the abelian case. We next require one more result for the main theorem:

Corollary 3.8.2: Dirichlet Density singleton Conjugacy

Let L/K be a finite abelian extension of number fields with galois group G. Then for every $\sigma \in G$, the Dirichlet density of the set S of primes $\mathfrak{p}$ of K unramified in L for which $\mathrm{Frob}_{\mathfrak{p}} = \{\sigma\}$ is

$$\frac{1}{\#G}$$

***Proof*:**
Let $\mathfrak{m} = \mathrm{cond}(L/K)$ be the conductor of the extension L/K; then L is a subfield of the ray class field $K(\mathfrak{m})$ and $\mathrm{Gal}(L/K) \cong \mathrm{Cl}_K^{\mathfrak{m}}/H$ for some subgroup H of the ray class group. For each unramified prime $\mathfrak{p}$ of K we have $\mathrm{Frob}_{\mathfrak{p}} = \{\sigma\}$ if and only if $\mathfrak{p}$ lies in one of the ray classes contained in the coset of H in $\mathrm{Cl}_K^{\mathfrak{m}}/H$ corresponding to σ. The Dirichlet density of the set of primes in each ray class is $1/\#\mathrm{Cl}_K^{\mathfrak{m}}$, by proposition 3.8.1, and there are $\#H$ ray classes in each coset of H; thus $d(S) = \#H/\#\mathrm{Cl}_K^{\mathfrak{m}} = 1/\#G$ as we sought to show.

Theorem 3.8.3: Chebotarev Density Theorem

Let L/K be a finite galois extension of number fields with galois group $G = \mathrm{Gal}_K(L)$. Let $C \subseteq G$ be a union of conjugacy classes, and let S be the set of primes$\mathfrak{p}$ of K unramified in L with $\mathrm{Frob}_{\mathfrak{p}} \subseteq C$. Then:

$$d(S) = \frac{\#C}{\#G}$$

Note that G is not assumed to be abelian, hence in this case $\mathrm{Frob}_{\mathfrak{p}}$ is a conjugacy class

***Proof*:**
It suffices to consider the case where C is a single conjugacy class, which we now assume; we can reduce to this case by partitioning C into conjugacy classes and summing Dirichlet densities (Dirichlet density is finitely additive, so the density of a union of disjoint conjugacy classes is the sum of the individual densities). Let S be the set of primes $\mathfrak{p}$ of K unramified in L for which $\mathrm{Frob}_{\mathfrak{p}}$ is the conjugacy class C.

Let $\sigma \in G$ be a representative of the conjugacy class C, let $H_\sigma := \langle \sigma \rangle \subseteq G$ be the subgroup it generates, and let $F_\sigma := L^{H_\sigma}$ be the corresponding fixed field. Let T_σ be the set of primes $\mathfrak{q}$ of F_σ unramified in L for which $\mathrm{Frob}_{\mathfrak{q}} = \{\sigma\} \subseteq \mathrm{Gal}(L/F_\sigma) \subseteq \mathrm{Gal}(L/K)$ (note that the Frobenius class $\mathrm{Frob}_{\mathfrak{q}}$ is a singleton because $\mathrm{Gal}(L/F_\sigma) = H_\sigma$ is abelian). We have $d(T_\sigma) = 1/\#H_\sigma$, since L/F_σ is abelian, by corollary 3.8.2

Restricting to degree-1 primes (primes whose residue field has prime order) does not change Dirichlet densities, so let us replace S and T_σ by their subsets of degree-1 primes, and define $T_\sigma(\mathfrak{p}) := \{\mathfrak{q} \in T_\sigma : \mathfrak{q}|\mathfrak{p}\}$ for each $\mathfrak{p} \in S$.

Claim: For each prime $\mathfrak{p} \in S$ we have $\#T_\sigma(\mathfrak{p}) = [G : H_\sigma]$.

Proof of claim: Let $\mathfrak{r}$ be a prime of L lying above $\mathfrak{q} \in T_\sigma(\mathfrak{p})$. Such an $\mathfrak{r}$ is unramified, since $\mathfrak{p}$ is, and we have $\mathrm{Frob}_{\mathfrak{r}} = \sigma$, since $\mathrm{Frob}_{\mathfrak{q}} = \{\sigma\}$. It follows that $\mathrm{Gal}(\mathbb{F}_{\mathfrak{r}}/\mathbb{F}_{\mathfrak{q}}) = \langle\overline{\sigma}\rangle \simeq H_\sigma$. Therefore $f_{\mathfrak{r}/\mathfrak{q}} = \#H_\sigma$ and $\#\{\mathfrak{r}|\mathfrak{q}\} = 1$, since $\#H_\sigma = [L : F_\sigma] = \sum_{\mathfrak{r}|\mathfrak{q}} e_{\mathfrak{r}/\mathfrak{q}} f_{\mathfrak{r}/\mathfrak{q}}$. We have $f_{\mathfrak{r}/\mathfrak{p}} = f_{\mathfrak{r}/\mathfrak{q}} f_{\mathfrak{q}/\mathfrak{p}} = f_{\mathfrak{r}/\mathfrak{q}} = \#H_\sigma$, since $f_{\mathfrak{q}/\mathfrak{p}} = 1$ for degree-1 primes $\mathfrak{q}|\mathfrak{p}$, and $e_{\mathfrak{r}/\mathfrak{p}} = 1$, thus

$$\#G = [L : K] = \sum_{\mathfrak{r}|\mathfrak{p}} e_{\mathfrak{r}/\mathfrak{p}} f_{\mathfrak{r}/\mathfrak{p}} = \#\{\mathfrak{r}|\mathfrak{p}\}\#H_\sigma = \#T_\sigma(\mathfrak{p})\#H_\sigma,$$

We now observe that

$$\sum_{\mathfrak{p}\in S} \mathrm{N}(\mathfrak{p})^{-s} = \sum_{\sigma\in C}\sum_{\mathfrak{p}\in S} \frac{1}{[G : H_\sigma]} \sum_{\mathfrak{q}\in T_\sigma(\mathfrak{p})} \mathrm{N}(\mathfrak{q})^{-s} = \frac{\#C}{[G : H_\sigma]} \sum_{\mathfrak{q}\in T_\sigma} \mathrm{N}(\mathfrak{q})^{-s}$$

since $\mathrm{N}(\mathfrak{q}) = \mathrm{N}(\mathfrak{p})$ for each degree-1 prime $\mathfrak{q}$ lying above a degree-1 prime $\mathfrak{p}$, and therefore

$$d(S) = \frac{\#C}{[G : H_\sigma]} d(T_\sigma) = \frac{\#C\#H_\sigma}{[G : H_\sigma]} = \frac{\#C}{\#G}.$$

completing the proof

4

What Next?

Class field theory settles the abelian case: the abelian extensions of a global or local field are classified by the arithmetic of the field itself. What it leaves open is everything nonabelian, together with the reformulations that make the nonabelian case tractable. A few natural directions follow.

The nonabelian reciprocity: automorphic forms and Langlands. Artin reciprocity identifies the one-dimensional representations of $\mathrm{Gal}_K(K^{\mathrm{ab}})$ with the characters of the idèle class group, equivalently with certain Hecke characters. The Langlands program is the conjectural extension to n-dimensional Galois representations, matched with automorphic representations of GL_n over the adeles; the abelian theory of this book is precisely the GL_1 case. The local half of the story for GL_2 is developed in *Langlands for* GL_2 [6]. For the global automorphic picture, Bump's *Automorphic Forms and Representations* and Gelbart's *Automorphic Forms on Adele Groups* are standard entry points.

Explicit class field theory and complex multiplication. The Kronecker-Weber theorem constructs $\mathbb{Q}^{\mathrm{ab}}$ explicitly from roots of unity. Hilbert's twelfth problem asks for an analogous explicit construction over an arbitrary number field; it is solved only for $\mathbb{Q}$ (cyclotomic values) and for imaginary quadratic fields, where the abelian extensions are generated by special values of the j-invariant and of elliptic and modular functions. This theory of complex multiplication is developed in *Elliptic Curves* [3]; Silverman's *Advanced Topics in the Arithmetic of Elliptic Curves* and Cox's *Primes of the Form* $x^2 + ny^2$ are the standard references.

Arithmetic geometry and Shimura varieties. The modular curve, whose points parametrize elliptic curves, is the first example of a Shimura variety, and its cohomology realizes the Galois representations attached to modular forms. Shimura varieties are the geometric arena in which class field theory generalizes toward the reciprocity laws of the Langlands program; they are the subject of *Shimura Varieties* [7]. Milne's course notes and Deligne's Corvallis articles are the classical sources.

The cohomological and geometric reformulations. The cohomological machinery of chapter 1 is the tip of a larger structure: Poitou-Tate duality, the Galois cohomology of number fields, and the étale-cohomological reformulation of class field theory over both number fields and function fields.

Neukirch-Schmidt-Wingberg's *Cohomology of Number Fields* is the comprehensive reference; for the geometric, function-field side, the étale-cohomological viewpoint is the natural continuation.

The through-line is the one the book began with: reciprocity is the statement that the arithmetic of a field and its Galois symmetries are two descriptions of a single object. Each direction above extends that statement to wider classes of fields, representations, and spaces.

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